

High Priority

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner: County of Hawaii
Contractor: Nan, Inc. Project No.: WW-4705R
Project Name: Hilo WWTP Phase 1 Submittal Number:
Submittal Title: For Information Only
TO:
From: Nan Inc.

Specification No. and Subject of Submittal / Equipment Supplier	
Spec:	Paragraph:
Authorized By:	Date Submitted:

Submittal Certification		
Check Either (A) or (B):		
☐ (A)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with <u>no exceptions</u> .	
☐ (B)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings <u>except</u> for the deviations listed.	
Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.		
General Contractor's Reviewer's Signature: 		
Printed Name and Title:		
In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.		
Firm:	Signature:	Date Returned:

PM/CM Office Use
Date Received GC to PM/CM:
Date Received PM/CM to Reviewer:
Date Received Reviewer to PM/CM:
Date Sent PM/CM to GC:

Nan, Inc

PROJECT: HILO WWTP REHABILITATION
AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY
THIS CONTRACTOR. IT IS CERTIFIED
CORRECT, COMPLETE, AND IN
COMPLIANCE WITH CONTRACT
DRAWINGS AND SPECIFICATIONS. ALL
AFFECTED CONTRACTORS AND
SUPPLIERS ARE AWARE OF, AND WILL
INTEGRATE THIS SUBMITTAL (UPON
APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED _____
SPECIFICATION SECTION # _____
SPECIFICATION _____
PARAGRAPH _____
DRAWING _____
SUBCONTRACTOR _____
SUPPLIER _____
MANUFACTURER _____

CERTIFIED BY CQCM or Designee:  Written Consent Required

PREVIOUS SUBMITTAL COMMENTS



Submittal Review Response

Project Name:	Hilo WWTP Rehabilitation and Replacement Project Phase 1
Submittal No.:	15211-001.0
Date:	9/23/2025

Client:	County of Hawai'i	Carollo Project No.:	203975
Contractor:	Nan, Inc.		
Submittal Name:	Ductile Iron Pipe-AWWA C151		
Reviewed By:	Mason Drapesa-Morse, Preston Merrell, Kyle Leonard		

SUBMITTAL REVIEW

Review is for general compliance with contract documents. No responsibility is assumed by Carollo for correctness of quantities, dimensions, and details. No deviation or variation is approved unless specifically addressed in these review comments. Refer to Section 01330 for additional requirements. The Contractor shall assume full responsibility for coordination with all other trades and deviations from contract requirements.

Approved	☐	No Exceptions
	☐	Make Corrections Noted - See Comments
	☐	Make Corrections Noted - Confirm
Not Approved	☒	Correct and Resubmit
	☐	Rejected - See Remarks
Receipt Acknowledged	☐	Filed for Record
	☐	With Comments - Resubmit

Review Comments:

- Contractor's variance request for mechanical joint fittings is denied. Contractor shall provide integrally restrained mechanical joints or integrally restrained push-on joints as called out in the Piping Schedule in Section 15052 and Section 15211-2.01.B.5 and 6.
- Per Section 15211-1.03.C, submit manufacturer qualifications showing at least 5 years' experience with the specified material.
- Per Section 15211-1.03.D, submit manufacturer's coating and lining Quality Assurance Manual.
- Per Section 15211-1.03.E, provide shop drawings with layout, alignment, valves, fittings, joints, and connections, plus photos and descriptions of fittings, gaskets, couplings, linings, and coatings. Submittal only includes a tabular product list and a few cut sheets. If the intent of this package is limited to manufacturer/product approval, note that shop drawings are still required and must be submitted and approved prior to fabrication.

5. Per Section 15211-1.03.F, confirm that the manufacturer's test reports for glass lining (holiday detection, compliance with material requirements, and coating rep reports) will be provided prior to shipment of materials.
6. Per Section 15211-2.01.A.1, pipe must be manufactured by American, McWane, SIP, or U.S. Pipe. Submittal lists American (acceptable) and C&B Piping (not listed). C&B is not identified in the spec as an approved pipe or specialty manufacturer. Confirm that all pipe barrels will come from an approved manufacturer and that C&B's role is limited to fabricating specials from approved pipe. If C&B is proposed as a pipe or fitting manufacturer, it is not compliant.
7. Per Section 15211-2.01.A.2, pipe shall meet the pressure/thickness class in the piping schedule (Section 15052). Submittal lists CL51 and CL53 but does not reference the schedule. Provide a cross-reference table confirming that the submitted classes match the required classes for each pipe size and service.
8. Per Section 15211-2.01.A.2.c, pipe must be manufactured from >90% recycled material. No confirmation provided in submittal.
9. Per Section 15211-2.01.A.3, flanged pipe must comply with AWWA C115, min Class 53. Submittal lists C&B Piping fabricated flanged joints, but no data proving C115 compliance.
10. Per Section 15211-2.01.B.1, flanged joints must be long hub ductile iron, machined after threading, with SBR gaskets. Submittal does not provide flange hub details, machining process, or gasket data. Provide manufacturer data verifying compliance with hub dimensions, machining requirements, and gasket material.
11. Per Section 15211-2.01.C.1, compact fittings conforming to AWWA C153 are not allowed; only full-body fittings per AWWA C110 are permitted. Submittal proposes C153 fittings which are not allowed. Revise to show only C110 full-body fittings.
12. Provide a cross-reference table confirming what linings and coatings will be used for each pipe size and service.
13. Per Section 15211-2.01.D.5, glass linings must be Ferrocl MEH-32 or Vitco SG-14. Submittal lists "glass lining" but does not identify the manufacturer or product. Provide manufacturer's technical data confirming the proposed glass lining complies with the approved products specified.
14. Per Section 15211-2.02.A-B, polyethylene encasement must be supplied by pipe manufacturer and be either 2-layer LLDPE, HDCLPE, or V-Bio. Submittal does not include any encasement product data.
15. The Piping Schedule in Section 15052 does not call out any unrestrained ductile iron piping or is it allowed in Section 15211. Therefore, submitted American Fastite joint pipe is not allowed to be used and shall be removed from the submittal.

16. The Piping Schedule in Section 15052 does not call out any standard mechanical fittings or are they allowed in Section 15211. Therefore, submitted mechanical joints and megaflange/foster adapter/megacouplings/megalugs/mid span restraint/non-restrained mechanical joint assembly are not allowed to be used and shall be removed from the submittal.
17. PVC pipe restraints shall be removed from the submittal and submitted under the appropriate specification section.

SUBMITTAL COMMENT RESPONSES



C&B PIPING (W), INC.

P.O. BOX 974

LEEDS, AL 35094

PHONE: (205) 699-0455

FAX: (205) 699-0773

www.cbpipe.com

DATE: 9/29/2025
CONTRACTOR: NAN
PROJECT: HILO WWTP REHAB AND REPLACEMENT PROJECT PHASE 1
REFERENCE: SUBMITTAL RESPONSE
JOB NO.

1 Contractor's variance request for mechanical joint fittings is denied. Contractor shall provide integrally restrained mechanical joints or integrally restrained push-on joints as called out in the Piping Schedule in Section 15052 and Section 15211-2.01.B.5 and 6

RESPONSE Confirmed. [As per OEC meeting 9.24.25 we will submit the variance again](#)

2 Per section 15211-1.03.C, submit manufacturer qualifications showing at least 5 years experience with the specified material.

RESPONSE Attached.

3 Per section 15211-1.03.D, submit manufacturer's coating and lining Quality Assurance Manual.

RESPONSE Glass lining Quality Assurance chart included. The two coatings that C&B is providing, asphaltic coating and Tnemec L140F, are shop coats that do not require a quality assurance manual.

4 Per Section 15211-1.03.E, provide shop drawings with layout, alignment, valves, fittings, joints, and connections, plus photos and descriptions of fittings, gaskets, couplings, linings, and coatings. Submittal only includes a tabular product list and a few cut sheets. If the intent of this package is limited to manufacturer/product approval, note that shop drawings are still required and must be submitted and approved prior to fabrication.

RESPONSE Drawings to be submitted at a later date.

5 Per Section 15211-1.03-F, confirm that the manufacturer's test reports for glass lining (holiday detection, compliance with material requirements, and coating rep reports) will be provided prior to shipment of materials.



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JOB NO.

RESPONSE Confirmed. To be provided prior to shipping.

6 Per Section 15211-2.01.A.1, pipe must be manufactured by American, McWane, SIP or U.S. Pipe. Submittal lists American (acceptable) and C&B Piping (not listed). C&B is not identified in the spec as an approved pipe or specialty manufacturer. Confirm that all pipe barrels will come from an approved manufacturer and that C&B's role is limited to fabricating specials from approved pipe. If C&B is proposed as a pipe or fitting manufacturer, it is not compliant.

RESPONSE C&B Piping is a fabricator and linings/coatings applicator. C&B Piping will be using American pipe barrels.

7 Per Section 15211-2.01.A.2, pipe shall meet the pressure/thickness class in the piping schedule (Section 15052). Submittal lists CL51 and CL53 but does not reference the schedule. Provide a cross-reference table confirming that the submitted classes match the required classes for each pipe size and service

RESPONSE The shop drawings will call out the classes of pipe in the bill of material.

8 Per Section 15211-2.01.A.2.c, pipe must be manufactured from >90% recycled material. No confirmation provided in submittal.

RESPONSE Please see the attached letter from American confirming.

9 Per Section 15211-2.01.A.3, flanged pipe must comply with AWWA C115, min Class 53. Submittal lists C&B Piping fabricated flanged joints, but no data proving C115 compliance.

RESPONSE Confirmed. C&B Piping will provide flanged pipe in accordance with AWWA C115.



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JOB NO.

10 Per Section 15211-2.01.B.1, flanged jointed must be long hub ductile iron, machined after threading, with SBR gaskets. Submittal does not provide flange hub details, machining process, or gasket data. Provide manufacturer data verifying compliance with hub dimensions, machining requirements and gasket material.

RESPONSE C&B uses high hub ductile iron flanges. Please verify the dimensions for long hub flanges. Flanges are machined after threading. Flange gasket submittal included separately.

11 Per Section 15211-2.01.C.1, compact fittings conforming to AWWA C153 are not allowed; only full-body fittings per AWWA C110 are permitted. Submittal proposes C153 fittings which are not allowed. Revise to show only C110 full-body fittings.

RESPONSE Please see attached revised submittal.

12 Provide a cross-reference table confirming what lining and coatings will be used for each pipe size and service.

RESPONSE Each spool/fitting will be called out on the drawings with the lining and coating required.

13 Per Section 15211-2.01.D.5, glass linings must be Ferrocl MEH-32 or Vitco SG-14. Submittal lists "glass lining" but does not identify the manufacturer or product. Provide Manufacturer's technical data confirming the proposed glass lining complies with the approved products specified.

RESPONSE This was included on page 106 of the original submittal. We are submitting on CBGL911.



C&B PIPING (W), INC.

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REFERENCE: SUBMITTAL RESPONSE
JOB NO.

14 Per Section 15211-2.02.A-B, polyethylene encasement must be supplied by pipe manufacturer and be either 2-layer LLDPE, HDCLPE, or V-Bio. Submittal does not include any encasement product data.

RESPONSE Attached.

15 The Piping Schedule in Section 15052 does not call out any unrestrained ductile iron piping or is it allowed in Section 15211. Therefore, submitted American Fastite joint pipe is not allowed to be used and shall be removed from the submittal.

RESPONSE This is removed.

16 The Piping Schedule in Section 15052 does not call out any standard mechanical fittings or are they allowed in Section 15211. Therefore, submitted mechanical joints and megaflange/foster adapter/megacouplings/mid span restraint/non-restrained mechanical joint assembly are not allowed to be used and shall be removed from the submittal.

RESPONSE Please confirm how we should tie-in and connect to existing lines without using megacouplings.

17 PVC pipe restraints shall be removed the submittal and submitted under the appropriate specification section.

RESPONSE Removed.

VARIANCE REQUEST FORM

Hilo Wastewater Treatment Plant Phase 1
Contract No. WW-4705R

VARIANCE

Request Form

Variance Request _____

**Clearly and Concisely describe the nature of your request by completing the following questions below. By submitting the Variance Request form, Contractor acknowledges that they understand and accept the limitation of this form and agree that the information provided does not guarantee the approval of the Variance request. The decision to grant or deny a variance is at the sole discretion of the relevant authority or entity responsible for reviewing such requests.*

Specification section 15211 – Ductile Iron Pipe: AWWA C151

1. What are the benefits the variance has to the Government?

County and Nan Inc will reap the benefits such as; meeting the expectations of the schedule by reduced lead time on the product proposed i.e., **Standard Mechanical joint fittings** as compared to the **Proprietary Restrained Joint fittings** and cost savings in terms of installation if potential field adjustments are required.

2. What are the positive and negative impacts to the project and Government if the variance is accepted?

Pursuing Standard Mechanical joints will significantly facilitate manufacturing, procurement, upkeeping schedule, minimal cost impact, smoother field adjustments, and possibly lessen change order cost.

3. Discuss how the variance is “equal to” or “better than” the specification requirement.

Standard Mechanical Joint Fittings comes with equal or rather more advantages, starting with the highly reduced lead time aspect, easement in installation, managing schedule, and the ability to be flexible with existing conditions.

4. What is the cost of the product/material that is originally specified?

Nan Inc is in the process of determining potential cost savings with the manufacturer.

SPECIFICATION SECTION 15211

SECTION 15211

DUCTILE IRON PIPE: AWWA C151

PART 1 GENERAL

1.01 SUMMARY

- A. Section includes: Ductile iron pipe, joints, fittings, gaskets, and pipe linings and coatings.

1.02 REFERENCES

- A. American Society of Mechanical Engineers (ASME):
 - 1. B16.1 - Gray Iron Pipe Flanges and Flanged Fittings: Classes 25, 125, and 250.
- B. American Water Works Association (AWWA):
 - 1. C104 - Standard for Cement-Mortar Lining for Ductile-Iron Pipe and Fittings.
 - 2. C105 - Polyethylene Encasement for Ductile-Iron Pipe Systems.
 - 3. C110 - Standard for Ductile-Iron and Gray-Iron Fittings.
 - 4. C111 - Standard for Rubber-Gasket Joints for Ductile-Iron Pressure Pipe and Fittings.
 - 5. C115 - Flanged Ductile Iron Pipe with Ductile-Iron or Gray-Iron Threaded Flanges.
 - 6. C150 - Standard for Thickness Design of Ductile-Iron Pipe.
 - 7. C151 - Standard for Ductile-Iron Pipe, Centrifugally Cast.
 - 8. C600 - Installation of Ductile Iron Water Mains and Their Appurtenances.
 - 9. C606 - Standard for Grooved and Shouldered Joints.
- C. American Welding Society (AWS):
 - 1. D11.2 - Guide for Welding Iron Castings.
- D. ASTM International (ASTM):
 - 1. A47 - Standard Specifications for Ferritic Malleable Iron Castings.
 - 2. A183 - Standard Specifications for Carbon Steel Track Bolts and Nuts.
 - 3. A536 - Standard Specifications for Ductile Iron Castings.
 - 4. C283 - Standard Test Methods for Resistance of Porcelain Enameled Utensils to Boiling Acid.
 - 5. D792 - Standard Test Methods for Density and Specific Gravity (Relative Density) of Plastics by Displacement.
- E. NACE International (NACE):
 - 1. SP0188 - Discontinuity (Holiday) Testing of New Protective Coatings on Conductive Substrates.
- F. National Association of Pipe Fabricators, Inc. (NAPF):
 - 1. 500-03 - Surface Preparation Standard for Ductile Iron Pipe and Fittings in Exposed Locations Receiving Special External Coatings and/or Special Internal Linings.

- G. Society for Protective Coatings (SSPC):
 - 1. PA-2 - Measurement of Dry Coating Thickness With Magnetic Gages.

1.03 SUBMITTALS

- A. Submit as specified in Section 01330 - Submittal Procedures.
- ✓ B. Product data: As specified in Section 15052 - Common Work Results for General Piping.
- C. Manufacturer's qualifications:
 - 1. Manufacturer qualifications and list of projects using the specified material: 5 years minimum.
- D. Manufacturer's Quality Assurance Manual:
 - 1. Submit manufacturer's coating and lining application quality assurance manual to Engineer prior to beginning coating application:
 - a. Strict conformance to the requirements of the manual will be required.
 - b. Deviation from the requirements of the manual will be grounds for the Engineer to reject the applied coating.
- E. Shop drawings:
 - 1. Detailed layout drawings showing alignment of pipes, location of valves, fittings, and appurtenances, types of joints, and connections to pipelines or structures.
 - 2. Photographs, drawings, and descriptions of fittings, gaskets, couplings, grooving of pipe and fittings, pipe linings, and coatings.
- F. Provide manufacturer's test reports:
 - 1. On glass lining certifying successful performance of holiday detection tests:
 - a. This documentation shall identify each piece by mark designation, and show the actual test results during the final inspection by the manufacturer prior to shipment.
 - b. Acceptance criteria for glass lining shall be as specified under Field Quality Control.
 - 2. On glass-lined pipe certifying compliance with specified material requirements for glass-lining.
 - 3. Include Coating Manufacturer's Technical Representative's reports.
 - 4. At the request of the Engineer, perform a maximum of 2 jobsite test verifications on 12 randomly selected segments per test verification.

1.04 QUALITY ASSURANCE

- A. Ductile iron pipe shall be supplied by a single manufacturer.
- B. Hydrostatically test each joint of ductile iron pipe in accordance with AWWA C151.
- C. Pre-installation meeting:
 - 1. Arrange for Coating Manufacturer's Technical Representative to attend preconstruction conferences, and to make periodic visits to factory or shop to inspect surface preparation of pipe, fittings, and accessories; and to inspect

application of linings to interior and coatings to exterior of pipe, fittings, and accessories.

1.05 DELIVERY, STORAGE, AND HANDLING

- A. Block piping and associated fittings for shipment to prevent damage to coatings and linings.
- B. Carefully handle piping and associated fittings during loading, unloading, and installation:
 - 1. Do not drop piping material from cars or trucks.
 - 2. Lower piping by mechanical means.
 - 3. Do not drop or pound pipe to fit grade.
- C. Glass lined pipe and fittings must be handled only from the outside:
 - 1. No forks, chains, straps, hooks, or other lifting device shall be placed inside the pipe or fittings for lifting, positioning, or laying.
- D. Protect gaskets and polyethylene encasement from long-term exposure to sunlight.
- E. Store piping, fittings, and other accessories such that they do not accumulate and hold rainwater, dirt, and debris.

1.06 WARRANTY

- A. Provide warranty as specified in Section 01783 - Warranties and Bonds.

PART 2 PRODUCTS

2.01 MANUFACTURED UNITS



- A. Ductile iron piping:
 - 1. Manufacturers: One of the following or equal:
 - a. American Cast Iron Pipe Co.
 - b. McWane Ductile.
 - c. SIP Industries.
 - d. U.S. Pipe.
 - 2. Typical type:
 - a. In accordance with AWWA C150 and AWWA C151.
 - b. Pressure class or special thickness class as indicated in the Piping Schedule provided in Section 15052 - Common Work Results for General Piping.
 - c. Manufactured from greater than 90 percent recycled material.
 - 3. Type with screw-on flanges:
 - a. In accordance with AWWA C115 with minimum special thickness Class 53 wall thickness unless otherwise specified or indicated on the Drawings.
 - b. Manufactured from greater than 90 percent recycled material.
 - 4. Type with grooved couplings:
 - a. Special thickness class as indicated in the Piping Schedule as specified in Section 15052 - Common Work Results for General Piping.

- b. Manufactured from greater than 90 percent recycled material.



B. Joints:

1. Flanged joints:
 - a. Screw-on flanges: Comply with the diameter, thickness, drilling, and other characteristics in accordance with ASME B16.1. In addition, comply with the following requirements:
 - 1) Ductile iron.
 - 2) Long hub, threaded, and specially designed for ductile iron pipe.
 - 3) After attaching to pipe, machine flange face to make pipe end and flange even and perpendicular to the axis of the pipe.
 - b. Bolt holes on flanges: 2-holed and aligned at both ends of pipe.
 - c. Cap screw or stud bolt holes: Tapped.
 - d. Bolts and nuts: As specified in Section 15052 - Common Work Results for General Piping.
 - e. Gaskets: Standard styrene butadiene copolymer (SBR) unless specified otherwise in Section 15052 - Common Work Results for General Piping.
2. Grooved joints: In accordance with AWWA C606, as complemented and modified below, radius-cut type, with following components:
 - a. Couplings: Rigid type, cast from ductile iron in accordance with ASTM A536, Grade 65-45-12, or malleable iron in accordance with ASTM A47, Grade 32510.
 - b. Bolts and nuts: In accordance with ASTM A183, Grade 2.
 - c. Gaskets: Capable of being applied on surface of piping with cavities to provide for an improved seal with the internal piping pressure. Material to be used for following services:
 - 1) For liquid service: NBR.
 - 2) For air service: Fluoroelastomer.
 - 3) For hot water service: EPDM.
 - d. Fittings: In accordance with AWWA C606, rigid radius-cut groove:
 - 1) Center-to-center dimensions: In accordance with AWWA C110.
 - 2) Wall thickness and other characteristics: In accordance with AWWA C606.
 - e. Flanged unit connections: Flanged to grooved joint adapters or a long enough spool with one end flanged and the other end grooved to prevent interference with the operation of adjacent valves, pumps, or other items.
3. Mechanical joints: In accordance with AWWA C111.
4. Push-on rubber gasket joints: In accordance with AWWA C111.
5. Integrally restrained mechanical joints:
 - a. Application:
 - 1) Where designation Mech Rest or RMJ is specified in the Piping Schedule provided in Section 15052 - Common Work Results for General Piping or indicated on the Drawings.
 - b. Design:
 - 1) Integral retainer weldment type or lugged type joint with Type 304 stainless steel rods and nuts.

- 2) Restrained mechanical joints of the configuration which utilizes a gripping or friction force for restraint will not be acceptable:
 - a) Exception: Mechanical wedge action joint restraints will be allowed only at connections to existing piping; EBAA Iron, Inc. Megalug® Series 1100 or equal.
 - (1) Materials:
 - (a) Gland body: Ductile iron in accordance with ASTM A536.
 - (b) Wedges and wedge actuating components: Ductile iron in accordance with ASTM A536. Wedges shall be heat treated to a minimum of 370 BHN.
 - (c) Actuating bolts and nuts: Ductile iron in accordance with ASTM A536. Provide torque-limiting twist off components to ensure proper installation.
 - (d) Gaskets: As specified in Section 15052 - Common Work Results for General Piping.
 - (2) Coatings:
 - (a) Provide manufacturer applied coating system.
 - (b) Manufacturers: EBAA Iron Inc., Mega-Bond®, or equal.
 - (3) Working pressure:
 - (a) Shall include a minimum safety factor of 2:1.
 - (b) For sizes 3- through 16-inch: 350 pounds per square inch.
 - (c) For sizes 18- through 48-inch: 250 pounds per square inch.
 - (4) Restraint shall consist of multiple gripping wedges incorporated into a follower gland meeting the requirements of AWWA C111.
 - (5) Restraint must be in accordance with applicable requirements of AWWA C110 and AWWA C111 for mechanical joints.
 - c. Manufacturers: Where restrained mechanical joints are required, use one of the following:
 - 1) American Cast Iron Pipe Company, MJ Coupled Joint.
 - 2) U.S. Pipe, Bolt-Lok.
6. Integrally restrained push-on joints:
 - a. Application:
 - 1) Where designation restrained push-on (RPO) as indicated in the Piping Schedule provided in Section 15052 - Common Work Results for General Piping or indicated on the Drawings.
 - b. Design:
 - 1) Restrained push-on joints of the configuration which utilizes a gripping or friction force for restraint will not be acceptable.
 - a) Exception: Push-on joint restraint harnesses will be allowed only at connections to existing piping; EBAA Iron, Inc. Megalug® Series 1700 or equal.
 - (1) Materials:
 - (a) Restraint and backup ring: Ductile iron in accordance with ASTM A536.

- (b) Wedges and wedge actuating components: Ductile iron in accordance with ASTM A536. Wedges shall be heat treated to a minimum of 370 BHN.
- (c) Actuating bolts and nuts: Ductile iron in accordance with ASTM A536. Provide torque-limiting twist off components to ensure proper installation.
- (d) Tie rods: Low alloy steel in accordance with AWWA C111.
- (e) Bolts and nuts, including T-bolts: As specified in 15052 - Common Work Results for General Piping.
- (2) Coatings:
 - (a) Provide manufacturer applied coating system.
 - (b) Manufacturers: EBAA Iron Inc., Mega-Bond®, or equal.
- (3) Working pressure:
 - (a) Shall include a minimum safety factor of 2:1.
 - (b) For sizes 3- through 16-inch: 350 pounds per square inch.
 - (c) For sizes 18- through 48-inch: 250 pounds per square inch.
- (4) Restraint shall consist of a backup ring behind the ductile iron bell and a restraint ring consisting of multiple gripping wedges connected with number and type of tie rods as recommended by the manufacturer.
- 2) Suitable for the following working pressures:
 - a) For 4- through 24-inch pipe: 350 pounds per square inch gauge.
 - b) For 30- through 54-inch pipe: 250 pounds per square inch gauge.
- c. Manufacturers: One of the following:
 - 1) U.S. Pipe, TR Flex.
 - 2) McWane Ductile, TR Flex.
 - 3) American Cast Iron Pipe Company, Flex Ring or Lok-Ring.
- d. Limit buried joints to half the manufacturer's published allowable angular joint deflection for purposes of pipeline alignment and elimination of fittings.

Both for approval;
AWWA 110 and
AWWA 153

✓ C. Fittings:

- 1. Ductile iron in accordance with AWWA C110. Compact fittings conforming to AWWA C153 are not allowed.
- 2. Joint type: Same as that of the associated piping as specified in Section 15052 - Common Work Results for General Piping.
- 3. Plain end-to-flanged joint connectors using setscrews are not acceptable.

✓ D. Pipe linings:

- 1. Cement-mortar lining:
 - a. In accordance with AWWA C104, apply cement-mortar on clean bare metal surfaces. Extend to faces of flanges, ends of spigots, and shoulders of hubs.
 - b. Minimum lining thickness: Standard in accordance with AWWA C104.
 - c. Type of cement: Type II.

2. Asphaltic seal coat:
 - a. Apply over cement mortar linings and to outside surface of pipes that will not receive another coating. Apply in accordance with AWWA C151.
3. Elastomeric polyurethane (100 percent solids) lining:
 - a. As specified in Section 09960 - High-Performance Coatings.
4. Ceramic epoxy lining:
 - a. Manufacturers: One of the following:
 - 1) PROTECTO 401.
 - 2) SP-2000W.
 - b. Material: Amine cured Novolac epoxy containing at least 20 percent by volume of ceramic quartz pigment.
 - c. Minimum dry film thickness (DFT): 40 mills.
 - d. Application:
 - 1) The lining shall only be applied by a manufacturer-authorized representative with no less than 5 years of experience in applying the specified material.
 - 2) The application of the lining shall be performed in accordance with manufacturer's published specifications.
 - 3) Pipe and fittings shall be delivered to application facility with no interior lining.
 - 4) Interior of pipe shall be abrasive blasted per manufacturer's specifications.
 - e. Coverage:
 - 1) Gasket and spigot ends-on joints: Provide 6 mils minimum and 10 mils maximum coverage using joint compound as specified by the manufacturer for the gasket area and spigot ends.
 - 2) Mechanical joints: Extend lining from spigot end to edge of gauging ring.
 - 3) Number of coats: As recommended by the lining manufacturer.
 - f. Source quality control:
 - 1) Test pipe and fitting lining with a magnetic film thickness gauge. Perform testing in accordance with the method outlined in SSPC PA-2 Film Thickness Rating.
 - 2) Test lining integrity of pipes using a holiday detection testing instrument set at the voltage as specified by the coating manufacturer:
 - a) Repair all holidays with joint compound in accordance with the recommendations of the coating manufacturer, and re-test.
 - 3) Discard piping or reline piping when pinholes or discontinuities are found.
- ✓ 5. Glass lining:
 - a. Manufacturers: One of the following:
 - 1) Water Works Manufacturing, Ferroch MEH-32 Lining.
 - 2) Vitco Corporation, SG-14 Lining.
 - b. Material: Special glasses and inorganic materials suited for lining of sewage, sludge, and scum piping with the following characteristics:
 - 1) Thickness: 0.008 to 0.012 inch.
 - 2) Hardness: 5 to 6 on the Mohs Scale.
 - 3) Density: 2.5 to 3.0 grams per cubic centimeter, measured in accordance with ASTM D792.

- 4) Thermal shock resistance: Capable of withstanding 350 degrees Fahrenheit change from 430 degrees Fahrenheit to 80 degrees Fahrenheit without crazing, blistering, or spalling.
- 5) Gloss retention: Capable of retaining gloss after immersion in an 8 percent sulfuric acid solution at 148 degrees Fahrenheit for 10 minutes.
- 6) Weight loss: Maximum 3 milligrams per square inch when tested in accordance with ASTM C283.
- c. Fabrication:
 - 1) Use piping that is suitable for glass lining with minimum Class 53 wall thickness after application of glass lining.
 - 2) Machine interior of pipe. Bore or grit blast in accordance with NAPF 500-03 prior to application of glass lining.
 - 3) Screw factory assembled flanges on pipe, align boltholes, and flange faces, unless otherwise specified.
 - 4) Apply lining to surfaces free of chemicals.
 - 5) Place piping in furnaces specially designed for heating piping until glass melts and fuses with an integral molecular bond to the base metal.



- E. Coatings:
1. Asphalt varnish: Factory applied.
 2. Primer:
 - a. Factory applied for field coating.
 - b. Compatible with materials as specified in Section 09960 - High-Performance Coatings.

2.02 POLYETHYLENE ENCASEMENT

- A. General:
1. Polyethylene encasement shall be supplied by the pipe manufacturer.
- B. Materials: Supply one of the following polyethylene encasements:
1. 2 layers of linear low-density polyethylene (LLDPE) film, minimum thickness of 8 mils in accordance with AWWA C105; or
 2. Single layer of high-density, cross-laminated polyethylene (HDCLPE) film, minimum thickness of 4 mils in accordance with AWWA C105.
 3. Single layer of V-Bio[®] enhanced polyethylene encasement (3 layers of co-extruded LLDPE film with anti-microbial additive and volatile corrosion inhibitor infused on the inside surface), meeting all requirements of AWWA C105.

PART 3 EXECUTION

3.01 INSTALLATION

- A. General:
1. Install ductile iron piping in accordance with AWWA C600, modified as specified in Section 15052 - Common Work Results for General Piping.

2. For underground piping, the trenching, backfill, and compaction: As specified in Section 02318 - Trenching.
- B. Polyethylene encasement:
1. Wrap all buried ductile iron pipe and fittings in 2 layers of loose low density polyethylene wrap or a single layer of high-density polyethylene wrap in accordance with AWWA C105.
 2. Polyethylene encasement shall be continuous and terminated neatly at connections to below grade equipment or structures.
 3. At wall penetrations, extend encasement to the wall and neatly terminate.
 4. At slab penetrations, extend encasement to 2 inches below the top of slab and neatly terminate.
 5. When rising vertically in unimproved areas, extend encasement 6 inches above existing grade and neatly terminate.
 6. Repair tears and make joints with 2 layers of plastic tape.
 7. All work shall be inspected prior to backfilling of pipe and associated items.
- C. Joints:
1. Install types of joints as specified in the piping schedule provided in Section 15052 - Common Work Results for General Piping.
 2. Mechanical joints are not acceptable in above ground applications.
 3. Grooved joints:
 - a. Install piping with grooved joints where specified in the piping schedule as specified in Section 15052 - Common Work Results for General Piping or indicated on the Drawings.
 - b. Assemble grooved joints in accordance with manufacturer's published instructions.
 - c. Support grooved-end pipe in accordance with manufacturer's published instructions:
 - 1) Install at least 1 support between consecutive couplings.
 - d. Grooved joints may be installed where flanged joints are scheduled, except under the following conditions:
 - 1) In underground and underwater installations.
 - 2) In piping subject to test pressure of 150 pounds per square inch gauge or more.
 - 3) In steam and in combustible gas piping systems.
 - 4) In sludge and scum piping designed to be steam cleaned as indicated on the Drawings.
 - 5) When wall thickness of pipe is less than the minimum recommended in published instructions by the manufacturer of the grooved end coupling and these Specifications.
 - e. Make connections to flanged valves, pumps, and piping appurtenances by either:
 - 1) Flanged-to-grooved joint adapters.
 - 2) Flanged-by-grooved end pipe spool of sufficient length to prevent interference with the operation of adjacent valve, pumps, or other items.
 - 3) Integrally cast flanged-by-grooved end pipe fittings.

- f. Except for closure pieces approved by the Engineer, field grooving of pipe will not be permitted:
 - 1) All field grooving will be done in the presence of and approved by the Engineer.

D. Tapping ductile iron pipe:

- 1. Direct tapping of ductile iron pipe may be performed but is limited to the following conditions:
 - a. Maximum allowable tap diameter by pipe diameter and pressure class:

Pipe Size (inches)	Pressure Class				
	150	200	250	300	350
	Maximum Allowable Direct Tap Size (inches)				
3	-	-	-	-	3/4
4	-	-	-	-	3/4
6	-	-	-	-	1
8	-	-	-	-	1
10	-	-	-	-	1
12	-	-	-	-	1-1/4
14	-	-	1-1/4	1-1/2	1-1/2
16	-	-	1-1/2	2	2
18	-	-	2	2	2
20	-	-	2	2	2
24	-	2	2	2	2

- b. The maximum allowable tap diameter for pipelines greater than 24 inches is 2 inches.
 - c. Two layers of 3-mil thread sealant are required to minimize the torque required to affect a watertight connection.
 - 2. Direct tapping of glass lined ductile iron pipe may be performed only when approved in writing by the Engineer. Direct tapping of glass lined pipe shall be performed in accordance with the above conditions for tapping ductile iron pipe in addition to the following conditions:
 - a. Drilling and tapping shall be performed using a hole saw:
 - 1) Use of a large drill bit is not acceptable.
 - b. As the hole saw approaches the glass lining, lessen the inward pressure to avoid excess chipping or cracking of the lining.
 - c. Minor chipping or spalling of the glass lining shall be repaired using an epoxy resin "glass repair kit" provided by the fabricator:
 - 1) Manufacturers: One of the following:
 - a) Devoe - Devran 224 HS.
 - b) Sherwin-Williams Co. - Sher-Tile High Solids Epoxy.
 - 2) Repair kit use is only allowed for areas of damage less than 1/2 inch in diameter:
 - a) Larger areas of damage will require replacement.

- 3) Surface shall be prepared and repair kit shall be applied in accordance with manufacturer and/or fabricator's instructions.

3.02 FIELD QUALITY CONTROL

- A. Testing ductile iron piping:
 1. Test as specified in Section 15052 - Common Work Results for General Piping and Section 15956 - Piping Systems Testing.
 2. Do not test sections longer than 1/2 mile in total pipe length.
- B. Repair damaged cement mortar lining to match quality, thickness, and bonding of original lining in accordance with AWWA C104:
 1. When lining cannot be repaired or repairs are defective, replace defective piping with undamaged piping.
- C. Verify that interior surfaces of glass lined pipe and fittings have continuous coverage:
 1. Verify with low voltage wet sponge holiday detector in accordance with NACE SP0188.
 2. Discard glass lined ductile iron piping and fittings with voids or casting anomalies. that exceed the maximum non-visible pinholes allowances below:

Diameter	Maximum Pinholes	
	Fittings	Pipe (per 20-foot length of pipe)
4- to 8-inch	3-5	10-12
10- to 18-inch	5-8	18-20
20-inch and Larger	8-10	25-28

3. Discard lined piping and fittings found to have pinholes, crazing, or fish scales, which expose the metal substrate.

END OF SECTION

C & B PIPING - DUCTILE IRON PIPE: AWWA C151



C&B PIPING (W), LLC.

1980 West Commerce Ave

Gilbert, AZ 85233

www.cb piping.com

C&B Piping Experience Letter

Contractor: Nan, Inc.
Project: Hilo Wastewater Treatment Plant
Location: Hilo, HI

C&B Piping was founded in 1986 and operates from three locations across the United States, providing full service project support; from planning and layout drawings to complete project management. We have been manufacturing ductile iron pipe fabrications ranging from 3" to 64" for nearly four decades, earning a reputation for dependable quality and service.

Since 2003, C&B has also been a leader in the application of porcelain glass lining to ductile iron pipe and fittings. Over the past two decades, we have developed expertise in this specialized process, becoming the largest volume producer of glass lined ductile iron products in the United States. Our capabilities allow us to serve both domestic and international markets, as well as act as an OEM glass lining provider for industry leaders such as DeZurik, Valmatic, InFact Corporation, and others.

Today, C&B Piping is recognized as the nation's largest direct to contractor manufacturer of ductile iron spools. We continue to provide a complete range of ductile iron products for plant work, delivering solutions that meet the requirements of projects of every size. Our commitment to quality and service ensures we remain a trusted partner in the marketplace.

Attached is a sample of some of the large scale projects where C&B has supplied glass lined ductile iron fabrications, pipe barrels, and fittings.

Sincerely,

Seth Fox

Seth Fox
Senior Project Manager
C&B Piping, Inc.



BUILDING

AMERICA

TOGETHER

www.cb piping.com

C & B PIPING, INC.
CBGL911 GLASS LINING
Major Projects

OWNER: Irvine Ranch Water District
FACILITY NAME: Michelson Water Recycling Plant
FACILITY LOCATION: Irvine, CA
ENGINEER: Black & Veatch
CONTRACTOR: J.R. Filanc / Balfour Beatty Joint Venture

OWNER: Metropolitan Council Environmental Services
FACILITY NAME: Metro Wastewater Treatment Plant
FACILITY LOCATION: St. Paul, MN
ENGINEER: CH2M Hill - Brown and Caldwell
CONTRACTOR: Shank Constructors, Inc.

OWNER: Metropolitan Council Environmental Services
FACILITY NAME: Empire Wastewater Treatment Facility
FACILITY LOCATION: Empire Township, MN
ENGINEER: Montgomery Watson Harza
CONTRACTOR: Shank Constructors, Inc.
Large Quantity of 30" pipe

OWNER: City of New York Department of Environmental Protection
FACILITY NAME: North River WPCP
FACILITY LOCATION: New York City
ENGINEER: Malcolm Pirnie
CONTRACTOR: Gottlieb Skanska

OWNER: City of New York Department of Environmental Protection
FACILITY NAME: Newtown Creek WPCP
FACILITY LOCATION: New York City
ENGINEER; Malcolm Pirnie
CONTRACTOR: WDF, Inc.

OWNER: Chandler Airport, AZ
FACILITY NAME: Chandler AWRF Expansion
FACILITY LOCATION: Chandler, AZ

C & B PIPING, INC.
CBGL911 GLASS LINING
Major Projects

ENGINEER: Wilson
CONTRACTOR NAME: McCarthy Building Co.
PROJECT MANAGER: BJ Peterson
CONTACT NUMBER: 602-828-4588

OWNER: South Essex Sewerage District
FACILITY NAME: South Essex WWTP
FACILITY LOCATION: Essex, MA
ENGINEER: Malcolm Pirnie and Tighe & Bond
CONTRACTOR: Methuen Construction Co., Inc.

OWNER: Loudoun County Sanitation Commission
FACILITY NAME: Broad Run WRF
FACILITY LOCATION: Loudoun County, VA
ENGINEER; CH2M Hill
CONTRACTOR: Pizzagalli Construction Co.

OWNER: Eastern Municipal Water District - CA
FACILITY NAME: Perris Valley RWRF Expansion Plant
FACILITY LOCATION: Riverside, CA
ENGINEER: Carollo
CONTRACTOR: Kiewit Pacific Construction

FACILITY NAME: Sanderson Lift Station
FACILITY LOCATION: Perris, CA
ENGINEER:
CONTRACTOR: A.E.S. Construction Co.

OWNER: City of Las Vegas
FACILITY NAME: Solids Dewatering Facility
FACILITY LOCATION: Las Vegas, NV
ENGINEER: Carollo
CONTRACTOR: MMC Construction

OWNER: City of Stratford
FACILITY NAME: Stratford WPCF
FACILITY LOCATION: Stratford, CT
ENGINEER: Camp Dresser & McKee

C & B PIPING, INC.
CBGL911 GLASS LINING
Major Projects

CONTRACTOR: C. H. Nickerson Contractors

OWNER: City of Des Moines
FACILITY NAME: WRF Combined Solids
FACILITY LOCATION: Des Moines, IA
ENGINEER: Black & Veatch Corporation
CONTRACTOR: John T. Jones Construction Company

OWNER: City of Calgary
FACILITY NAME: Pinecreek WWTP
FACILITY LOCATION: Alberta, Canada
ENGINEER: Stantec Consulting, LTD
CONTRACTOR: Balzers Canada, Inc

OWNER: City of Toronto
FACILITY NAME: Humber WTF
FACILITY LOCATION: Toronto, Ontario, Canada
ENGINEER: CH2M Hill
CONTRACTOR: Sutherland – Schultz, Inc.

OWNER: City of Toronto
FACILITY NAME: Ashbridges WTF
FACILITY LOCATION: Toronto, Ontario, Canada
ENGINEER: CH2M Hill
CONTRACTOR: Sutherland – Schultz, Inc.

OWNER: City of Courtice
FACILITY NAME: Courtice WPCP
FACILITY LOCATION: Courtice, Ontario, Canada
ENGINEER: CH2M Hill
CONTRACTOR: Sutherland – Schultz, Inc.

OWNER: City of Toronto
FACILITY NAME: Duffin Creek WPCP
FACILITY LOCATION: York and Durham, Ontario, Canada
ENGINEER: CH2M Hill / Earthtech
CONTRACTOR: North American Construction LTD

OWNER: City of Saskatoon
FACILITY NAME: H.M. Weir WPCP

C & B PIPING, INC.
CBGL911 GLASS LINING
Major Projects

FACILITY LOCATION: Saskatoon, Saskatchewan, Canada
ENGINEER: Stantec Consulting LTD
CONTRACTOR: Suer & Pollon Mechanical

OWNER: City of Calgary
FACILITY NAME: Bonnybrook WWTP
FACILITY LOCATION: Calgary, Alberta, Canada
ENGINEER: Stantec Consulting LTD
CONTRACTOR: Maple Reinders

OWNER: City of Philadelphia
FACILITY NAME: Southwest WPCP
FACILITY LOCATION: Philadelphia, PA
ENGINEER: City of Philadelphia Water Department
CONTRACTOR: Spectraserv, Inc.

OWNER: City of St. John, New Brunswick
FACILITY NAME: Eastern WWTP
FACILITY LOCATION: St. John, New Brunswick, Canada
ENGINEER: City of St. John Municipal Operations & Engineering
CONTRACTOR: Pro Pipe Construction

OWNER: City of Keswick, Ontario, Canada
FACILITY NAME: Keswick WPCP
FACILITY LOCATION: Keswick, Ontario, Canada
ENGINEER: Hatch, Mott MacDonald
CONTRACTOR: W.A. Stephenson

OWNER: City of Strathmore, Alberta, Canada
FACILITY NAME: Strathmore WWTP
FACILITY LOCATION: Strathmore, Alberta, Canada
ENGINEER: AECOM
CONTRACTOR: Maple Reinders

OWNER: City of South Windsor, CT
FACILITY NAME: South Windsor WPCF
FACILITY LOCATION: South Windsor, CT
ENGINEER: Camp Dresser & McKee
CONTRACTOR: C.H.Nickerson





July 7, 2025

To Whom It May Concern,

American Ductile Iron Pipe (A division of American Cast Iron Pipe Company), is a leading manufacturer of ductile iron pipe and fittings for the waterworks and power industries. We employ approximately 2,000 people nationwide. Our headquarters in Birmingham, AL is on a 2,100-acre site that measures three miles in length, with 60 acres of plant under roof. It's the largest individual iron pipe casting plant in the United States.

In reference to LEED category Materials and Resources - Credit 4.1 and 4.2 Recycled Content, our ductile iron pipe contains approximately 93% recycled metal (post-consumer). Therefore, approximately 7% is pre-consumer material, which consists of 3% Carbon from coke, 2% Silicon, and 2% from pig iron. In reference to LEED category Materials and Resources – Credit 5.1 and 5.2 Regional Materials, all scrap materials used in the production of this ductile iron pipe are sourced, inventoried and consumed at our facility in Birmingham, Alabama. These materials are acquired from sources within an approximate 250 mile radius of our manufacturing facility in Birmingham, Alabama.

Sincerely,

AMERICAN Ductile Iron Pipe

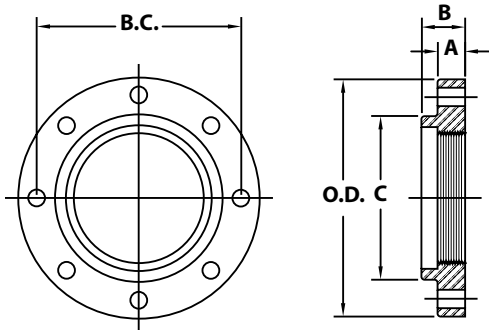
K. Scott Summers

K. Scott Summers
Director, Project Management
(205) 325-1934
ssummers@american-usa.com



Flanges

ANSI/AWWA C115/A21.15

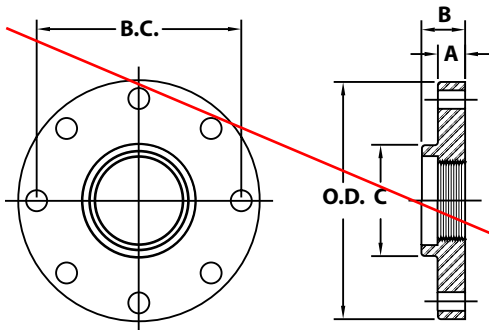


GENERAL SPECIFICATIONS

MATERIAL:	Ductile Iron per ASTM A536
PRESSURE:	250 PSI
DRILLING:	In accordance with ANSI/AWWA C110/21.10, ASME B16.1 Class 125 and ASME B16.42 Class 150
DIMENSIONS:	Per ANSI/AWWA C115/A21.15 Flange O.D. & thickness per ANSI/AWWA C110/21.10
FLANGE THREADS:	Taper pipe thread per ASME B1.20.1 adapted to ductile iron pipe O.D.

HI HUB THREADED 125 LB. FLANGES FOR DUCTILE PIPE (DUCTILE IRON)

NOM. SIZE	OD PIPE	FLANGE THK. A	FLANGE THK. B	THREAD LENGTH	OD FLANGE O.D.	HUB DIA. C	BOLT CIRCLE B.C.	BOLT HOLE DIA.	BOLT DIA. & LENGTH	NO. OF BOLT HOLES	WT (LBS.)
2	2.50	0.62	1.25	1.00	6.00	3.06	4.75	3/4	5/8 x 2 1/4	4	4
3	3.96	0.75	1.69	1.25	7.50	4.45	6.00	3/4	5/8 x 2 1/2	4	7
4	4.80	0.94	1.81	1.37	9.00	5.32	7.50	3/4	5/8 x 3	8	12
6	6.90	1.00	2.00	1.53	11.00	7.56	9.50	7/8	3/4 x 3 1/2	8	17
8	9.05	1.12	2.25	1.75	13.50	9.69	11.75	7/8	3/4 x 3 1/2	8	26
10	11.10	1.19	2.44	1.88	16.00	12.00	14.25	1	7/8 x 4	12	37
12	13.20	1.25	2.56	2.12	19.00	14.06	17.00	1	7/8 x 4	12	55
14	15.30	1.38	2.62	2.12	21.00	16.25	18.75	1 1/8	1 x 4 1/2	12	70
16	17.40	1.44	2.75	2.12	23.50	18.44	21.25	1 1/8	1 x 4 1/2	16	80
18	19.50	1.56	3.00	2.25	25.00	20.53	22.75	1 1/4	1 1/8 x 5	16	85
20	21.60	1.69	3.25	2.44	27.50	22.63	25.00	1 1/4	1 1/8 x 5	20	106
24	25.80	1.88	3.50	2.75	32.00	26.82	29.50	1 3/8	1 1/4 x 5 1/2	20	147
30	32.00	2.12	3.75	3.37	38.75	32.75	36.00	1 3/8	1 1/4 x 6 1/2	28	224
36	38.30	2.38	5.00	4.00	46.00	39.12	42.75	1 5/8	1 1/2 x 7	32	337
42	44.50	2.62	5.12	4.50	53.00	46.00	49.50	1 5/8	1 1/2 x 7 1/2	36	495
48	50.80	2.75	5.50	5.12	59.50	52.25	56.00	1 5/8	1 1/2 x 8	44	622
54	57.56	3.00	6.45	4.75	66.25	58.75	62.75	2	1 3/4 x 8 1/2	44	711
60	61.61	3.12	6.75	5.00	73.00	63.00	69.25	2	1 3/4 x 9	52	1084
64	65.67	3.38	7.06	5.25	80.00	67.18	76.00	2	1 3/4 x 9	52	1575



GENERAL SPECIFICATIONS

MATERIAL:	Ductile Iron per ASTM A536
PRESSURE:	250 PSI
DRILLING:	In accordance with ANSI/AWWA C110/21.10, ASME B16.1 Class 125 and ASME B16.42 Class 150
DIMENSIONS:	Per ANSI/AWWA C115/A21.15 Flange O.D. & thickness per ANSI/AWWA C110/21.10
FLANGE THREADS:	Taper pipe thread per ASME B1.20.1 adapted to ductile iron pipe O.D.

REDUCING 125 LB. FLANGES FOR DUCTILE PIPE (DUCTILE IRON)

NOM. SIZE	OD PIPE	FLANGE THK. A	FLANGE THK. B	THREAD LENGTH	OD FLANGE O.D.	HUB DIA. C	BOLT CIRCLE B.C.	BOLT HOLE DIA.	BOLT DIA. & LENGTH	NO. OF BOLT HOLES	WT (LBS.)
4 x 3	3.96	0.94	1.37	0.94	9.00	4.45	7.50	3/4	5/8 x 3	8	16
6 x 4	4.80	1.00	1.56	1.00	11.00	5.32	9.50	7/8	3/4 x 3 1/2	8	25
8 x 4	4.80	1.12	1.75	1.00	13.50	5.32	11.75	7/8	3/4 x 3 1/2	8	40
8 x 6	6.90	1.12	1.75	1.16	13.50	7.56	11.75	7/8	3/4 x 3 1/2	8	35
10 x 8	9.05	1.19	2.00	1.38	16.00	9.69	14.25	1	7/8 x 4	12	50
12 x 8	9.05	1.25	2.19	1.38	19.00	9.69	17.00	1	7/8 x 4	12	85
12 x 10	11.10	1.25	2.19	1.38	19.00	12.00	17.00	1	7/8 x 4	12	65



8 Mil Linear Low Density Polyethylene

Polycase LLDPE encasement film is a carefully engineered linear low density virgin polyethylene resin film designed to meet the requirements of the ANSI/AWWA C105/A21.5 standard.

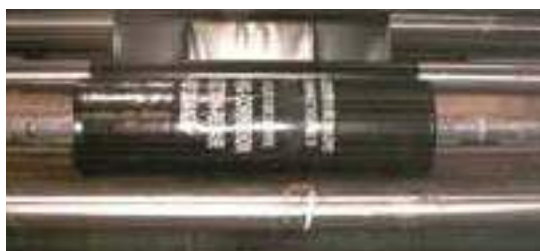
Carefully controlled molten resin is forced under high pressure through a die-head producing a bubble of polyethylene in a vertical column. The resulting tube is cooled and gathered onto spooling equipment at the top of the column.

Film strength characteristics are referred to as strength in the machine direction (direction of travel through the die-head), and the transverse direction (perpendicular to the machine direction). Minimum acceptable test values should consider both MD and TD.



Specifications:

Physical Attribute	Test Direction	ANSI/AWWA C105/A21.5 MINIMUM REQUIREMENT	SIGMA LLDP TYPICAL TEST VALUES
TENSILE STRENGTH ASTM D882	MACHINE DIRECTION	3600psi	4635psi
	TRANSVERSE DIRECTION	3600psi	4216psi
ELONGATION ASTM D882	MACHINE DIRECTION	800%	948%
	TRANSVERSE DIRECTION	800%	1012%
DIELECTRIC STRENGTH ASTM D149 (VOLTS / MIL)	n/a	800 volts / mil	1786 volts / mil
IMPACT RESISTANCE ASTM D1079 (grams)	n/a	600 grams	928 grams
PROPAGATION TEAR RESISTANCE ASTM D1922 (gf)	MACHINE DIRECTION	2550 grams/force	4082 grams/force
	TRANSVERSE DIRECTION	2550 grams/force	6159 grams/force



Years of effort have gone into the establishment of the ANSI/AWWA C105/A21.5 American National Standard for polyethylene encasement for ductile-iron pipe systems. This quality polyethylene encasement film product that meets this minimum standard does not “just happen”. The required design parameters must be defined. Product engineers must determine what raw materials and processes are necessary to meet the design characteristics. We know, from our own experience and testing, that only quality, virgin, materials that meet the criteria of the standard; coupled with proper manufacturing processes, will yield the required finished film physical properties. We are proud to offer certification per Section 5.1 of the standard.

www.sigmaco.com



POLYETHYLENE ENCASUREMENT PRODUCTS

8 Mil Linear Low Density Polyethylene

Quality Management

Polycase LLDPE encasement film is continuously monitored for thickness and tube size. Each roll contains control identification traceable to actual mill test reports, virgin resin tests, and applicable ASTM test data.

High speed printing plates provide printing as required by Sect. 4.3.1 of ANSI/AWWA C105.

8 mil Stock Tube Sizes

(BASED ON ANSI/AWWA C105/A21.5 Table 1)

Item No.	DI PIPE SIZE	LAYFLAT TUBE SIZE
47-PPR16	4" - 6"	16" X 500 FT.
47-PPR20	8"	20" X 500 FT.
47-PPR24	10"	24" X 500 FT.
47-PPR27	12"	27" X 500 FT.
47-PPR30	14"	30" X 500 FT.
47-PPR34	16"	34" X 500 FT.
47-PPR37	18"	37" X 500 FT.
47-PPR41	20"	41" X 500 FT.
47-PPR54	24"	54" X 500 FT.
47-PPR67	30"	67" X 440 FT.
47-PPR81	36" & 42"	81" X 360 FT.
47-PPR95	48"	95" X 220 FT.
47-PPR108	54" & 60"	108" X 220 FT.
47-PPR121	64"	121" X 110 FT.

Custom LLDP Products

For product not shown in our standard ANSI/AWWA C105/A21.5 Stock tube sizes we can provide film designed to meet your most demanding requirements .

Specialty Polyethylene Products Specifiacion/ Ordering Guidelines

THICKNESS	10-12-15-20-24-30 Mils.
TUBE SIZE	Based on Manageable Weight
Colors	Options are available
Minimum Order Qty.	Based on Requirements
Lead Time	Approximately 6 Weeks

Contact SIGMA For More Information or to Place an Order

Birmingham, AL
(205) 380-2333
oem-m79@sigmaco.com

Alexander City, AL
(800) 824-4513
rps-sales@sigmaco.com

Sauk Village, IL
(888) 999-0420
chi-sales@sigmaco.com

Cream Ridge, NJ
(800) 999-2550
crm-sales@sigmaco.com

Houston, TX
(800) 999-0109
htn-sales@sigmaco.com

Ontario, CA
(800) 688-6230
ont-sales@sigmaco.com

www.sigmaco.com



C&B PIPING (W), INC.

P.O. BOX 974

LEEDS, AL 35094

PHONE: (205) 699-0455

FAX: (205) 699-0773

www.cbpipling.com

DATE: 7/15/25

CONTRACTOR: NAN, INC.

PROJECT: HILO WWTP

BID DATE: 8/15/2024

AIS DOMESTIC MATERIAL

C&B PIPING WILL PROVIDE LAYOUT DRAWING SUBMITTALS SHOWING PIPE SYSTEMS AND ACTUAL PIPE SIZES, CLASSES, JOINTS PROPOSED FOR SUBMITTAL APPROVAL. THE DRAWINGS WILL BE SUBMITTED BY SYSTEM AS THEY ARE COMPLETED. THE SUBMITTAL PRODUCT LISTING AND ATTACHED CUT SHEETS ARE FOR REFERENCE TO COVER THE ITEMS THAT WILL BE SHOWN ON THE PIPING LAYOUT DRAWINGS.

DUCTILE IRON PIPE

RESTRAINED JOINT DUCTILE IRON PIPE

1) FLEX-RING JOINT; CL51 4" -24" CEMENT LINED AMERICAN

2) FLEX-RING JOINT; CL53 4"-24" GLASS LINED AMERICAN

HIGH DEFLECTION HIGH PRESSURE RESTRAINED PIPE

NOTE: SUPPLIED SBR RUBBER GASKET, AWWA C111

FABRICATED DUCTILE IRON PIPE

1) MJ COUPLE-JOINT, CL53 THICKNESS FOR 04" THRU 08" AMERICAN

2) FLANGED JOINT, CL53 THICKNESS FOR 04" THRU 12" C&B PIPING INC

3) GROOVED JOINT, CL53 THICKNESS FOR 04" THRU 24" C&B PIPING INC

DUCTILE IRON FITTINGS

MECHANICAL JOINT DUCTILE IRON FITTINGS

1) AWWA C110, FULL BODY TYPE PRIMARY ---> TYLER UNION

BACKUP ---> STAR

PROPRIETARY RESTRAINED DUCTILE IRON FITTINGS

AMERICAN

1) FLEX-RING RESTRAINED FITTINGS 04"- 36"

FLANGED JOINT DUCTILE IRON FITTINGS, C110 FULL BODY TYPE

PRIMARY ---> TYLER UNION

1) AWWA C110, FULL BODY TYPE

BACKUP ---> STAR

GROOVED DUCTILE IRON FITTINGS

1) VICTAULIC, 3"-24" SIZES VICTAULIC

LINING(S) FOR DUCTILE IRON PIPE AND FITTINGS

CEMENT MORTAR LINED

1) TYPE II, ANSI/AWWA C104/A21.4 (STD THICKNESS) AWWA C104



C&B PIPING (W), INC.

P.O. BOX 974

LEEDS, AL 35094

PHONE: (205) 699-0455

FAX: (205) 699-0773

www.cbpipe.com

DATE: 7/15/25

CONTRACTOR: NAN, INC.

PROJECT: HILO WWTP

BID DATE: 8/15/2024

AIS DOMESTIC MATERIAL

GLASS LINED

- 1) CBGL911 (10MIL DFT)

C&B PIPING INC

COATING(S) FOR DUCTILE IRON PIPE AND FITTINGS

ASPHALTIC VARNISH; 1 MIL DFT

- 1) ANSI/AWWA C104/A21.4 (1 MIL DFT)

AWWA STANDARDS

SHOP COAT PRIMER

- 1) TNEMEC L140F-1211 (3-5 MIL NOMINAL DFT)

TNEMEC COATINGS

DUCTILE IRON RESTRAINT FOR MECHANICAL JOINT

RESTRAINED FLANGED COUPLING ADAPTER

- 1) EBAA IRON MEGAFLANGE SERIES 2100

EBAA IRON

NOTE: SUPPLIED SBR RUBBER GASKET, AWWA C111

MJ BOLT THROUGH ADAPTER

- 1) FOSTER ADAPTER

INFACOT CORP

NOTE: SUPPLIED SBR RUBBER GASKET, AWWA C111

NOTE: SUPPLIED 316SS HARDWARE, AWWA C111

MEGACOUPLINGS

- 1) EBAA IRON MEGACOUPLING SERIES 3800

EBAA IRON

NOTE: SUPPLIED SBR RUBBER GASKET, AWWA C111

NOTE: SUPPLIED 316SS HARDWARE, AWWA C111

DUCTILE IRON MID-SPAN RESTRAINT

DUCTILE IRON RESTRAINT ANCHOR POINT FOR EXISTING DIP

- 1) EBAA IRON SERIES 1100SDB

EBAA IRON

NOTE: SUPPLIED 316SS HARDWARE, AWWA C111

GROOVED END DUCTILE IRON PIPE ACCESSORIES

COUPLING FOR GROOVE END PIPE/FTGS

- 1) AWWA SERIES 31, GRV X GRV

VICTAULIC

NOTE: SUPPLIED "S" RUBBER GASKET, AWWA C111

NOTE: SUPPLIED PLATED HARDWARE, AWWA C111



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CONTRACTOR: NAN, INC.

PROJECT: HILO WWTP

BID DATE: 8/15/2024

AIS DOMESTIC MATERIAL

2) AWWA SERIES 341, GRV X FLG ADAPTER

VICTAULIC

NOTE: SUPPLIED "S" RUBBER GASKET, AWWA C111

DUCTILE IRON PIPE X DI FITTINGS ACCESSORY SET

NON-RESTRAINED MJ ACCY SET

1) TYLER UNION C153 MECHANICAL JOINT ACCY SET

TYLER UNION

NOTE: SUPPLIED SBR RUBBER GASKET, AWWA C111

NOTE: SUPPLIED 316SS HARDWARE, AWWA C111



AMERICAN Ductile Iron Flex-Ring® Joint Pipe

Centrifugally Cast for Water, Sewage, or Other Liquids

4"-12" Flex-Ring® Joint



AMERICAN Flex-Ring® Restrained Joint Ductile Iron pipe, utilizing the sealing features of the time-proven Fastite® Joint and a boltless restrained connection, provides flexible, easily assembled, positive restraint against endwise separation due to thrust.

The patented Flex-Ring Joint is designed for a working pressure equal to that of the pipe or up to 350 psi in sizes 4"-24" and up to 250 psi in sizes 30" through 60". The joint has been thoroughly factory tested to withstand dead-end thrust resulting from more than twice those working pressures.

Flex-Ring® joint pipe with its positive, flexible joint restraint may also be used in trenchless applications such as **horizontal directional drilling** and pipe bursting. With spigot ahead, the low-profile Flex-Ring® bell assembles quickly and offers a smooth transition during pipe pull-back. AMERICAN offers a Flex-Ring® pulling bell assembly specifically designed for this installation method.



Pulling Bell Assembly

14"-60" Flex-Ring® Joint



For 4"-12" sizes, a beveled ductile iron, welded-on retainer ring and a yellow painted ductile iron split flex-ring, assembled behind the retainer ring, provide the means of restraint. After the plain end of the pipe is assembled into the Flex-Ring bell, the split flex-ring is inserted and springs into the socket locking groove. The flex-ring is securely positioned behind the welded-on retainer ring and in the socket locking groove on the inside of the pipe bell. This provides the flexible restraint.

For 14"-60" sizes, a shop-applied weld bead and a rubber-backed ring, containing yellow painted ductile iron segments, provide the means of restraint. As the plain end of the pipe is fully assembled into the bell, the ductile iron segments automatically close on the pipe behind the weld bead. The enclosure of the segments between the weld bead, spigot, and the sloped inner lip of the bell provides the flexible restraint.

The Flex-Ring Joint can be safely deflected after assembly to the limits shown in Table Nos. 9-1 and 9-2. This liberal deflection facilitates installation, decreases the number of necessary fittings, and accommodates settlement.

The Flex-Ring Joint is Underwriters Laboratories listed and Factory Mutual approved in sizes 4"-12". This UL listing and FM approval applies to all pressure classes and special thickness classes of ductile iron pipe. The only joint components needed to assemble the Flex-Ring Joint are a gasket and a single ring.



AMERICAN Ductile Iron Flex-Ring® Joint Pipe

Centrifugally Cast for Water, Sewage, or Other Liquids

No loose lugs, heavy wedges, rubber tubes, etc. are necessary. Also, there is no need to orient bells to ensure proper installation, though for convenience, most installers orient the split locking ring ends in 4"-12" sizes away from the very bottom of the joints. Just follow the simple instructions shown on the following pages, and positive restraint is ensured.

Flex-Rings, Flex-Ring segments, and retainer rings are manufactured of ductile iron compatible with pipe. Welds and weld beads (if used) are nickel-iron, proven desirably cathodic to the ductile iron pipe, and welding is performed using welders qualified to produce high-quality, dependable welds.

Fittings for use with 16"- 48" Flex-Ring pipe are ductile iron and meet or exceed the applicable performance and manufacturing requirements of ANSI /AWWA C110/A21.10 or ANSI/AWWA C153/A21.53. These are rated

for the same working pressures shown for like fittings in C110 and C153. Fittings in these sizes are also available in both bell-bell and bell-Flex-Ring spigot configurations for installation versatility and economy.

AMERICAN Flex-Ring® pipe and fittings are normally furnished with standard asphaltic coating outside and cement lined in accordance with ANSI/AWWA C104/A21.4. Special coatings and linings can be furnished when specified.

Field closures or other restraint can normally be securely made by using AMERICAN's Amarillo Fast-Grip® gasket, which is available in 4"-30" sizes. (See page 9-2 for details of the Amarillo Fast-Grip gasket.) The Fast-Grip® gasket restraint closure is UL listed and FM approved for use in Flex-Ring and Fastite bells in 4"-16" sizes. Field closures or other restraint in 14"-36" sizes can also be made in Flex-Ring bells only by using AMERICAN's Field Flex-Ring®. (See page 9-23.)



AMERICAN DUCTILE IRON PIPE

AMERICAN Ductile Iron Flex-Ring® Joint Pipe Standard Dimensions and Pressure Ratings

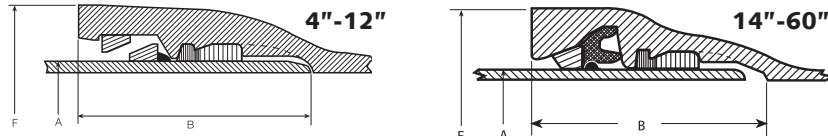


Table No. 9-1

4"-12"

Size in.	Working Pressure* psi	Nominal Laying Length** ft.	A O.D. in.	B Socket Depth in.	F Bell O.D.† in.	Allowable Pulling Load lb.††	Allowable Deflection degree	Offset per 20' Length in.	Radius of Curve^ ft.	Empty Pipe Buoyancy in Water (lb/ft)^^
4	350	20	4.80	5.62	7.06	10,000	5	21	230	-5
6	350	20	6.90	5.62	9.19	20,000	5	21	230	-2
8	350	20	9.05	5.74	11.33	30,000	5	21	230	3
10	350	20	11.10	6.72	13.56	45,000	5	21	230	11
12	350	20	13.20	6.72	15.74	60,000	5	21	230	19

* Working pressure is the maximum pressure rating of the joint and is based on its capability to resist thrust due to internal pressure. If higher working pressure is required, check AMERICAN.

** Laying length is nominal 20'. Where exact lengths are required, contact AMERICAN. Minimum laying lengths for Flex-Ring & Flex-Ring End pipe is 1'-0" and for Flex-Ring End & Flex-Ring End pipe is 2'-0".

† Dimensions subject to change at our option. Check AMERICAN if smaller or exact dimensions are required.

†† Intended for Horizontal Directional Drilling (HDD) applications, so these pulling loads are for a fully deflected position. Flex-Ring pipe may be available for greater pulling loads than indicated in the tabulated values. **Contact AMERICAN when higher pulling loads are required.**

^ Approximate radius of curve produced by a succession of 20' lengths of pipe fully deflected.

^^ Based on weight of empty (full of air) Pressure Class 350 Flex-Ring pipe with standard cement lining immersed in water. Positive numbers indicate such pipe will float.

Table No. 9-2

14"-60"

Size in.	Working Pressure* psi	Nominal Laying Length** ft.	A O.D. in.	B Socket Depth in.	F Bell O.D.† in.	Allowable Pulling Load lb.††	Allowable Deflection degree	Offset per 20' Length in.	Radius of Curve^ ft.	Empty Pipe Buoyancy in Water (lb/ft)^^
14	350	20	15.30	7.38	19.31	75,000	4	17	285	27
16	350	20	17.40	7.38	21.08	95,000	3 3/4	16	305	38
18	350	20	19.50	8.20	23.70	120,000	3 3/4	16	305	52
20	350	20	21.60	8.20	25.37	150,000	3 1/2	15	327	69
24	350	20	25.80	8.96	29.88	210,000	3	12	380	104
30	250	20	32.00	9.63	36.34	220,000	2 1/2	10	458	175
36	250	20	38.30	9.63	43.10	310,000	2	8	570	266
42	250	20	44.50	10.84	49.92	390,000	2	8	570	359
48	250	20	50.80	12.37	56.36	500,000	2	8	570	484
54	250	20	57.56	12.74	63.90	650,000	1 1/2	6	760	632
60	250	20	61.61	13.13	68.16	745,000	1 1/2	6	760	781

* Working pressure is the maximum pressure rating of the joint and is based on its capability to resist thrust due to internal pressure. If higher working pressure is required, check AMERICAN. Pressure rating of the joint is limited by the pressure rating of the parent pipe.

** Laying length is nominal 20'. Where exact lengths are required, contact AMERICAN. See below for minimum laying lengths for 14"-60" Flex-Ring.

† Dimensions subject to change at our option. Check AMERICAN if smaller or exact dimensions are required.

†† Intended for Horizontal Directional Drilling (HDD) applications, so these pulling loads are for a fully deflected position. The tabulated values are based on Pressure Class 350 pipe thickness. **Contact AMERICAN when it may be desirable to use lesser pressure class pipe or when higher pulling loads are required.** Flex-Ring pipe may be available for greater pulling loads than indicated in the tabulated values.

^ Approximate radius of curve produced by a succession of 20' lengths of pipe fully deflected.

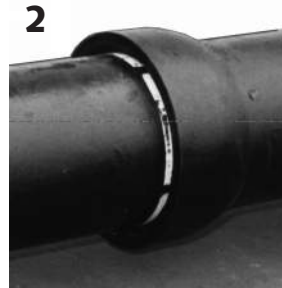
^^ Based on weight of empty (full of air) Pressure Class 350 Flex-Ring pipe with standard cement lining immersed in water. Positive numbers indicate such pipe will float.

Minimum Laying Lengths

Size in.	Flex-Ring & Flex-Ring End	Flex-Ring End & Flex-Ring End
14	1'-6"	2'-0"
16	1'-6"	2'-0"
18	1'-6"	2'-0"
20	1'-6"	2'-0"
24	2'-0"	2'-6"
30	2'-0"	2'-6"
36	2'-0"	2'-6"
42	2'-0"	3'-0"
48	2'-6"	3'-0"
54	3'-0"	3'-6"
60	3'-6"	4'-0"



AMERICAN Ductile Iron Flex-Ring® Joint Pipe Assembly Instructions 4"-12"



Prior to joint assembly, remove the packing material holding the split flex-ring onto the pipe. (See "Field Assembly of Flex-Ring" if split flex-ring is shipped separately.) Thoroughly clean the socket locking groove as well as the Fastite gasket recess and pipe plain end. In accordance with standard Fastite joint assembly instructions, insert the gasket and lubricate the pipe plain end, bevel, and inside surface of the gasket. With the pipe in essentially straight alignment, assemble the plain end into the Flex-Ring socket until the spigot stripe disappears into the bell. The orientation of the spigot stripe relative to the bell face is an indication of pipe alignment.

1. Tap the flex-ring into the socket beginning with one end of the flex-ring and progressing around the joint as shown in Photo 1. This

operation is made easier by holding one end of the flex-ring inside the bell as the remainder of the ring is caulked into the socket. Correct seating is generally ensured by a snapping noise as the flex-ring springs into position. This should be accompanied by visual or tactile inspection (the flex-ring is painted yellow to aid in this inspection). (Note: When a visual inspection to determine the flex-ring position is not practical, such as in an underwater installation, a feeler gauge can be used to ensure the correct positioning of the flex-ring in the socket locking groove. It may be necessary to move the entering pipe slightly to improve alignment if the ring does not readily spring into the socket locking groove.)

2. The completed joint.



This bridge crossing illustrates design/construction advantages, including the deflection capabilities of AMERICAN Flex-Ring Joint Pipe.



AMERICAN Ductile Iron Flex-Ring® Joint Pipe Assembly Instructions 4"-12"



FIELD ASSEMBLY OF FLEX-RING

If the split flex-ring is shipped separately, assemble it onto the pipe spigot by spreading the Flex-Ring ends as shown above. Be sure that the flex-ring is oriented so that the small end is toward the pipe plain end.



DISASSEMBLY OF 4"-12" FLEX-RING

If disassembly of the joint is required, it may be accomplished by inserting pins or nails into the drilled holes furnished in the flex-ring ends and compressing the flex-ring firmly onto the pipe as shown above. If desired, steel pins can be field welded onto the ends of common adjustable pliers, if such a disassembly tool is more desirable to the user. If axial movement or joint extension has occurred in the joint prior to disassembly, it may be necessary to move the spigot completely to the rear of the socket in straight alignment to allow the Flex-Ring to be compressed for removal.

THE FOLLOWING INFORMATION PERTAINS TO 4"-60" JOINTS:

NOTE: The AMERICAN Flex-Ring Joint allows for joint take-up and flexibility after installation. In most underground installations, including most restrained bend locations, this feature is advantageous in that increased thrust-resisting soil forces are generated. Also, expansion and contraction due to temperature variations may be accommodated without excessive stress in the pipe members.

In any application where axial or lateral movement may be undesirable, such as certain bridge crossings, certain exposed or unburied piping applications, or certain connections of restrained pipe sections to rigid piping, special provisions, including effective joint extension, may be necessary to control unacceptable pipe-

line movement. (See also Section 7, Pipe-On-Supports, etc.)

Depending on job conditions and restrained pipe length, cumulative joint take-up can be substantial, particularly in exposed or unburied piping applications. In this regard, joints may be extended after assembly to minimize further joint take-up in test or service. This will not prevent proper joint deflection.

The amount of joint take-up or line movement in buried restrained pipelines is substantially limited by the surrounding soil. Therefore, system security and safety is maximized by filling and testing restrained sections of pipelines after back-filling as recommended by ANSI/AWWA C600, [Installation of Ductile Iron Water Mains and Their Appurtenances](#) and AWWA M41.



AMERICAN Ductile Iron Flex-Ring® Joint Pipe Assembly Instructions 14"-60"

1) Cleaning and Fastite gasket insertion

Thoroughly clean the socket restraining groove (nearest the bell end), the Fastite gasket recess, and the pipe plain end, removing dirt, sand, ice, mud, or any other material which could prevent the proper placement of the Fastite gasket and flex-ring. As in normal Fastite joint assembly, insert the gasket into the gasket socket groove. (1).

Important: A Fastite gasket must also be used, because the rubber backed flex-ring does not perform any sealing function.

2) Placement of the the flex-ring in socket and joint lubrication

Remove the flex-ring from its container and place it in the socket restraining groove in gasket-like fashion (Photo 2).

The yellow restraining segments of the flex-ring must be oriented toward the entering spigot. This may be done by first placing the flex-ring in the socket groove by forming one or more inward or lateral loops in the rubber backed ring (Photo 3). The 48-inch, 54-inch, and 60-inch diameter rubber backed flex-ring has an area with an extended rubber section. When placing the 48-inch, 54-inch,

and 60-inch rubber backed flex-ring into the socket, this extended rubber section should be oriented at the top of the joint.

Work all inward or lateral loops fully outward and planar such that each metal segment fits reasonably flush against the wedging surface of the socket, and no rubber bulges or twists remain (Photo 4).

Lubricate the inside surface of the gasket and the first four inches of the spigot including the beveled nose end of the pipe. It is not necessary to lubricate the rubber backed flex-ring or the yellow segments. Do not allow the lubricated spigot end of the pipe to contact the ground prior to insertion.

3) Initial placement of Flex-Ring spigot end into socket

With the spigot in reasonably straight alignment and centered within the flex-ring (Photo 5), insert the spigot until it contacts the back of the socket per normal Fastite joint assembly procedure. (See Section 2 for additional detail on Fastite assembly procedures.) When the weld bead is in proper assembled position fully beyond the yellow Flex-Ring segments, every segment will be trapped firmly between the weld bead, the spigot, and the wedging surface of the socket.

Verify the correct positioning of the yellow Flex-Ring segments by visual inspection (or by "feeler" gauge if installed in conditions of poor visibility).

The segments will normally snap directly into the correct assembled position. However, if any segment should not come down firmly onto the pipe at any location, deflect the pipe slightly in that direction, thereby allowing the segment to seat itself correctly.

After joint assembly, the joint may be extended and then deflected within the range of allowable joint deflection for the size of pipe being assembled.

4) Assembly of fittings

Flex-Ring pipe and fitting joints can generally be assembled with the same tools and methods used for many years with Fastite joints. When using a field-cut pipe to locate a fitting, it may be advantageous to use an uncut flex-ring spigot end (with factory weld bead) and a standard Flex-Ring in the fitting socket rather than using a field-cut plain end and Field Flex-Ring with

Photo 1



Photo 2



Photo 3



Photo 4



Photo 5





AMERICAN Ductile Iron Flex-Ring® Joint Pipe Assembly Instructions 14"-60"

black-toothed gripping segments. A Field Flex-Ring and cut pipe plain end may then be used in the nearest pipe socket on either side of the fitting. When possible, the use of standard flex-ring with yellow segments and factory spigots with weld beads in the sockets of a fitting may allow easier orientation or rotation of the fitting relative to the pipe after assembly, if this is needed. (See Section 4 for additional detail on the assembly of Fastite fittings.)

5) Joint extension after installation

The 14" - 60" Flex-Ring locking mechanism allows approximately one inch of free axial movement and also provides substantial flexibility after installation. However, the joints may be extended after assembly to minimize this joint takeup in test or service conditions.

In most underground installations (including most restrained bend locations), joint take-up is advantageous in that increased thrust-resisting forces are generated. Also, expansion and contraction due to temperature variations may be accommodated without excessive stress in pipe members. The amount of joint take-up or line movement in buried restrained pipelines is substantially limited by the surrounding soil. There-

fore, system security and safety is maximized by filling and testing restrained sections of pipelines **after** backfilling as recommended by ANSI/AWWA C600, Installation of Ductile Iron Water Mains and Their Appurtenances and AWWA M41.

In any application where axial or lateral movement may be undesirable, such as certain bridge crossings, certain exposed or unburied piping applications, or certain connections of restrained pipe sections to rigid piping, special provisions, including effective joint extension, may be necessary to control unacceptable pipeline movement.

Depending on job conditions and restrained pipe length, cumulative joint take-up can obviously be substantial, particularly in exposed piping applications. Where joint pre-extension is necessary in a piping system, it may be accomplished by pulling or jacking the spigot away from the socket until firm resistance is encountered. This will not limit joint flexibility. See "Restrained Joint Pipe Assembly Extension Procedure" in this section of the Pipe Manual for more information concerning joint extension.



The versatile performance capabilities of AMERICAN Flex-Ring Joint Pipe are perfectly suited for projects containing a variety of conditions such as the hilly, rocky terrain shown in this photo.



AMERICAN Ductile Iron Flex-Ring® Joint Pipe Disassembly Instructions for 14"-60" Flex-Ring Joints

Flex-Ring joints may be disassembled if required using sharp wedges and 3/16"-1/4" thick disassembly shims. Flex-Ring disassembly sets are available from AMERICAN and are suggested for disassembly. These disassembly sets include two sharp steel starter wedges and an appropriate number of "L"-shaped shims. The wedges are used to start the separation of the yellow Flex-Ring joint locking segments outward from the spigot while it is in the bell of an already assembled joint. The "L"-shaped shims are then hammered between the spigot and each locking segment. The thicker shims lift the locking segments entirely away from the spigot when fully inserted, and allow the spigot weld bead to pass under the locking segments generally located as shown in Figs. 1 and 2. Step-by-step instructions follow:

1. First straighten the joint as much as possible and push or pull the spigot back into the bell until it "bottoms out" in the rear of the socket. (Fig. 3)

2. Hammer a starter wedge under a yellow locking segment until an approximately 1/8" gap is seen between the segment and the spigot. (Fig. 4)

3. Hammer a second wedge (if necessary to start the shims) under the other end of the locking segment as in step 2.

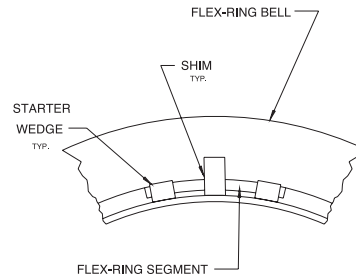


Figure 1
Starter shim and wedge arrangement
for 14", 18", and 20" sizes.

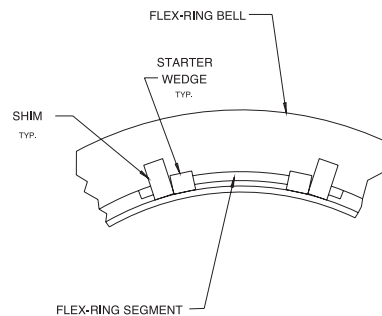


Figure 2
Shim and wedge arrangement for
16", 24", 30", 42", 48", 54" and 60" sizes.

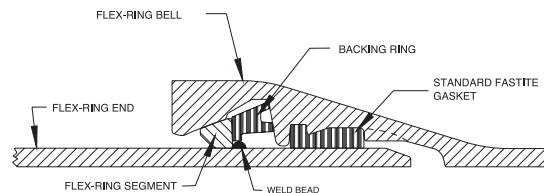


Figure 3

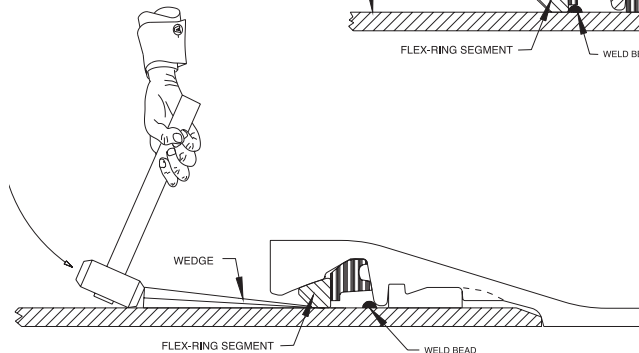
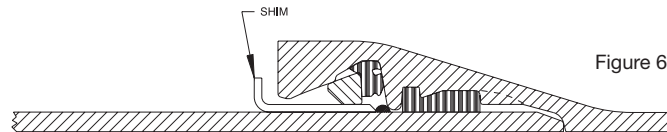
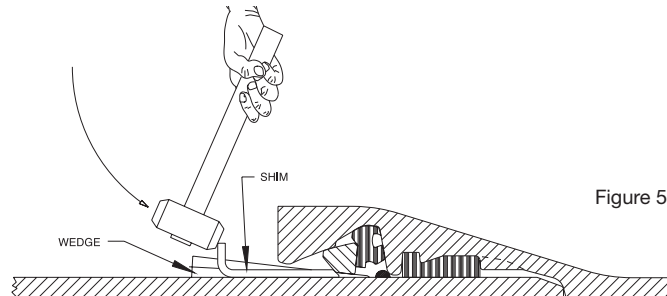


Figure 4



4. With a large hammer (such as a six pounder) vigorously drive one or two “L”-shaped shims under the locking segment until the ends of the shims firmly contact the spigot weld bead. (Fig. 5) Shims and wedges can be safely and firmly held against the pipe as they are hammered using a block of wood or a board. Safety precautions such as the wearing of safety glasses and keeping clear of the hammer during striking should always be taken to avoid injury.

5. Remove the starter wedges from between the locking segment and spigot, leaving the shim(s) in place. (Fig. 6) Note that the wedges are reused for each locking segment.

6. Drive wedges and shims under all locking segments as shown in steps 2-5. (See photo.) Check to be sure that the inner surface of all segments will not interfere with the spigot weld bead during joint separation after inserting shims.

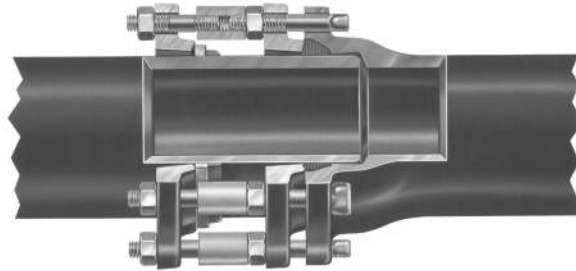
7. Separate the joint. During joint separation, it is generally best to pull the spigot straight out of the socket. Extreme back and forth deflecting motions of the spigot during joint separation can cause shims to fall out of the joint and/or relocking to occur. If the joint does not readily come apart, check to see if one or more of the segments is in locking contact with the spigot weld bead. If so, push or deflect the spigot back in that location and add or replace shims as required.



Disassembly kits accompanied by instructions for use can be furnished by AMERICAN upon request.



AMERICAN Ductile Iron MJ Coupled Joint Pipe



The MJ Coupled Joint offers a method of providing joint restraint with a standard Mechanical Joint socket of a pipe, fitting, valve, or other product. It is available in sizes 4"-12" with a pressure rating of 350 psi and deflection capability up to 2 1/2°, and in 14"-54" with a pressure rating of 250 psi intended for installation in essentially straight alignment. Check AMERICAN if larger sizes or greater deflections are desired. The MJ Coupled Joint is commonly used as a method of joint restraint for Mechanical Joint all bell fittings and valves.

The joint material consists of a standard MJ gasket and a standard ductile iron joint gland. The restraint gland* is ductile iron and is assembled on

the pipe behind a welded-on 1/2" square alloy steel retaining ring. Coupling nuts, studs, teehead bolts and nuts, and retaining ring are all alloy steel. Tee-head bolts are special-length bolts.

* The restraint gland for 30"-54" sizes is a substantially rectangular cross-section ductile iron gland.

Note: It is sometimes possible to provide for field-cutting and positive, even "drop-in" field-adaptable restraint in restrained piping areas by using modified MJ Coupled joints where the weld ring and restraint gland are intentionally moved farther away from the joint, and the standard short studs of the joint are replaced by longer threaded rods in the field. Threaded rods, of course, can also be cut to suit in the field. Similar effect can also be accomplished utilizing standard fabricated, stuffing-box couplings and similarly modified "Coupling Gland Ends" (see pg 9-42). Contact AMERICAN for additional information on this.

Assembly Instructions

1. Assemble the Mechanical Joint according to standard MJ joint procedures with the pipe and/or fitting in essentially straight alignment. Note that the bolts for this joint are not standard length; special-length bolts are furnished for this joint by AMERICAN. Where tapped holes are encountered in fittings or in valves, use studs of the same special length as specified for the THHN Bolts.

2. Thread the coupling nuts onto the tee-head bolts and hand tighten against the hex nuts.** Install short, threaded end of studs into the coupling nuts.

3. Slide the gland into position and install outer hex nuts. (Note: If deflection is desired, it should be taken at this point prior to tightening the outer hex nuts.) Tighten to 20 ft-lb in order to draw gland substantially against welded-on ring on pipe. Joint is now complete.

** In 20" and 24" sizes only, the coupling nut is to be positioned not against the hex nut, but rather so that there is a minimum of 3/4" thread engagement with both the tee-head as well as stud threads.



AMERICAN Ductile Iron MJ Coupled Joint

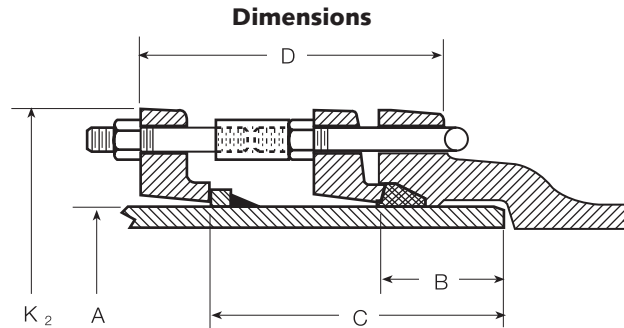


Table No. 9-13

4"-24"

Size in.	A Outside Diameter in.	B Socket Depth in.	C Plain End to Gland in.	K2 Gland O.D. in.	Bolts or Studs No.	Tee-Head Bolt in.	Stud in.	MJCJE & MJCJE Minimum Laying Length (ft.)
4	4.80	2.50	6.34	9.12	4	3/4 x 4	3/4 x 3 1/2	2'-3"
6	6.90	2.50	6.50	11.12	6	3/4 x 4	3/4 x 3 1/2	2'-3"
8	9.05	2.50	6.64	13.37	6	3/4 x 4 1/2	3/4 x 3 1/2	2'-3"
10	11.10	2.50	6.71	15.69	8	3/4 x 4 1/2	3/4 x 3 1/2	2'-3"
12	13.20	2.50	6.77	17.94	8	3/4 x 4 1/2	3/4 x 3 1/2	2'-3"
14	15.30	3.50	8.88	20.31	10	3/4 x 5	3/4 x 5	2'-9"
16	17.40	3.50	8.88	22.56	12	3/4 x 5	3/4 x 5	2'-9"
18	19.50	3.50	8.88	24.83	12	3/4 x 5	3/4 x 5	3'-0"
20	21.60	3.50	8.88	27.08	14	3/4 x 6	3/4 x 5	3'-0"
24	25.80	3.50	8.88	31.58	16	3/4 x 6	3/4 x 5	3'-0"

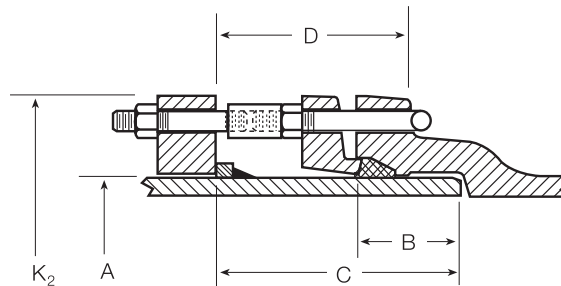
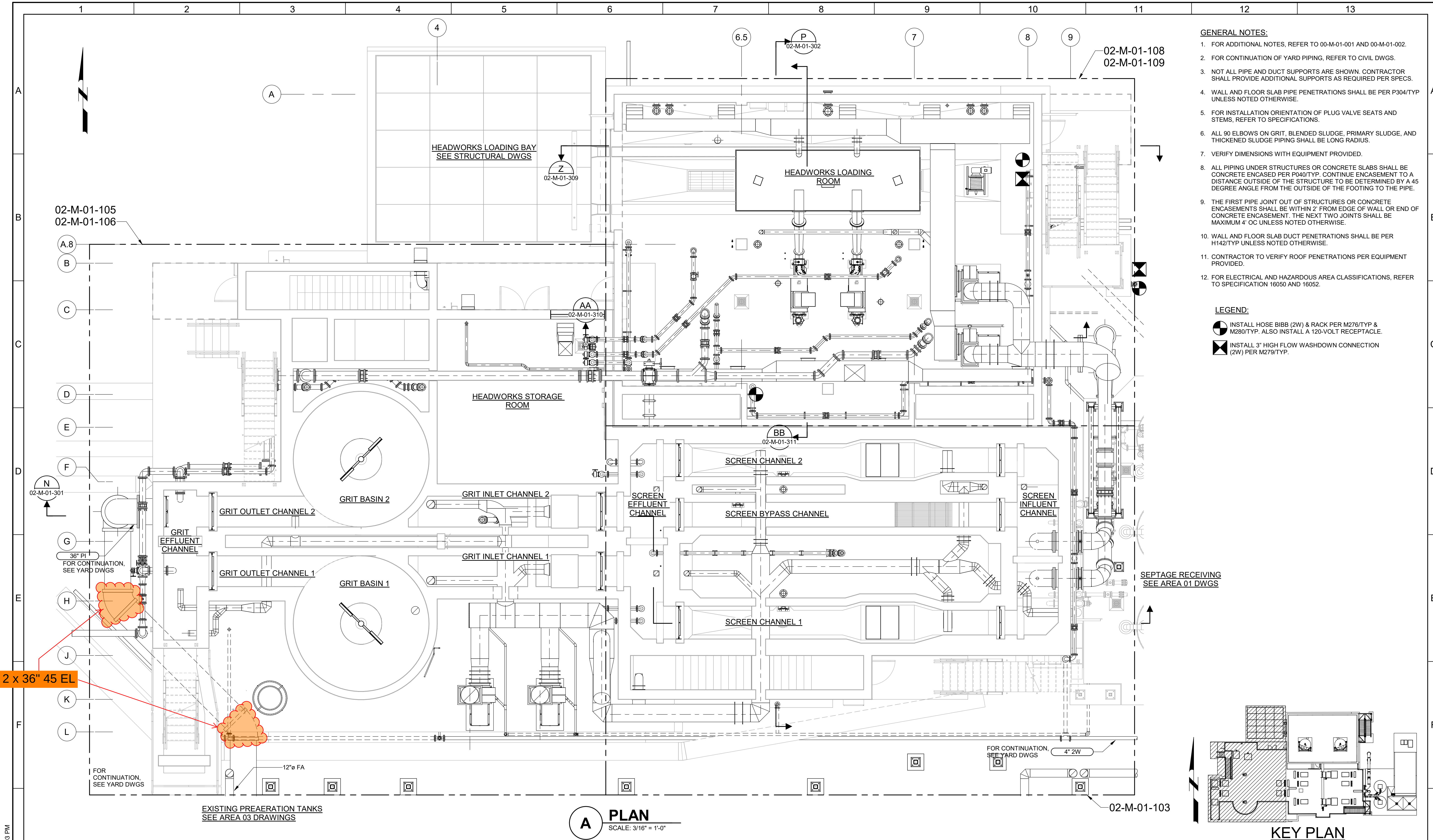


Table No. 9-14

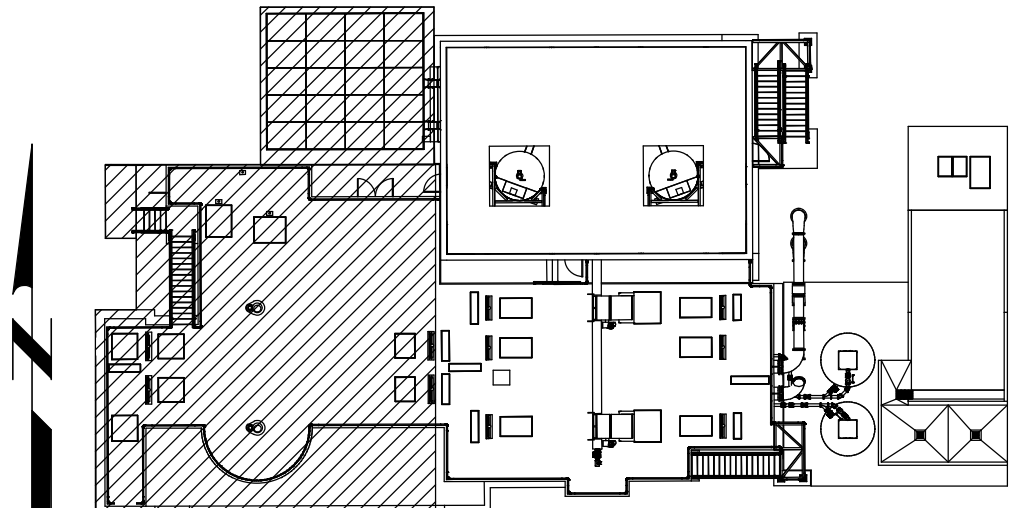
30"-48"

Size in.	A Outside Diameter in.	B Socket Depth in.	C Plain End to Gland in.	K2 Gland O.D. in.	Bolts or Studs No.	Tee-Head Bolt in.	Stud in.	MJCJE & MJCJE Minimum Laying Length (ft.)
30	32.00	4.00	10.88	39.12	20	1 x 7	1 x 6	3'-6"
36	38.30	4.00	10.88	46.00	24	1 x 7	1 x 6	3'-6"
42	44.50	4.00	11.63	53.12	28	1 1/4 x 7	1 1/4 x 7	4'-0"
48	50.80	4.00	11.63	60.00	32	1 1/4 x 7	1 1/4 x 7	4'-0"
54	57.56	4.00	17.50	66.88	36	1 1/4 x 7	1 1/4 x 14	5'-6"



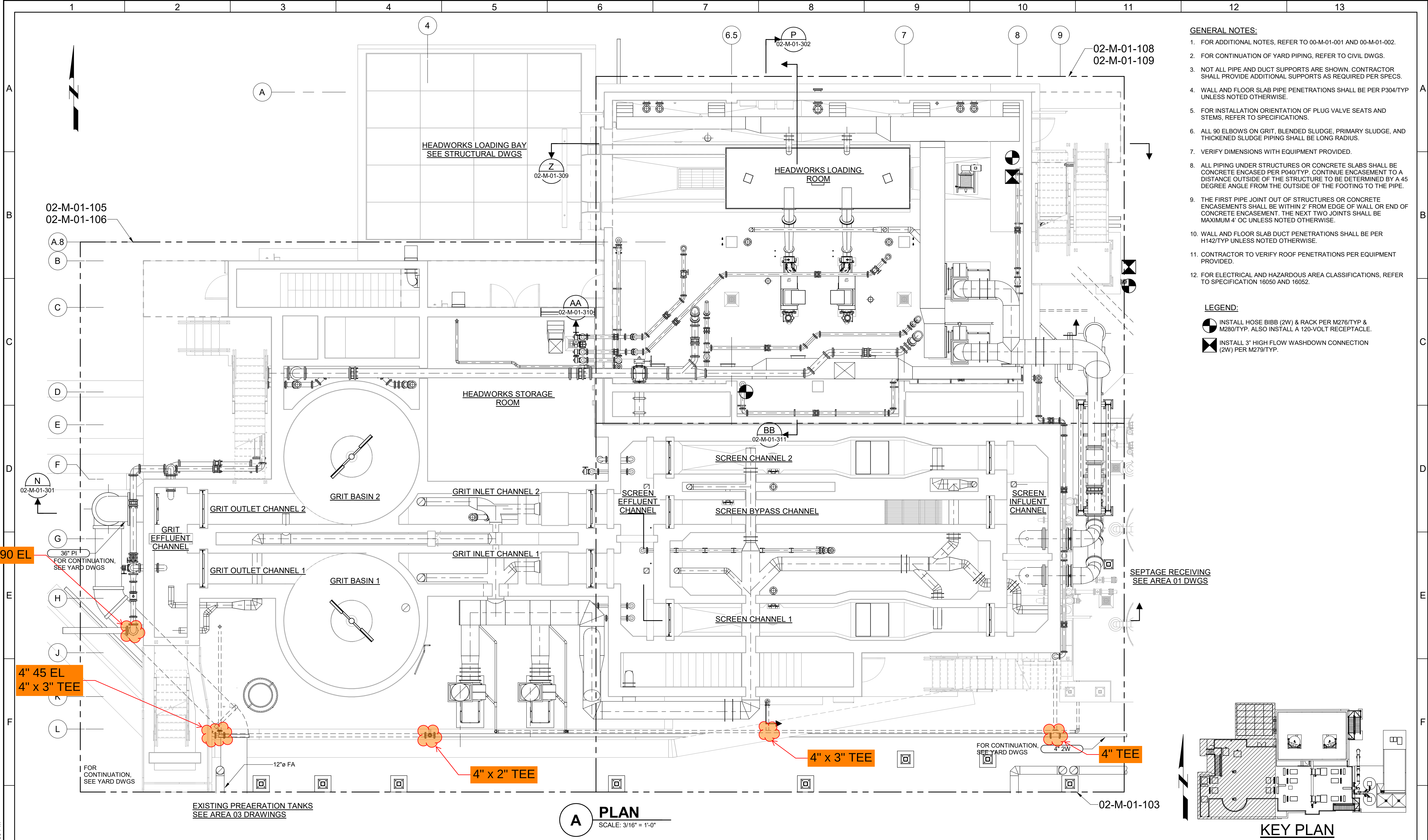
- GENERAL NOTES:**
- FOR ADDITIONAL NOTES, REFER TO 00-M-01-001 AND 00-M-01-002.
 - FOR CONTINUATION OF YARD PIPING, REFER TO CIVIL DWGS.
 - NOT ALL PIPE AND DUCT SUPPORTS ARE SHOWN. CONTRACTOR SHALL PROVIDE ADDITIONAL SUPPORTS AS REQUIRED PER SPECS.
 - WALL AND FLOOR SLAB PIPE PENETRATIONS SHALL BE PER P304/TYP UNLESS NOTED OTHERWISE.
 - FOR INSTALLATION ORIENTATION OF PLUG VALVE SEATS AND STEMS, REFER TO SPECIFICATIONS.
 - ALL 90 ELBOWS ON GRIT, BLENDED SLUDGE, PRIMARY SLUDGE, AND THICKENED SLUDGE PIPING SHALL BE LONG RADIUS.
 - VERIFY DIMENSIONS WITH EQUIPMENT PROVIDED.
 - ALL PIPING UNDER STRUCTURES OR CONCRETE SLABS SHALL BE CONCRETE ENCASED PER P040/TYP. CONTINUE ENCASEMENT TO A DISTANCE OUTSIDE OF THE STRUCTURE TO BE DETERMINED BY A 45 DEGREE ANGLE FROM THE OUTSIDE OF THE FOOTING TO THE PIPE.
 - THE FIRST PIPE JOINT OUT OF STRUCTURES OR CONCRETE ENCASEMENTS SHALL BE WITHIN 2' FROM EDGE OF WALL OR END OF CONCRETE ENCASEMENT. THE NEXT TWO JOINTS SHALL BE MAXIMUM 4' OC UNLESS NOTED OTHERWISE.
 - WALL AND FLOOR SLAB DUCT PENETRATIONS SHALL BE PER H142/TYP UNLESS NOTED OTHERWISE.
 - CONTRACTOR TO VERIFY ROOF PENETRATIONS PER EQUIPMENT PROVIDED.
 - FOR ELECTRICAL AND HAZARDOUS AREA CLASSIFICATIONS, REFER TO SPECIFICATION 16050 AND 16052.

- LEGEND:**
- INSTALL HOSE BIBB (2W) & RACK PER M276/TYP & M280/TYP. ALSO INSTALL A 120-VOLT RECEPTACLE.
 - INSTALL 3" HIGH FLOW WASHDOWN CONNECTION (2W) PER M279/TYP.



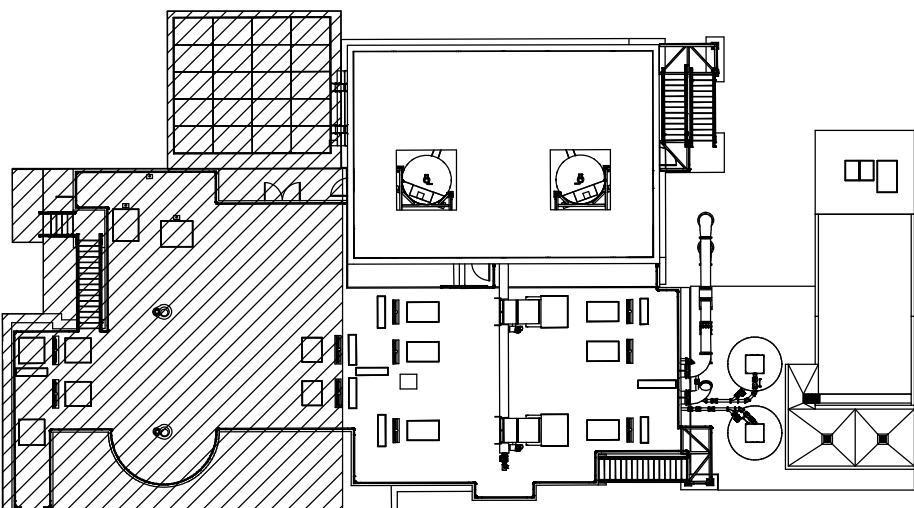
DESIGNED XHk/TH				CONFORMED DOCUMENT	VERIFY SCALES BAR IS ONE INCH ON ORIGINAL DRAWING 0 1" IF NOT ONE INCH ON THIS SHEET, ADJUST SCALES ACCORDINGLY		COUNTY OF HAWAII			JOB NO. 201400	
DRAWN LV							HILO WWTP REHABILITATION AND REPLACEMENT PROJECT PHASE 1			DRAWING NO. 02-M-01-101	
CHECKED MES							MECHANICAL			SHEET NO.	
DATE APR 2024							HEADWORKS OVERALL CHANNEL PLAN			369 of 1000	
REV	DATE	BY	DESCRIPTION					10	11	12	13
1											

FILE NAME: R:\d1361566\201400_02M0801.rvt



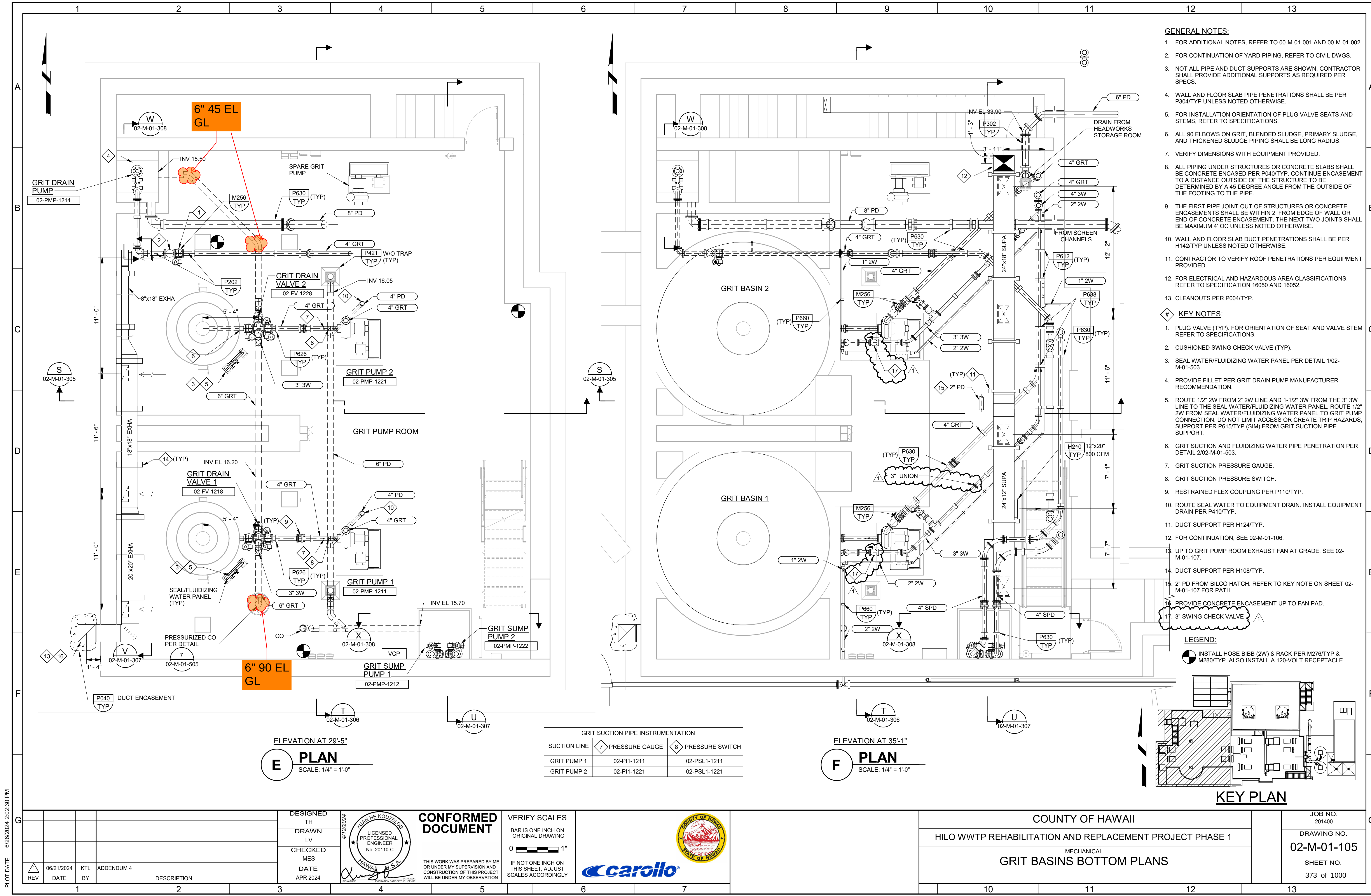
- GENERAL NOTES:**
- FOR ADDITIONAL NOTES, REFER TO 00-M-01-001 AND 00-M-01-002.
 - FOR CONTINUATION OF YARD PIPING, REFER TO CIVIL DWGS.
 - NOT ALL PIPE AND DUCT SUPPORTS ARE SHOWN. CONTRACTOR SHALL PROVIDE ADDITIONAL SUPPORTS AS REQUIRED PER SPECS.
 - WALL AND FLOOR SLAB PIPE PENETRATIONS SHALL BE PER P304/TYP UNLESS NOTED OTHERWISE.
 - FOR INSTALLATION ORIENTATION OF PLUG VALVE SEATS AND STEMS, REFER TO SPECIFICATIONS.
 - ALL 90 ELBOWS ON GRIT, BLENDED SLUDGE, PRIMARY SLUDGE, AND THICKENED SLUDGE PIPING SHALL BE LONG RADIUS.
 - VERIFY DIMENSIONS WITH EQUIPMENT PROVIDED.
 - ALL PIPING UNDER STRUCTURES OR CONCRETE SLABS SHALL BE CONCRETE ENCASED PER P040/TYP. CONTINUE ENCASEMENT TO A DISTANCE OUTSIDE OF THE STRUCTURE TO BE DETERMINED BY A 45 DEGREE ANGLE FROM THE OUTSIDE OF THE FOOTING TO THE PIPE.
 - THE FIRST PIPE JOINT OUT OF STRUCTURES OR CONCRETE ENCASEMENTS SHALL BE WITHIN 2' FROM EDGE OF WALL OR END OF CONCRETE ENCASEMENT. THE NEXT TWO JOINTS SHALL BE MAXIMUM 4' OC UNLESS NOTED OTHERWISE.
 - WALL AND FLOOR SLAB DUCT PENETRATIONS SHALL BE PER H142/TYP UNLESS NOTED OTHERWISE.
 - CONTRACTOR TO VERIFY ROOF PENETRATIONS PER EQUIPMENT PROVIDED.
 - FOR ELECTRICAL AND HAZARDOUS AREA CLASSIFICATIONS, REFER TO SPECIFICATION 16050 AND 16052.

- LEGEND:**
- INSTALL HOSE BIBB (2W) & RACK PER M276/TYP & M280/TYP. ALSO INSTALL A 120-VOLT RECEPTACLE.
 - INSTALL 3" HIGH FLOW WASHDOWN CONNECTION (2W) PER M279/TYP.



KEY PLAN

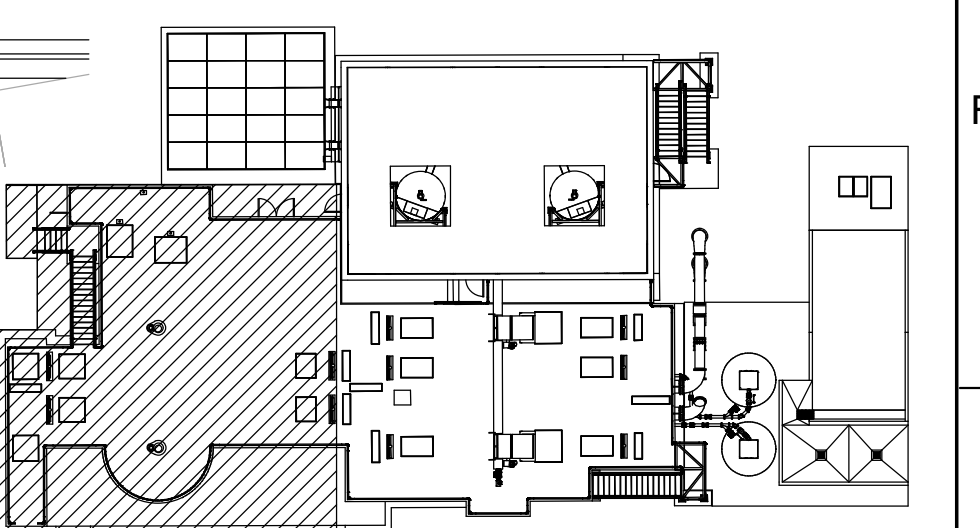
DESIGNED XH/K/TH DRAWN LV CHECKED MES DATE APR 2024				4/12/2024 XUAN HE KOUZELIS LICENSED PROFESSIONAL ENGINEER No. 20110-C HAWAII THIS WORK WAS PREPARED BY ME OR UNDER MY SUPERVISION AND CONSTRUCTION OF THIS PROJECT WILL BE UNDER MY OBSERVATION		CONFORMED DOCUMENT VERIFY SCALES BAR IS ONE INCH ON ORIGINAL DRAWING 0 1" IF NOT ONE INCH ON THIS SHEET, ADJUST SCALES ACCORDINGLY		COUNTY OF HAWAII HILO WWTP REHABILITATION AND REPLACEMENT PROJECT PHASE 1 MECHANICAL HEADWORKS OVERALL CHANNEL PLAN		JOB NO. 201400 DRAWING NO. 02-M-01-101 SHEET NO. 369 of 1000	
REV DATE BY DESCRIPTION				1 2 3 4		5 6 7 8 9 10 11 12 13		10 11 12 13		10 11 12 13	



- GENERAL NOTES:**
- FOR ADDITIONAL NOTES, REFER TO 00-M-01-001 AND 00-M-01-002.
 - FOR CONTINUATION OF YARD PIPING, REFER TO CIVIL DWGS.
 - NOT ALL PIPE AND DUCT SUPPORTS ARE SHOWN. CONTRACTOR SHALL PROVIDE ADDITIONAL SUPPORTS AS REQUIRED PER SPECS.
 - WALL AND FLOOR SLAB PIPE PENETRATIONS SHALL BE PER P304/TYP UNLESS NOTED OTHERWISE.
 - FOR INSTALLATION ORIENTATION OF PLUG VALVE SEATS AND STEMS, REFER TO SPECIFICATIONS.
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 - WALL AND FLOOR SLAB DUCT PENETRATIONS SHALL BE PER H142/TYP UNLESS NOTED OTHERWISE.
 - CONTRACTOR TO VERIFY ROOF PENETRATIONS PER EQUIPMENT PROVIDED.
 - FOR ELECTRICAL AND HAZARDOUS AREA CLASSIFICATIONS, REFER TO SPECIFICATION 16050 AND 16052.
 - CLEANOUTS PER P004/TYP.

- # KEY NOTES:**
- PLUG VALVE (TYP). FOR ORIENTATION OF SEAT AND VALVE STEM REFER TO SPECIFICATIONS.
 - CUSHIONED SWING CHECK VALVE (TYP).
 - SEAL WATER/FLUIDIZING WATER PANEL PER DETAIL 1/02-M-01-503.
 - PROVIDE FILLET PER GRIT DRAIN PUMP MANUFACTURER RECOMMENDATION.
 - ROUTE 1/2" 2W FROM 2" 2W LINE AND 1-1/2" 3W FROM THE 3" 3W LINE TO THE SEAL WATER/FLUIDIZING WATER PANEL. ROUTE 1/2" 2W FROM SEAL WATER/FLUIDIZING WATER PANEL TO GRIT PUMP CONNECTION. DO NOT LIMIT ACCESS OR CREATE TRIP HAZARDS. SUPPORT PER P615/TYP (SIM) FROM GRIT SUCTION PIPE SUPPORT.
 - GRIT SUCTION AND FLUIDIZING WATER PIPE PENETRATION PER DETAIL 2/02-M-01-503.
 - GRIT SUCTION PRESSURE GAUGE.
 - GRIT SUCTION PRESSURE SWITCH.
 - RESTRAINED FLEX COUPLING PER P110/TYP.
 - ROUTE SEAL WATER TO EQUIPMENT DRAIN. INSTALL EQUIPMENT DRAIN PER P410/TYP.
 - DUCT SUPPORT PER H124/TYP.
 - FOR CONTINUATION, SEE 02-M-01-106.
 - UP TO GRIT PUMP ROOM EXHAUST FAN AT GRADE. SEE 02-M-01-107.
 - DUCT SUPPORT PER H108/TYP.
 - 2" PD FROM BILCO HATCH. REFER TO KEY NOTE ON SHEET 02-M-01-107 FOR PATH.
 - PROVIDE CONCRETE ENCASEMENT UP TO FAN PAD.
 - 3" SWING CHECK VALVE

- LEGEND:**
- INSTALL HOSE BIBB (2W) & RACK PER M276/TYP & M280/TYP. ALSO INSTALL A 120-VOLT RECEPTACLE.



KEY PLAN

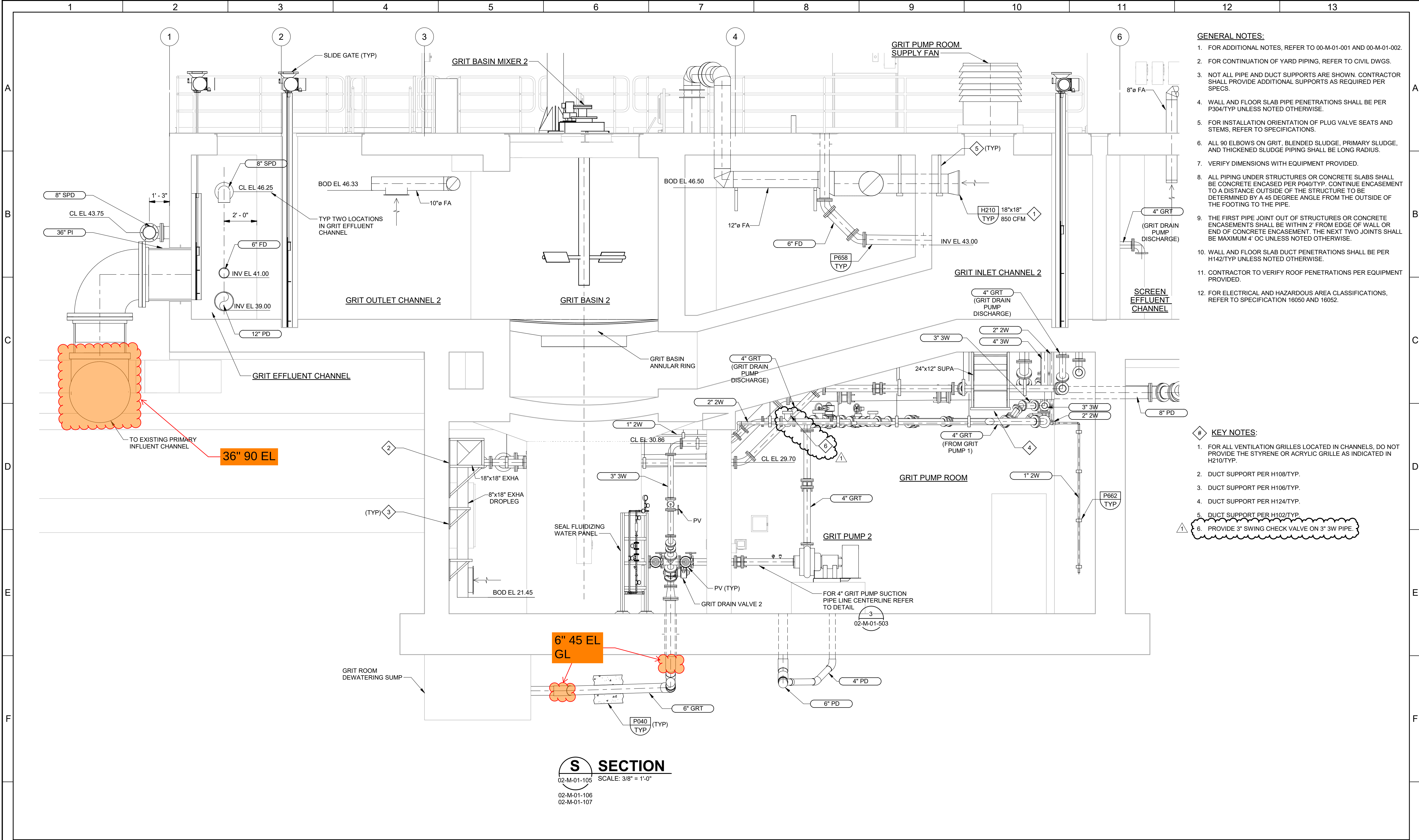
GRIT SUCTION PIPE INSTRUMENTATION			
SUCTION LINE	7 PRESSURE GAUGE	8 PRESSURE SWITCH	
GRIT PUMP 1	02-PI1-1211	02-PSL1-1211	
GRIT PUMP 2	02-PI1-1221	02-PSL1-1221	

E PLAN
ELEVATION AT 29'-5"
SCALE: 1/4" = 1'-0"

F PLAN
ELEVATION AT 35'-1"
SCALE: 1/4" = 1'-0"

DESIGNED TH				CONFORMED DOCUMENT	VERIFY SCALES BAR IS ONE INCH ON ORIGINAL DRAWING 0 1" IF NOT ONE INCH ON THIS SHEET, ADJUST SCALES ACCORDINGLY		COUNTY OF HAWAII			JOB NO. 201400					
DRAWN LV							HILO WWTP REHABILITATION AND REPLACEMENT PROJECT PHASE 1								
CHECKED MES							MECHANICAL								
DATE APR 2024							GRIT BASINS BOTTOM PLANS								
REV			DESCRIPTION						DRAWING NO. 02-M-01-105						
1			2			3			SHEET NO. 373 of 1000						

PLOT DATE: 6/26/2024 2:02:30 PM

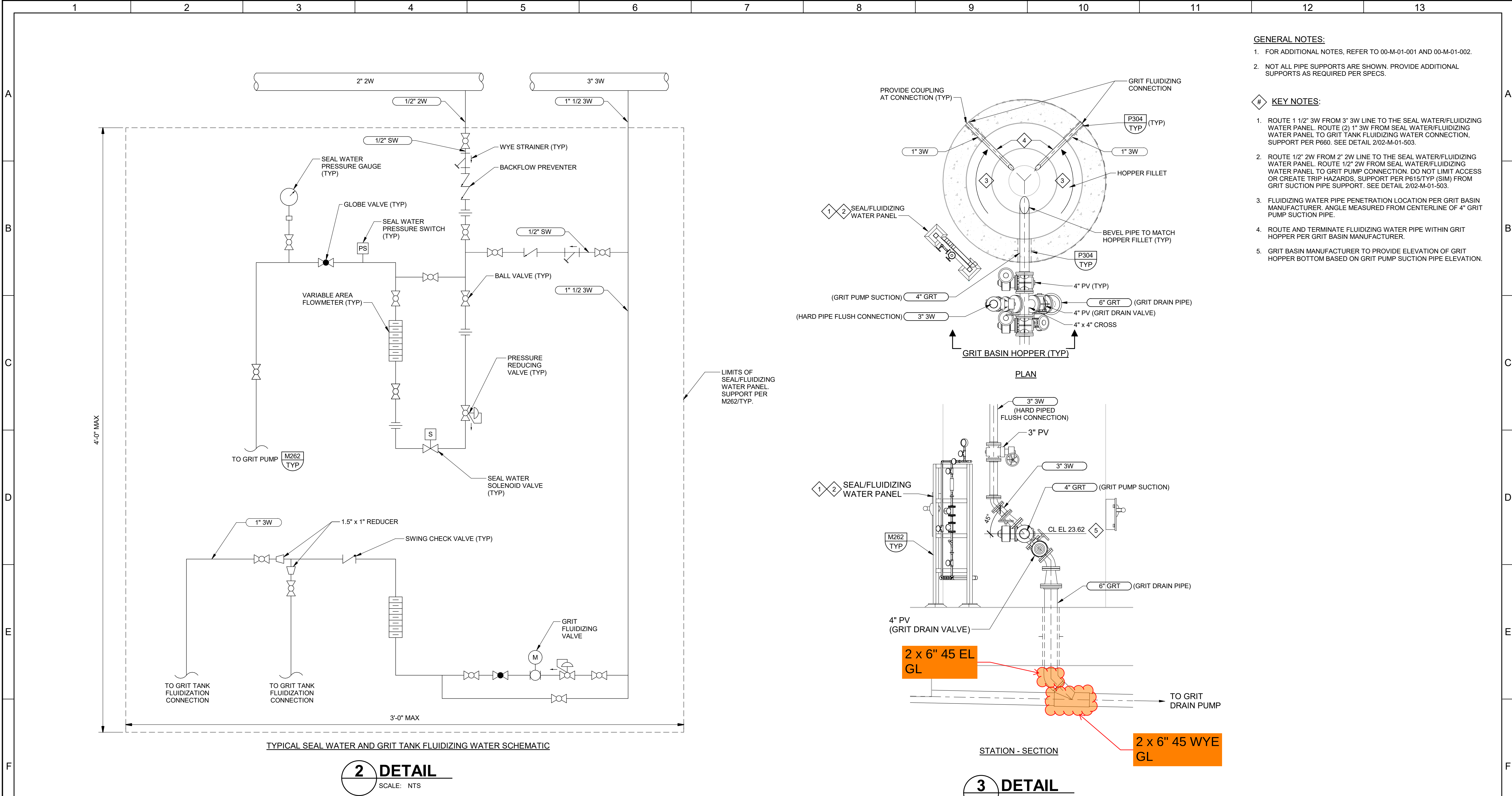


S SECTION

02-M-01-105 SCALE: 3/8" = 1'-0"

02-M-01-106
02-M-01-107

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- GENERAL NOTES:**
- FOR ADDITIONAL NOTES, REFER TO 00-M-01-001 AND 00-M-01-002.
 - NOT ALL PIPE SUPPORTS ARE SHOWN. PROVIDE ADDITIONAL SUPPORTS AS REQUIRED PER SPECS.
- # KEY NOTES:**
- ROUTE 1 1/2" 3W FROM 3" 3W LINE TO THE SEAL WATER/FLUIDIZING WATER PANEL. ROUTE (2) 1" 3W FROM SEAL WATER/FLUIDIZING WATER PANEL TO GRIT TANK FLUIDIZING WATER CONNECTION, SUPPORT PER P660. SEE DETAIL 2/02-M-01-503.
 - ROUTE 1/2" 2W FROM 2" 2W LINE TO THE SEAL WATER/FLUIDIZING WATER PANEL. ROUTE 1/2" 2W FROM SEAL WATER/FLUIDIZING WATER PANEL TO GRIT PUMP CONNECTION. DO NOT LIMIT ACCESS OR CREATE TRIP HAZARDS, SUPPORT PER P615/TYP (SIM) FROM GRIT SUCTION PIPE SUPPORT. SEE DETAIL 2/02-M-01-503.
 - FLUIDIZING WATER PIPE PENETRATION LOCATION PER GRIT BASIN MANUFACTURER, ANGLE MEASURED FROM CENTERLINE OF 4" GRIT PUMP SUCTION PIPE.
 - ROUTE AND TERMINATE FLUIDIZING WATER PIPE WITHIN GRIT HOPPER PER GRIT BASIN MANUFACTURER.
 - GRIT BASIN MANUFACTURER TO PROVIDE ELEVATION OF GRIT HOPPER BOTTOM BASED ON GRIT PUMP SUCTION PIPE ELEVATION.

2 DETAIL
SCALE: NTS

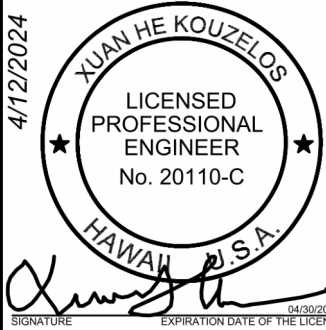
3 DETAIL
SCALE: NTS

GRIT BASIN FLUIDIZING WATER	
GRIT BASIN	FLUIDIZING VALVE
1	02-FV-1216
2	02-FV-1226

GRIT PUMP SEAL WATER			
GRIT PUMP	SOLENOID VALVE	PRESSURE SWITCH	PRESSURE GAUGE
1	02-SV-1211	02-PSL3-1211	02-PI3-1211
2	02-SV-1221	02-PSL3-1221	02-PI3-1221

REV	DATE	BY	DESCRIPTION

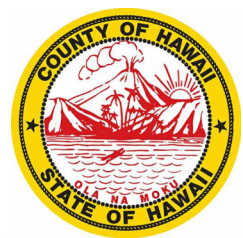
DESIGNED TH	4/12/2024
DRAWN LV	
CHECKED MES	
DATE APR 2024	



CONFORMED DOCUMENT

THIS WORK WAS PREPARED BY ME OR UNDER MY SUPERVISION AND CONSTRUCTION OF THIS PROJECT WILL BE UNDER MY OBSERVATION

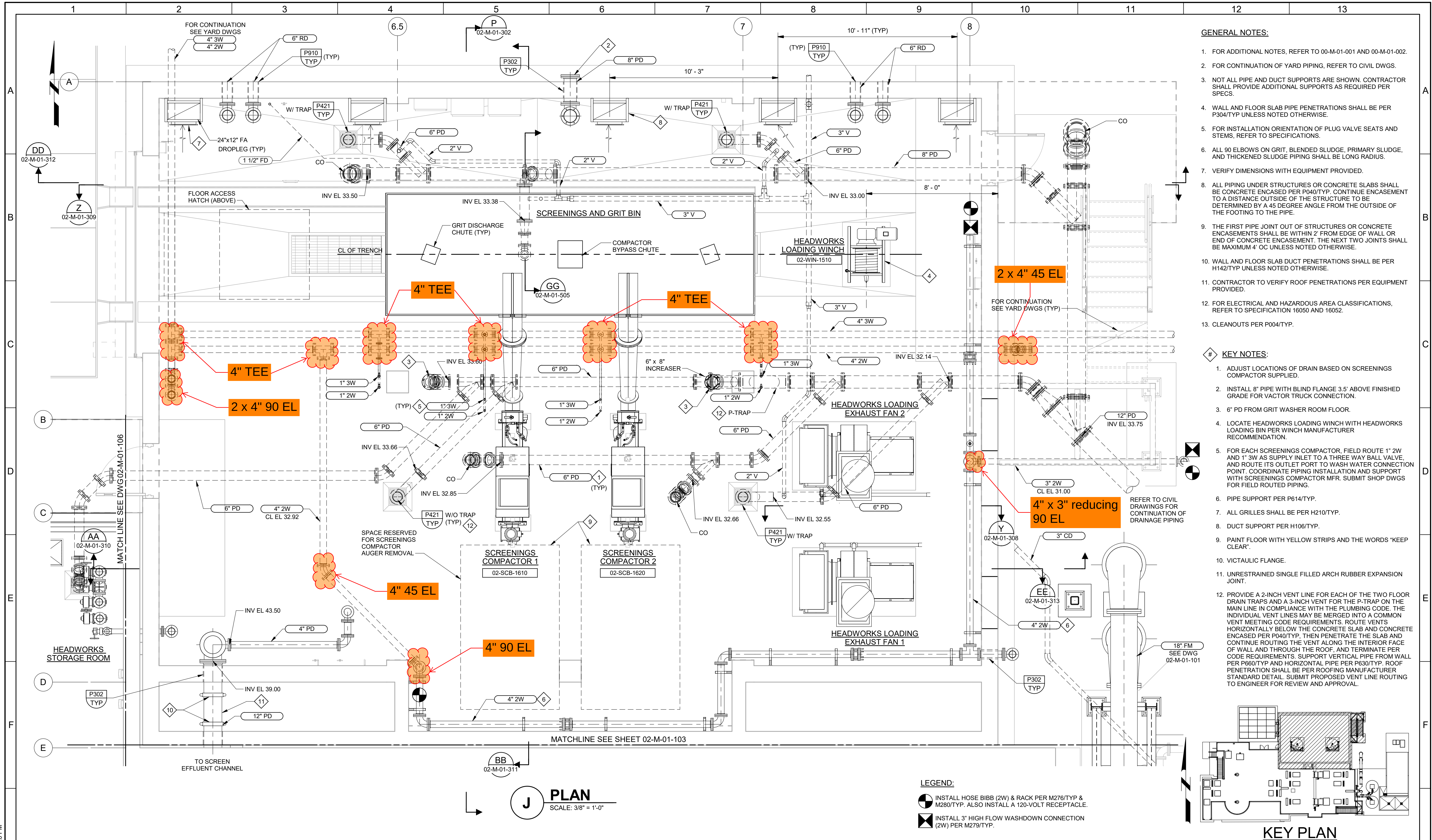
VERIFY SCALES
BAR IS ONE INCH ON ORIGINAL DRAWING
0 1"
IF NOT ONE INCH ON THIS SHEET, ADJUST SCALES ACCORDINGLY



COUNTY OF HAWAII
HILO WWTP REHABILITATION AND REPLACEMENT PROJECT PHASE 1
MECHANICAL
GRIT BASIN DETAILS

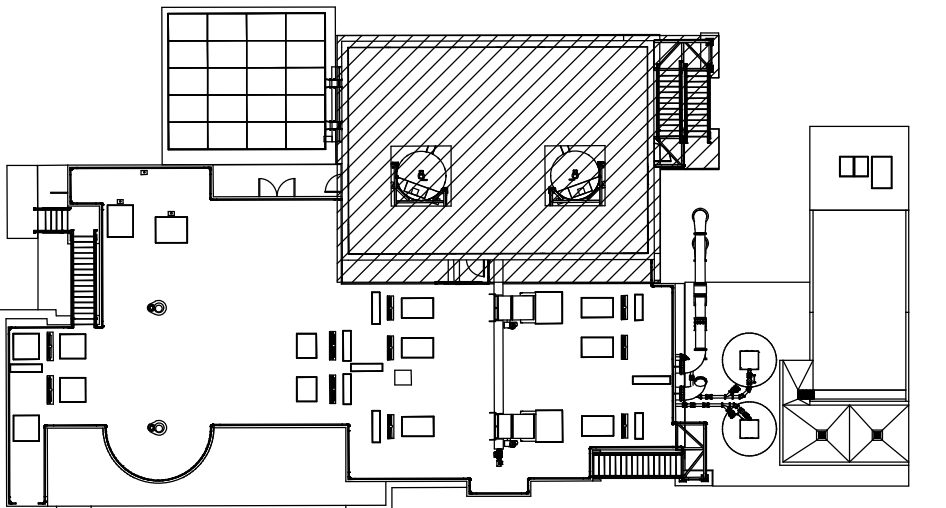
JOB NO.
201400
DRAWING NO.
02-M-01-503
SHEET NO.
394 of 1000

PLOT DATE: 3/29/2024 5:35:35 PM



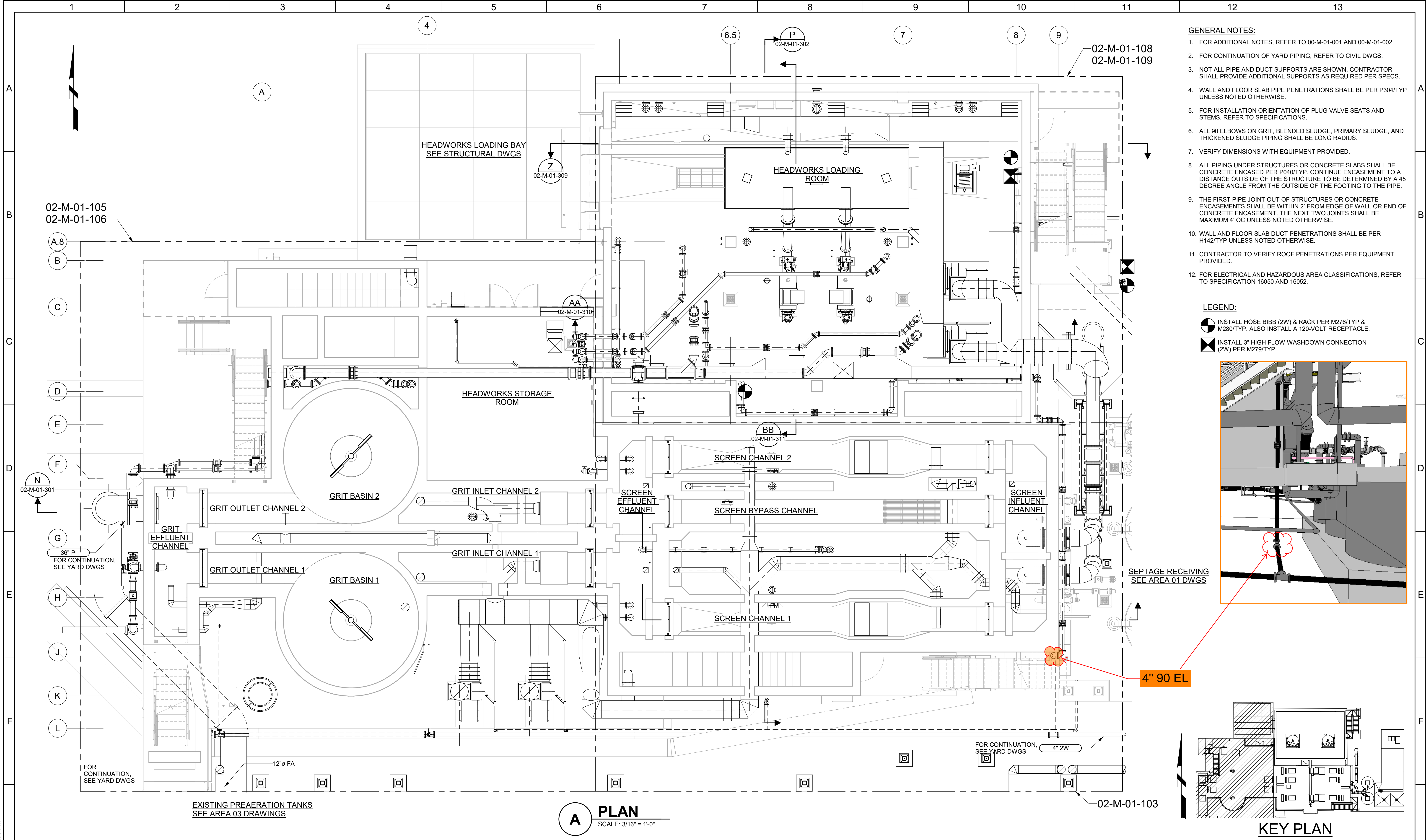
- GENERAL NOTES:**
1. FOR ADDITIONAL NOTES, REFER TO 00-M-01-001 AND 00-M-01-002.
 2. FOR CONTINUATION OF YARD PIPING, REFER TO CIVIL DWGS.
 3. NOT ALL PIPE AND DUCT SUPPORTS ARE SHOWN. CONTRACTOR SHALL PROVIDE ADDITIONAL SUPPORTS AS REQUIRED PER SPECS.
 4. WALL AND FLOOR SLAB PIPE PENETRATIONS SHALL BE PER P304/TYP UNLESS NOTED OTHERWISE.
 5. FOR INSTALLATION ORIENTATION OF PLUG VALVE SEATS AND STEMS, REFER TO SPECIFICATIONS.
 6. ALL 90 ELBOWS ON GRIT, BLENDED SLUDGE, PRIMARY SLUDGE, AND THICKENED SLUDGE PIPING SHALL BE LONG RADIUS.
 7. VERIFY DIMENSIONS WITH EQUIPMENT PROVIDED.
 8. ALL PIPING UNDER STRUCTURES OR CONCRETE SLABS SHALL BE CONCRETE ENCASED PER P040/TYP. CONTINUE ENCASEMENT TO A DISTANCE OUTSIDE OF THE STRUCTURE TO BE DETERMINED BY A 45 DEGREE ANGLE FROM THE OUTSIDE OF THE FOOTING TO THE PIPE.
 9. THE FIRST PIPE JOINT OUT OF STRUCTURES OR CONCRETE ENCASEMENTS SHALL BE WITHIN 2' FROM EDGE OF WALL OR END OF CONCRETE ENCASEMENT. THE NEXT TWO JOINTS SHALL BE MAXIMUM 4' OC UNLESS NOTED OTHERWISE.
 10. WALL AND FLOOR SLAB DUCT PENETRATIONS SHALL BE PER H142/TYP UNLESS NOTED OTHERWISE.
 11. CONTRACTOR TO VERIFY ROOF PENETRATIONS PER EQUIPMENT PROVIDED.
 12. FOR ELECTRICAL AND HAZARDOUS AREA CLASSIFICATIONS, REFER TO SPECIFICATION 16050 AND 16052.
 13. CLEANOUTS PER P004/TYP.

- KEY NOTES:**
1. ADJUST LOCATIONS OF DRAIN BASED ON SCREENINGS COMPACTOR SUPPLIED.
 2. INSTALL 8" PIPE WITH BLIND FLANGE 3.5' ABOVE FINISHED GRADE FOR VACTOR TRUCK CONNECTION.
 3. 6" PD FROM GRIT WASHER ROOM FLOOR.
 4. LOCATE HEADWORKS LOADING WINCH WITH HEADWORKS LOADING BIN PER WINCH MANUFACTURER RECOMMENDATION.
 5. FOR EACH SCREENINGS COMPACTOR, FIELD ROUTE 1" 2W AND 1" 3W AS SUPPLY INLET TO A THREE WAY BALL VALVE, AND ROUTE ITS OUTLET PORT TO WASH WATER CONNECTION POINT. COORDINATE PIPING INSTALLATION AND SUPPORT WITH SCREENINGS COMPACTOR MFR. SUBMIT SHOP DWGS FOR FIELD ROUTED PIPING.
 6. PIPE SUPPORT PER P614/TYP.
 7. ALL GRILLES SHALL BE PER H210/TYP.
 8. DUCT SUPPORT PER H106/TYP.
 9. PAINT FLOOR WITH YELLOW STRIPS AND THE WORDS "KEEP CLEAR".
 10. VICTAULIC FLANGE.
 11. UNRESTRAINED SINGLE FILLED ARCH RUBBER EXPANSION JOINT.
 12. PROVIDE A 2-INCH VENT LINE FOR EACH OF THE TWO FLOOR DRAIN TRAPS AND A 3-INCH VENT FOR THE P-TRAP ON THE MAIN LINE IN COMPLIANCE WITH THE PLUMBING CODE. THE INDIVIDUAL VENT LINES MAY BE MERGED INTO A COMMON VENT MEETING CODE REQUIREMENTS. ROUTE VENTS HORIZONTALLY BELOW THE CONCRETE SLAB AND CONCRETE ENCASED PER P040/TYP, THEN PENETRATE THE SLAB AND CONTINUE ROUTING THE VENT ALONG THE INTERIOR FACE OF WALL AND THROUGH THE ROOF, AND TERMINATE PER CODE REQUIREMENTS. SUPPORT VERTICAL PIPE FROM WALL PER P660/TYP AND HORIZONTAL PIPE PER P630/TYP. ROOF PENETRATION SHALL BE PER ROOFING MANUFACTURER STANDARD DETAIL. SUBMIT PROPOSED VENT LINE ROUTING TO ENGINEER FOR REVIEW AND APPROVAL.



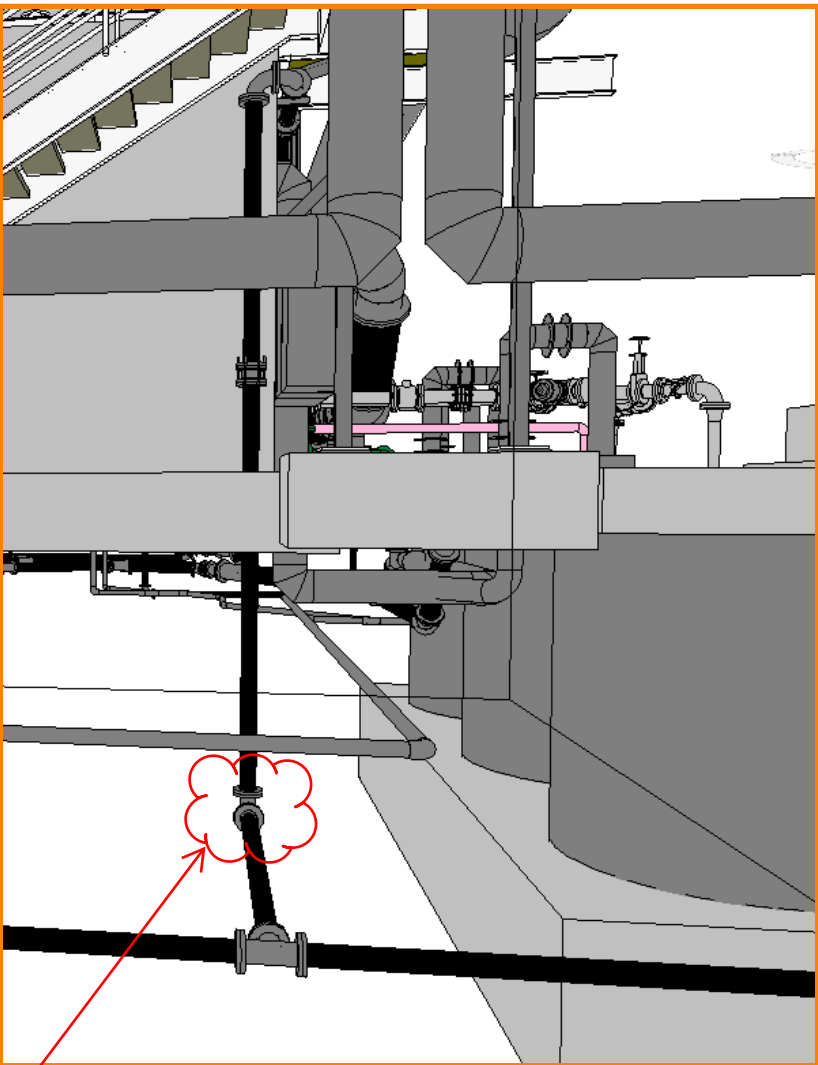
DESIGNED XH/K/TH				DRAWN LV				CHECKED MES				DATE APR 2024			
CONFORMED DOCUMENT				VERIFY SCALES BAR IS ONE INCH ON ORIGINAL DRAWING 0 1"				COUNTY OF HAWAII				HILO WWTP REHABILITATION AND REPLACEMENT PROJECT PHASE 1			
THIS WORK WAS PREPARED BY ME OR UNDER MY SUPERVISION AND CONSTRUCTION OF THIS PROJECT WILL BE UNDER MY OBSERVATION				carollo				MECHANICAL				SCREENINGS AND GRIT HANDLING BUILDING			
SCALE: 3/8" = 1'-0"				PLAN				KEY PLAN				02-M-01-108			
REV				DATE				BY				DESCRIPTION			
1				2				3				4			
5				6				7				8			
9				10				11				12			
13				14				15				16			

FILE NAME: R:\d1361566\201400_02M0801.rvt



- GENERAL NOTES:**
- FOR ADDITIONAL NOTES, REFER TO 00-M-01-001 AND 00-M-01-002.
 - FOR CONTINUATION OF YARD PIPING, REFER TO CIVIL DWGS.
 - NOT ALL PIPE AND DUCT SUPPORTS ARE SHOWN. CONTRACTOR SHALL PROVIDE ADDITIONAL SUPPORTS AS REQUIRED PER SPECS.
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 - CONTRACTOR TO VERIFY ROOF PENETRATIONS PER EQUIPMENT PROVIDED.
 - FOR ELECTRICAL AND HAZARDOUS AREA CLASSIFICATIONS, REFER TO SPECIFICATION 16050 AND 16052.

- LEGEND:**
- INSTALL HOSE BIBB (2W) & RACK PER M276/TYP & M280/TYP. ALSO INSTALL A 120-VOLT RECEPTACLE.
 - INSTALL 3" HIGH FLOW WASHDOWN CONNECTION (2W) PER M279/TYP.



DESIGNED XHk/TH DRAWN LV CHECKED MES DATE APR 2024				4/12/2024 XUAN HE KOUZELOS LICENSED PROFESSIONAL ENGINEER No. 20110-C HAWAII, U.S.A.		CONFORMED DOCUMENT THIS WORK WAS PREPARED BY ME OR UNDER MY SUPERVISION AND CONSTRUCTION OF THIS PROJECT WILL BE UNDER MY OBSERVATION		VERIFY SCALES BAR IS ONE INCH ON ORIGINAL DRAWING 0 1" IF NOT ONE INCH ON THIS SHEET, ADJUST SCALES ACCORDINGLY		COUNTY OF HAWAII HILO WWTP REHABILITATION AND REPLACEMENT PROJECT PHASE 1 MECHANICAL HEADWORKS OVERALL CHANNEL PLAN		JOB NO. 201400 DRAWING NO. 02-M-01-101 SHEET NO. 369 of 1000																			
REV				DATE		BY		DESCRIPTION		1		2		3		4		5		6		7		10		11		12		13	

C110 DUCTILE IRON FULL BODY FITTINGS

SAMPLE SPECIFICATIONS

(Current ANSI/AWWA revisions apply)

Mechanical joint watermain fittings with accessories 2" through 48" shall be manufactured from ductile iron in accordance with and meet all applicable terms and provisions of standards ANSI/AWWA C110/A21.10 and ANSI/AWWA C111/A21.11. Ductile iron mechanical joint fittings 2" through 24" shall be rated for 350 psi working pressure. All ductile iron mechanical joint fittings 30" through 48" shall be rated for 250 psi working pressure. Flanged ductile iron fittings in 24" (610 mm) and smaller sizes may be rated for 350 psi (2,413 kPa) with the use of special (annular ring or comparable) gaskets.

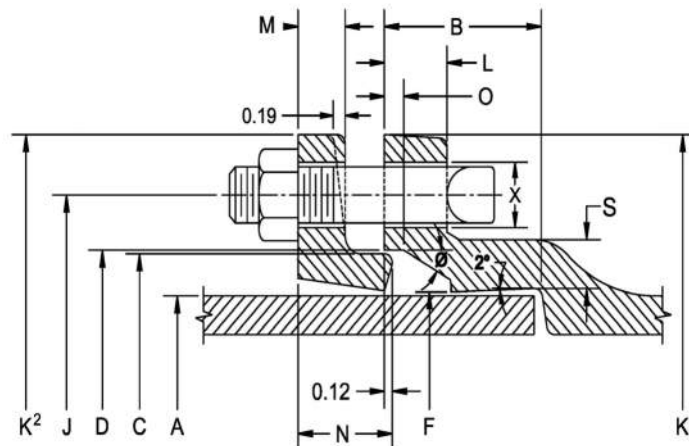
NOTE: EXCEPTIONS: Mechanical joint fittings with flanged branches and 14" and larger caps and plugs are rated for water pressure of 250 PSI.

NOTE: Installation per AWWA C600 and AWWA C651, current revision.

NOTE: Fittings are cement lined and seal coated in accordance with ANSI/AWWA C104/A21.4. Fittings are also available double cement lined, bare or epoxy coated. Coated and lined fittings meet requirements of NSF-61, NSF-372 & Annex G.

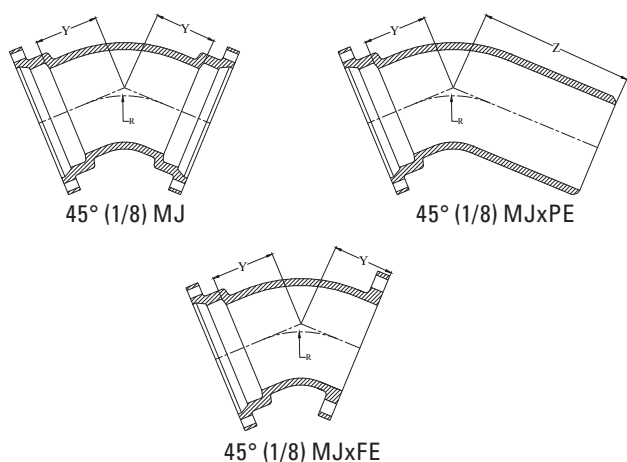
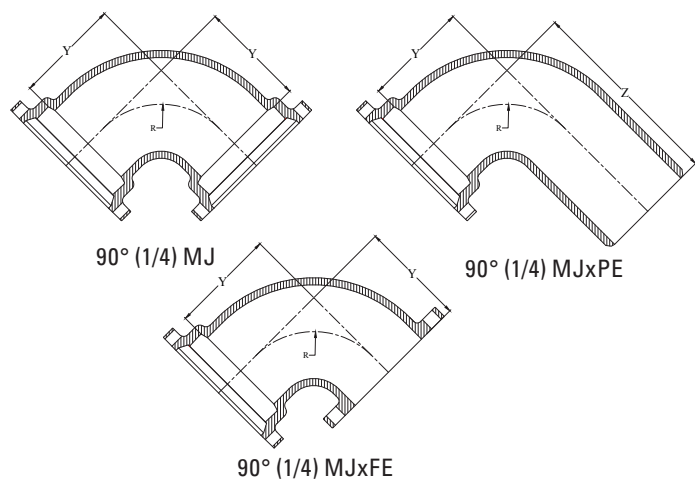
Nominal Joint Dimensions in Inches															
Size	A	B	C	D	F	ø	X	J	K1	K2	L	M	N	O	S
*2	2.50	2.50	3.39	3.50	2.61	28°	3/4	4.75	6.25	6.25	0.75	0.62	1.12	0.31	0.44
3	3.96	2.50	4.84	4.94	4.06	28°	3/4	6.19	7.69	7.69	0.94	0.62	1.37	0.31	0.52
4	4.80	2.50	5.92	6.02	4.90	28°	7/8	7.50	9.12	9.12	1.00	0.75	1.50	0.31	0.65
6	6.90	2.50	8.02	8.12	7.00	28°	7/8	9.50	11.12	11.12	1.06	0.88	1.63	0.31	0.70
8	9.05	2.50	10.17	10.27	9.15	28°	7/8	11.75	13.37	13.37	1.12	1.00	1.75	0.31	0.75
10	11.10	2.50	12.22	12.34	11.20	28°	7/8	14.00	15.69	15.62	1.19	1.00	1.75	0.31	0.80
12	13.20	2.50	14.32	14.44	13.30	28°	7/8	16.25	17.94	17.88	1.25	1.00	1.75	0.31	0.85
14	15.30	3.50	16.40	16.54	15.44	28°	7/8	18.75	20.31	20.25	1.31	1.25	2.00	0.31	0.89
16	17.40	3.50	18.50	18.64	17.54	28°	7/8	21.00	22.56	22.50	1.38	1.31	2.06	0.31	0.97
18	19.50	3.50	20.60	20.74	19.64	28°	7/8	23.25	24.83	24.75	1.44	1.38	2.13	0.31	1.05
20	21.60	3.50	22.70	22.84	21.74	28°	7/8	25.50	27.08	27.00	1.50	1.44	2.19	0.31	1.12
24	25.80	3.50	26.90	27.04	25.94	28°	7/8	30.00	31.58	31.50	1.62	1.56	2.31	0.31	1.22
30	32.00	4.00	33.29	33.46	32.17	20°	1 1/8	36.88	39.12	39.12	1.81	2.00	2.75	0.38	1.50
36	38.30	4.00	39.59	39.76	38.47	20°	1 1/8	43.75	46.00	46.00	2.00	2.00	2.75	0.38	1.80
42	44.50	4.00	45.79	45.96	44.67	20°	1 3/8	50.62	53.12	53.12	2.00	2.00	2.75	0.38	1.95
48	50.80	4.00	52.09	52.26	50.97	20°	1 3/8	57.50	60.00	60.00	2.00	2.00	2.75	0.38	2.20

*Not included in AWWA C110.

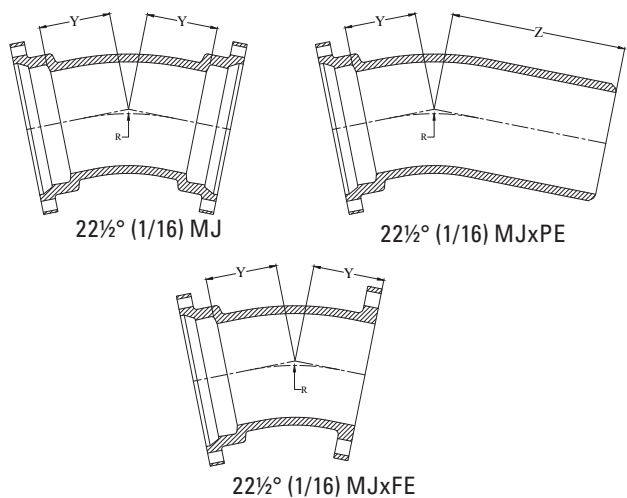


ANSI/AWWA C110 Mechanical Joint Fittings

C110 DUCTILE IRON FULL BODY FITTINGS



*Not included
in ASSA C110.



90° (1/4) BENDS

Domestic							Import	
Size	R	Y	Z	Weight			Weight	
				MJxMJ	MJxPE	MJxFE	MJ	MJxPE
*2	2.30	3.30	—	16	—	—	—	—
3	4.00	5.50	13.50	26	36	—	35	35
4	4.50	6.50	14.50	56	53	47	55	50
6	6.00	8.00	16.00	88	80	75	88	97
8	7.00	9.00	17.00	123	119	118	136	153
10	9.00	11.00	19.00	182	181	170	190	190
12	10.00	12.00	20.00	280	252	246	255	255
14	11.50	14.00	22.00	380	—	—	400	—
16	12.50	15.00	23.00	552	—	465	480	410
18	14.00	16.50	24.50	625	600	591	641	577
20	15.50	18.00	26.00	862	775	—	725	650
24	18.50	22.00	30.00	1423	1301	1150	1020	985
30	21.50	25.00	33.00	1942	1920	—	1843	1585
36	24.50	28.00	36.00	2629	2310	—	2513	2310
42	27.50	31.00	—	3410	—	—	3410	—
48	30.50	34.00	—	4595	—	—	4595	—

45° (1/8) BENDS

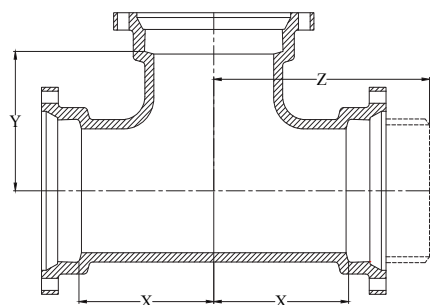
Domestic							Import	
Size	R	Y	Z	Weight			Weight	
				MJ	MJxFE	MJxPE	MJ	MJxPE
*2	1.96	1.80	—	12	—	—	—	—
3	3.62	3.00	11.00	30	—	—	30	30
4	4.81	4.00	12.00	53	48	45	49	48
6	7.25	5.00	13.00	77	60	69	77	81
8	8.44	5.50	13.50	110	107	111	117	123
10	10.88	6.50	14.50	156	148	167	155	168
12	13.25	7.50	15.50	214	215	196	223	215
14	12.06	7.50	15.50	300	—	—	270	—
16	13.25	8.00	16.00	391	—	349	335	320
18	14.50	8.50	16.50	527	416	455	467	395
20	16.88	9.50	17.50	631	543	537	527	500
24	18.12	11.00	19.00	880	1099	825	754	715
30	27.75	15.00	23.00	1898	—	1510	1451	1275
36	35.00	18.00	26.00	2372	—	1930	2176	1930
42	42.25	21.00	—	3020	—	—	2955	—
48	49.50	24.00	—	4170	—	—	4080	—

22 1/2° (1/16) BENDS

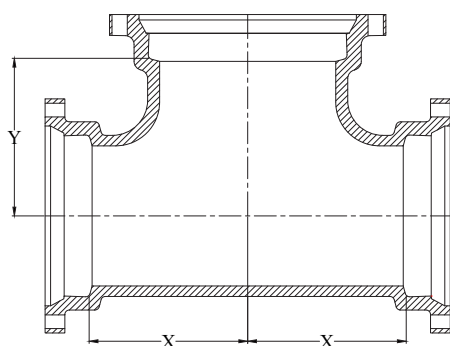
Domestic							Import	
Size	R	Y	Z	Weight			Weight	
				MJ	MJxFE*	MJxPE	MJ	MJxPE
3	7.56	3.00	11.00	30	—	—	30	—
4	10.06	4.00	12.00	52	—	—	51	45
6	15.06	5.00	13.00	77	71	70	75	70
8	17.62	5.50	13.50	110	107	163	108	108
10	22.62	6.50	14.50	156	155	163	159	160
12	27.62	7.50	15.50	214	215	212	199	220
14	25.12	7.50	15.50	300	—	—	275	—
16	27.62	8.00	16.00	391	344	334	318	325
18	30.19	8.50	16.50	527	422	423	430	405
20	35.19	9.50	17.50	631	—	575	545	505
24	37.69	11.00	19.00	880	800	930	758	725
30	57.81	15.00	23.00	1898	—	1540	1400	1400
36	72.88	18.00	26.00	2372	—	1970	2121	1970
42	88.00	21.00	—	3020	—	—	3020	—
48	103.06	24.00	—	4170	—	—	4170	—

Tyler Union does not recommend the use of wedge action restraints on plain end fittings

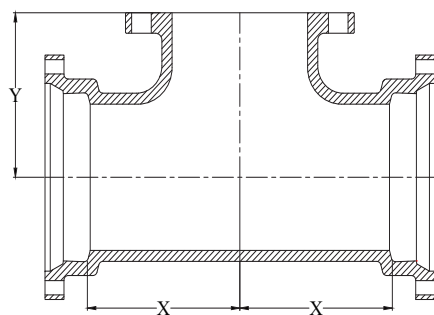
C110 DUCTILE IRON FULL BODY FITTINGS



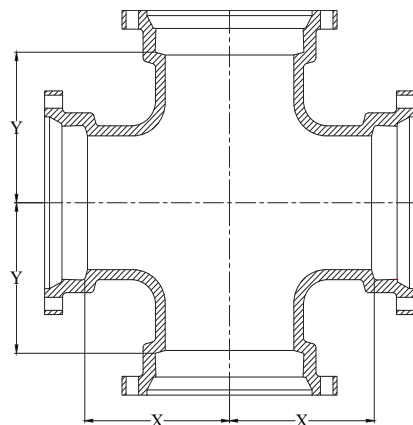
MJ Tees and MJxPExMJ Tees



MJ Bullhead Tees



MJxFE Tees



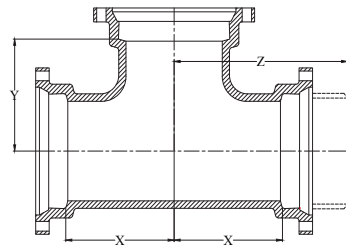
Cross

TEES AND CROSSES

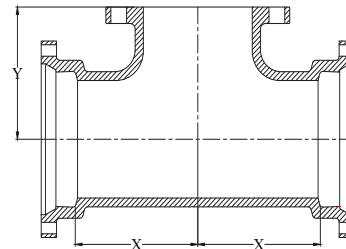
Size Run	Branch	X	Y	Z	MJ	Weight		
						MJxPExMJ	MJxMJxFE	Cross
2	2	3.25	3.25	—	21	—	—	—
3	2	3.25	3.25	—	45	—	—	—
3	3	5.50	5.50	13.50	58	—	—	—
4	2	4.80	4.80	14.50	68	—	49	—
4	3	6.50	6.50	14.50	77	—	—	—
4	4	6.50	6.50	14.50	78	75	76	—
4	6	8.00	8.00	—	112	—	—	—
6	2	8.00	8.00	—	78	—	—	—
6	3	8.00	8.00	16.00	98	—	—	—
6	4	8.00	8.00	16.00	110	—	109	—
6	6	8.00	8.00	16.00	119	120	141	160
6	8	9.00	9.00	—	158	—	—	—
8	3	9.00	9.00	17.00	155	—	—	—
8	4	9.00	9.00	17.00	157	—	150	185
8	6	9.00	9.00	17.00	175	170	182	205
8	8	9.00	9.00	17.00	199	180	194	255
10	4	11.00	11.00	19.00	—	—	229	—
10	6	11.00	11.00	19.00	258	—	264	285
10	8	11.00	11.00	19.00	268	—	245	310
10	10	11.00	11.00	19.00	300	250	—	380
12	4	12.00	12.00	20.00	318	315	323	—
12	6	12.00	12.00	20.00	325	325	335	360
12	8	12.00	12.00	20.00	335	335	372	371
12	10	12.00	12.00	20.00	392	390	—	486
12	12	12.00	12.00	20.00	396	396	476	—
14	12	14.00	14.00	22.00	540	560	—	—
14	14	14.00	14.00	22.00	585	570	—	779
16	4	15.00	15.00	23.00	580	580	575	—
16	6	15.00	15.00	23.00	615	590	605	650
16	8	15.00	15.00	23.00	625	605	615	675
16	10	15.00	15.00	23.00	645	620	—	—
16	12	15.00	15.00	23.00	627	640	651	—
16	16	15.00	15.00	23.00	740	720	730	895
18	6	13.00	15.50	—	710	—	707	—
18	8	13.00	15.50	—	659	—	675	775
18	12	13.00	15.50	—	749	—	733	860
18	18	16.50	16.50	—	945	—	953	1140
20	6	14.00	17.00	—	849	—	—	—
20	8	14.00	17.00	—	892	—	859	951
20	12	14.00	17.00	—	896	—	—	977
20	16	18.00	18.00	—	1095	—	—	1245
20	20	18.00	18.00	—	1258	—	1168	1440
24	6	15.00	19.00	—	1233	—	1228	—
24	8	15.00	19.00	—	1234	—	1242	1244
24	12	15.00	19.00	—	1256	—	1165	1326
24	14	15.00	19.00	—	1220	—	—	—
24	16	15.00	19.00	—	1245	—	—	1479
24	18	22.00	22.00	—	1735	—	—	—
24	20	22.00	22.00	—	1720	—	1795	1965
24	24	22.00	22.00	—	1947	—	—	2192
30	6	18.00	23.00	—	2050	—	—	2085
30	8	18.00	23.00	—	2060	—	—	—
30	10	18.00	23.00	—	2075	—	—	—
30	12	18.00	23.00	—	2090	—	—	2165
30	16	18.00	23.00	—	2145	—	—	—
30	18	18.00	23.00	—	2170	—	—	—
30	20	18.00	23.00	—	2205	—	—	—
30	24	25.00	25.00	—	2880	—	3080	3180
30	30	25.00	25.00	—	2275	—	2430	3640

Tyler Union does not recommend the use of wedge action restraints on plain end fittings

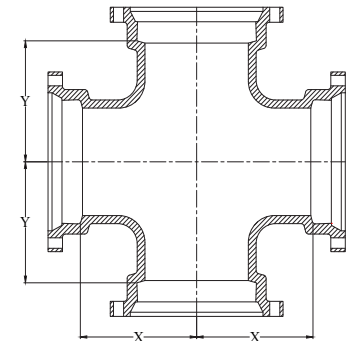
C110 DUCTILE IRON FULL BODY FITTINGS



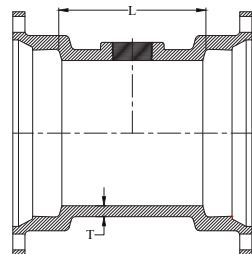
MJ Tees AND MJxPExMJ Tees



MJxFE Tees



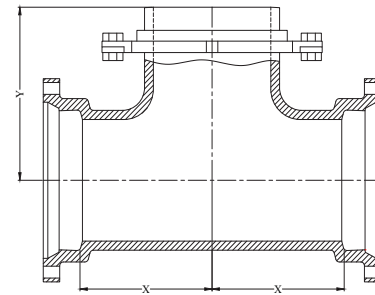
Cross



MJ Tapped Tees (2" TAP)

TEES AND CROSSES (Continued)

Size Run	Branch	X	Y	Z	MJ	Weight		
						MJXPEXMJ	MJxMJxFE	Cross
36	6	20.00	26.00	—	2439	—	—	—
36	8	20.00	26.00	—	2444	—	—	—
36	10	20.00	26.00	—	2535	—	2550	—
36	12	20.00	26.00	—	2541	—	—	—
36	14	20.00	26.00	—	2570	—	2450	—
36	16	20.00	26.00	—	2585	—	—	4370
36	18	20.00	26.00	—	2610	—	—	3420
36	20	20.00	26.00	—	2635	—	2660	3455
36	24	20.00	26.00	—	2792	—	—	3495
36	30	28.00	28.00	—	3545	—	—	3535
36	36	28.00	28.00	—	3450	—	—	3590
42	12	23.00	30.00	—	3555	—	—	3690
42	14	23.00	30.00	—	3575	—	—	4815
42	16	23.00	30.00	—	3595	—	—	6430
42	18	23.00	30.00	—	3615	—	—	6920
42	20	23.00	30.00	—	3640	—	—	4665
42	24	23.00	30.00	—	3690	—	—	4695
42	30	31.00	31.00	—	4650	—	—	4735
42	36	31.00	31.00	—	4880	—	—	4775
42	42	31.00	31.00	—	6320	—	—	4825
48	12	26.00	34.00	—	4870	—	—	—
48	14	26.00	34.00	—	4885	—	—	—
48	16	26.00	34.00	—	4905	—	—	—
48	18	26.00	34.00	—	4925	—	—	—
48	20	26.00	34.00	—	4950	—	—	—
48	24	26.00	34.00	—	4995	—	—	4920
48	30	26.00	34.00	—	5140	—	—	5210
48	36	34.00	34.00	—	6280	—	—	6500
48	42	34.00	34.00	—	8130	—	—	8530
48	48	34.00	34.00	—	8420	—	—	9095



MJxMJxSwivel Tees

MJ TAPPED TEE (2" TAP)

Domestic				Import			
size	L	Max. Tap	Weight	T	L	Max. Tap	Weight
3	8.00	2	40	0.33	6.00	2	18
4	8.00	2	51	0.34	6.00	2	24
6	8.00	2	73	0.36	6.00	2	42
8	8.00	2	104	0.38	6.00	2	52
10	8.00	2	130	0.40	6.00	2	64
12	8.00	2	180	0.42	6.00	2	81

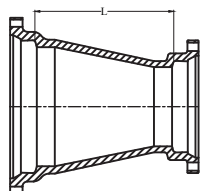
Tyler Union does not recommend the use of wedge action restraints on plain end fittings

MJ X MJ X SWIVEL TEE

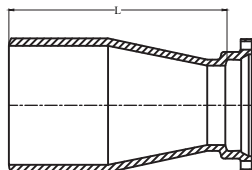
Size	X	Y	Weight
6	8.00	10.50	150
8x6	9.00	11.50	199
8	9.00	11.50	210
10x6	11.00	13.50	267
12x6	12.00	14.50	346
16x6	15.00	17.50	619
16x8	15.00	17.50	649
30x6	18.00	24.50	2070

All weights shown include the swivel gland.

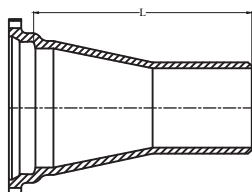
C110 DUCTILE IRON FULL BODY FITTINGS



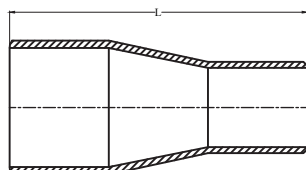
MJ x MJ



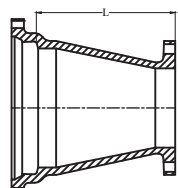
PE x MJ-SEB



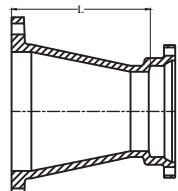
MJ-LEB x PE



PE x PE



MJ x FE

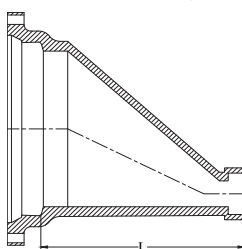


FE x MJ

REDUCERS

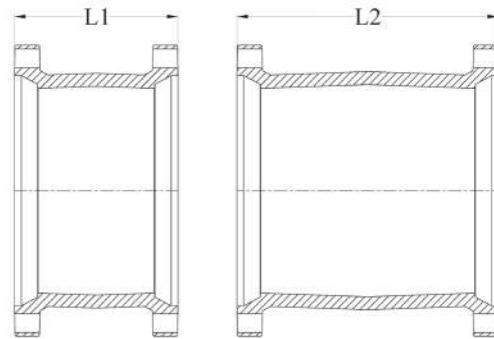
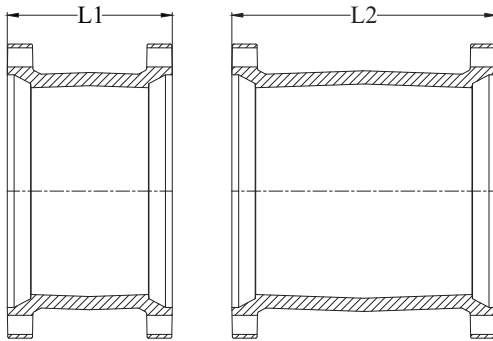
Size	Laying Length (L)						Weight					
	MJ	SEB	MJ-LEB	PE x PE	FE x MJ	MJ x FE	MJ	MJ-SEB	MJ-LEB	PE x PE	FE x MJ	MJ x FE
* 3x2	6	14	14	—	—	—	24	24	24	—	—	—
* 4x2	7	15	15	—	—	—	31	30	31	—	—	—
4x3	7	15	15	23	7	7	37	38	37	34	34	35
* 6x2	9	17	17	—	—	—	46	43	47	—	—	—
6x3	9	17	17	25	—	9	55	50	55	—	—	50
6x4	9	17	17	25	9	9	56	60	59	57	53	62
8x3	11	19	19	27	—	—	84	77	70	—	—	—
8x4	11	19	19	27	11	11	84	82	84	—	73	75
8x6	11	19	19	27	11	11	94	90	93	96	84	80
10x6	12	20	20	28	12	12	115	116	117	—	100	105
10x8	12	20	20	28	12	12	142	135	130	135	130	130
12x4	14	22	22	30	—	—	139	131	—	—	—	—
12x6	14	22	22	30	14	12	148	150	153	—	145	130
12x8	14	22	22	30	14	12	173	168	165	168	170	175
12x10	14	22	22	30	14	12	194	190	178	185	188	190
14x6	16	24	24	32	—	16	—	—	—	—	—	195
14x8	16	24	24	32	—	16	—	—	—	—	—	215
14x12	16	24	24	32	—	16	—	—	—	—	—	270
16x6	18	26	26	34	—	—	250	—	—	—	—	—
16x8	18	26	26	34	—	—	288	248	—	—	—	—
16x10	18	26	26	34	—	—	300	—	—	—	—	—
16x12	18	26	26	34	18	18	330	304	325	—	305	325
16x14	18	26	26	34	—	—	370	—	—	—	—	—
18x8	19	27	27	35	—	19	320	—	—	—	—	300
18x10	19	27	27	35	—	—	388	—	—	—	—	—
18x12	19	27	27	35	—	19	380	355	—	—	—	405
18x14	19	27	27	35	—	—	450	—	—	—	—	—
18x16	19	27	27	35	—	19	476	—	—	—	—	445
20x10	20	28	28	36	—	—	410	—	—	—	—	—
20x12	20	28	28	36	—	—	515	420	—	—	—	—
20x16	20	28	28	36	—	20	578	525	510	—	—	510
20x18	20	28	28	36	—	—	575	—	—	—	—	—
24x12	24	32	32	40	—	24	610	570	—	—	—	455
24x16	24	32	32	40	—	—	705	665	753	—	—	—
24x18	24	32	32	40	—	—	789	720	—	—	—	—
24x20	24	32	32	40	—	—	815	775	804	—	—	—
*30x16	30	38	38	46	—	—	1150	1040	—	1015	—	—
30x18	30	38	38	46	—	—	1160	1050	—	1025	—	—
30x20	30	38	38	46	—	—	1225	1120	—	1090	—	—
30x24	30	38	38	46	—	—	1360	1255	1320	1215	—	—
36x20	36	44	44	52	—	—	1495	—	1466	—	—	—
36x24	36	44	44	52	—	—	1580	—	1535	1389	—	—
36x30	36	44	44	52	—	—	1919	1721	—	1585	—	—
42x24	42	50	50	58	—	—	2060	—	—	—	—	—
42x30	42	50	50	58	—	—	2370	—	—	—	—	—
42x36	42	50	50	58	—	—	2695	—	—	—	—	—
48x30	48	56	56	64	—	—	3005	—	—	—	—	—
48x36	48	56	56	64	—	—	3370	—	—	—	—	—
48x42	48	56	56	64	—	—	3750	—	—	—	—	—

Tyler Union does not recommend the use of wedge action restraints on plain end fittings



MJxIPT Eccentric Reducer		
Size	L	Weight
6x2	13	51
8x2	15	71

C110 DUCTILE IRON FULL BODY FITTINGS



SOLID SLEEVES					
Size	Pipe O.D.	L1	Weight	L2	Weight
*2	2.50	8.00	13	12.00	18
3	4.00	7.50	25	12.00	36
4	4.80	7.50	35	12.00	47
6	6.90	7.50	45	12.00	65
8	9.10	7.50	65	12.00	90
10	11.10	7.50	85	12.00	108
12	13.20	7.50	120	12.00	136
16	17.40	9.50	206	15.00	289
18	19.50	9.50	246	15.00	282
20	21.60	9.50	275	15.00	336
24	25.80	9.50	360	15.00	459
30	32.00	15.00	745	24.00	1220
36	38.30	15.00	1047	24.00	1502
42	44.50	15.00	1312	24.00	1550
48	50.80	15.00	1585	24.00	1940

*DUO PURPOSE SLEEVE					
Size	Pipe O.D.	L1	Weight	L2	Weight
4	4.80/5.00	7.50	33	12.00	44
6	6.90/7.10	7.50	46	12.00	67
8	9.05/9.30	7.50	65	12.00	88
10	11.10/11.40	—	—	12.00	111
†12	13.20/13.50	—	—	12.00	221
†16	17.40/17.80	—	—	15.00	385

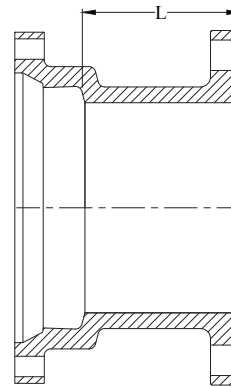
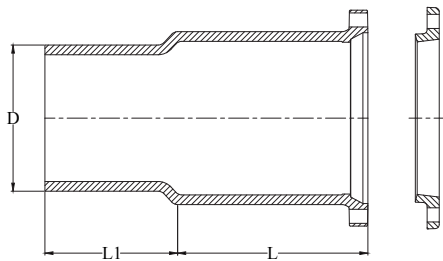
All sizes use MJ DUO purpose Gland.

* Not included in AWWA C110.

† 12" and 16" are sold assembled.

NOTE: Sizes 4"–10" Use standard MJ Gasket;

12" and 16" require special gaskets.

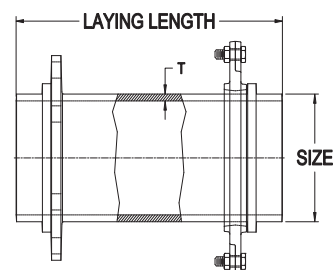
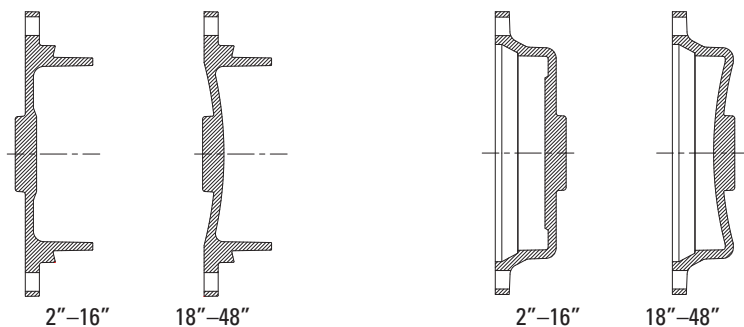


* MJ x PE DUAL-PURPOSE CUTTING-IN SLEEVE						
Size	For use on pipe OD	L	L1	D	Gland only	Gland and sleeve
4	4.80–5.00	12.00	8.00	4.80	6.0	72
6	6.90–7.10	12.00	8.00	6.90	10.0	94
8	9.05–9.30	12.00	8.00	9.05	16.0	122
10	11.10–11.40	12.00	8.00	11.10	25.0	175
12	13.20–13.50	12.00	8.00	13.20	30.0	235

Tyler Union does not recommend the use of wedge action restraints on plain end fittings

ADAPTERS MJxFE		
Size	L	Weight
3	8.00	30
4	8.00	42
6	8.00	57
8	8.00	88
10	8.00	120
12	8.00	150
16	8.00	257
18	8.00	304
20	8.00	372
24	8.00	488
30	10.00	682
36	10.00	1070

C110 DUCTILE IRON FULL BODY FITTINGS



PLUGS	
Size	Weight
*2	5
3	9
4	13
6	15
8	45
10	66
12	79
14	120
16	147
†18	192
†20	220
†24	338
†30	660
†36	838
†42	1180
†48	1455

† Dished

* Not included in AWWA C110.

Note: Optional 2" tap in the center.

Testing Purposes

CAPS	
Size	Weight
*2	6
3	10
4	18
6	34
8	46
10	58
12	86
16	178
†18	215
†20	250
†24	370
†30	680
†36	850
†42	1180
†48	1595

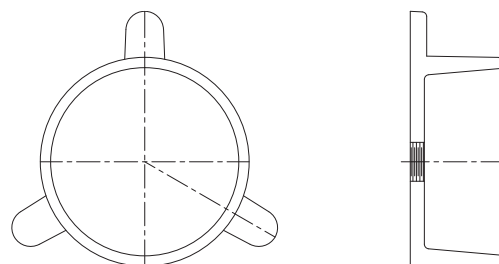
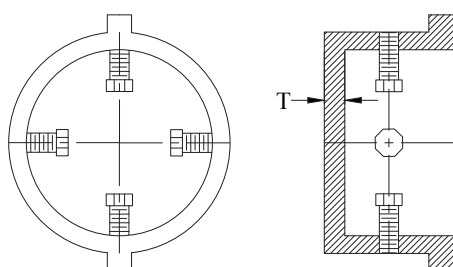
† Dished

* Not included in AWWA C110.

Note: Optional 2" tap in the center.

SWIVEL x SOLID ADAPTER		
Size X Laying Length	Pipe O.D.	L
4 X 13	0.52	42
6 X 12	0.55	61
6 X 18	0.55	89
6 X 24	0.55	108
6 X 36	0.55	156
8 X 13	0.60	97
12 X 13	0.75	164

* Weights with Gland.



TYTON® PLUG** SOLID OR TAPPED			
Size	Tap	T	Weight**
4	2	0.60	18
6	2	0.65	25
8	2	0.70	46
10	2	0.75	70
12	2	0.75	95

*Weights do not include accessories.

** Not included in AWWA C110.

TYTON is a registered trademark of U.S Pipe and Foundry company.

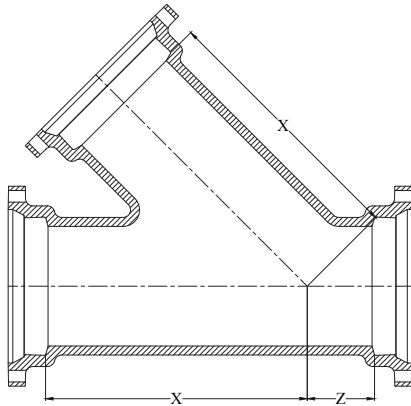
PUSH-IN PLUG WITH EARS		
Size	Tap	Weight
14	2	101
16	2	137
18	2	177
†20	2	239
†24	2	311

† Dished – Not flat as shown.

NOTE: To be used with all push in pipe and fittings.

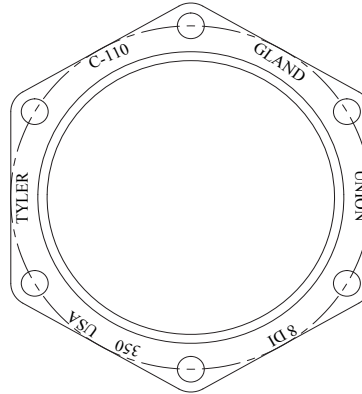
NOTE: Blocking still required ears for assembly only.

C110 DUCTILE IRON FULL BODY FITTINGS



WYES / LATERALS				
Size		Dimensions		Weight
Run	Branch	X	Z	
3	3	10.00	3.00	60
4	4	12.00	3.00	90
6	4	14.50	3.50	130
6	6	14.50	3.50	145
8	4	17.50	4.50	190
8	6	17.50	4.50	205
8	8	17.50	4.50	230
10	6	20.50	5.00	330
10	8	20.50	5.00	310
10	10	20.50	5.50	435
12	8	24.50	5.50	505
12	12	24.50	5.50	490
14	6	27.00	6.00	626
16	16	30.00	6.50	1079
18	8	32.00	7.00	1073
18	10	32.00	7.00	975
18	12	32.00	7.00	1015
18	16	32.00	7.00	1135
18	18	32.00	7.00	1130
20	10	35.00	8.00	1220
20	12	35.00	8.00	1260
20	16	35.00	8.00	1375
20	20	35.00	8.00	1525
24	24	40.50	9.00	2372
30	30	49.00	10.00	3670
36	24	54.00	15.00	5390
36	36	60.00	19.50	6335
42	24	60.00	12.00	6810
42	30	63.00	12.00	7210
42	36	66.00	12.00	8355
42	42	71.00	15.00	9900
48	48	77.00	16.00	13150

* Not included in AWWA C-110.



MJ GLANDS	
Size	Weight
2	3
3	4
4	6
6	10
8	16
10	19
12	26
14	34
16	54
18	52
20	73
24	91
30	90
36	127
42	279
48	341

MJ ACCESSORY KITS AND WEIGHTS						
size	No.	Bolt Size	Bolt Length	Bolt Torque ft/lbs	Wt. of Gland, Bolts and Gasket	Pipe Barrel O.D.
*2	2	5/8	3	45-60	5	2.50
3	4	5/8	3	45-60	7	3.96
4	4	3/4	3 1/2	75-90	10	4.80
6	6	3/4	3 1/2	75-90	16	6.90
8	6	3/4	4	75-90	25	9.05
10	8	3/4	4	75-90	30	11.10
12	8	3/4	4	75-90	40	13.20
14	10	3/4	4 1/2	75-90	45	15.30
16	12	3/4	4 1/2	75-90	55	17.40
18	12	3/4	4 1/2	75-90	65	19.50
20	14	3/4	4 1/2	75-90	85	21.60
24	16	3/4	5	75-90	105	25.80
30	20	1	6	100-120	220	32.00
36	24	1	6	100-120	301	38.30
42	28	1 1/4	6 1/2	120-150	389	44.50
48	32	1 1/4	6 1/2	120-150	477	50.80



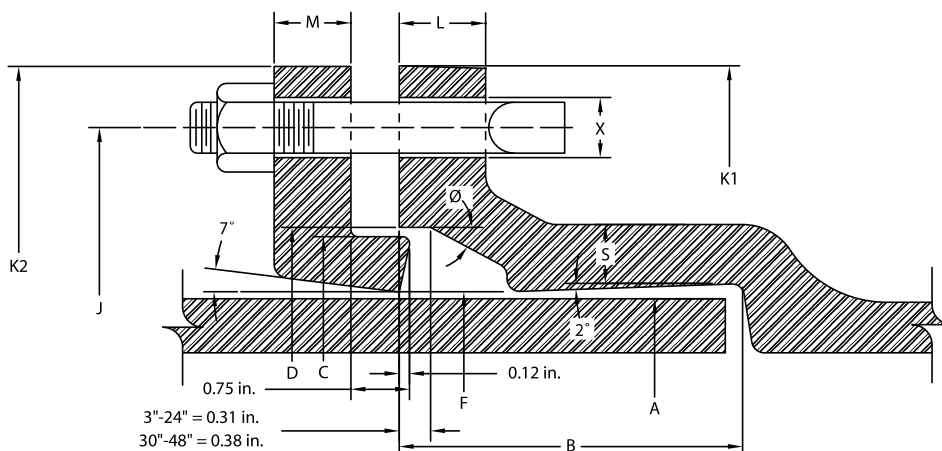
Full Body MJ Fittings

ANSI/AWWA C110/A21.10

2" - 54" DUCTILE IRON MECHANICAL JOINT FULL BODY FITTINGS

GENERAL SPECIFICATIONS

MATERIAL:	Ductile iron per ASTM A536
PRESSURE:	350 PSI rating for 2" - 24" sizes, 250 PSI rating for 30" - 48" and 150 PSI for 54" or as noted
TESTING:	In accordance with ANSI/AWWA C110/A21.10
LAYING LENGTH:	In accordance with ANSI/AWWA C110/A21.10 (fittings not listed in ANSI/AWWA have dimensions per Star design as noted in the catalog)
WEIGHTS:	Are in pounds, unless noted otherwise and do not include accessories, cement lining and coating
FLANGES:	Flanged ends on fittings match ANSI/AWW C110/A21.10 and ANSI B16.1 class 125 flanges
CEMENT LINING:	In accordance with ANSI/AWWA C104/A21.4 -- sizes 2" - 3" single thickness and sizes 4" - 54" double thickness
COATING:	Asphaltic seal coat inside and out in accordance with ANSI/AWWA C104/A21.4
GASKETS:	SBR in accordance with ANSI/AWWA C111/A21.11 (see pg. 17)
T-BOLTS/NUTS:	Low alloy steel in accordance with ANSI/AWWA C11/A21.11 (see pg. 16)
APPROVALS:	Sizes 3" and greater are UL/NSF-61
DIMENSIONS:	All dimensions are in inches unless noted otherwise

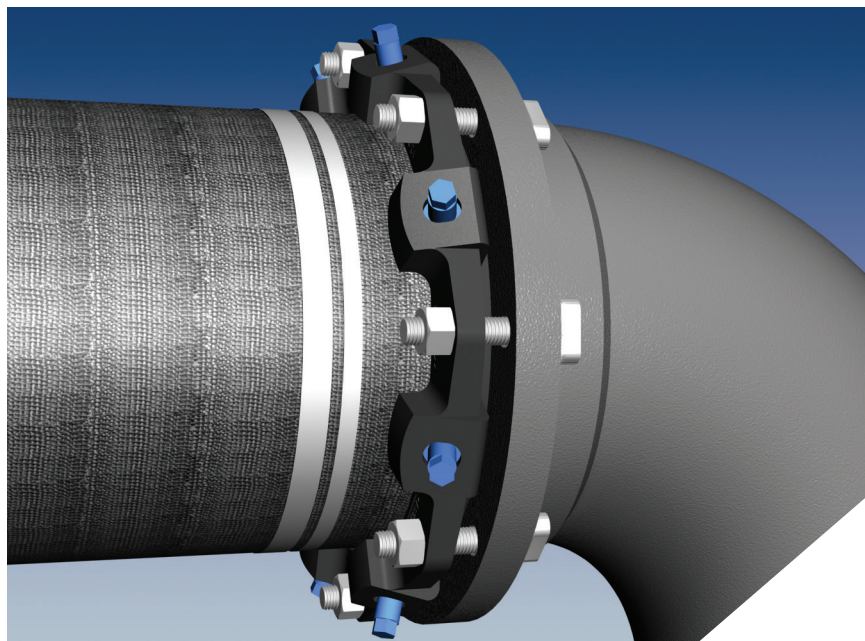


MECHANICAL JOINT DIMENSIONS															
NOM. SIZE	A DIA.	B	C DIA.	D DIA.	F DIA.	J DIA.	K1 DIA.	K2 DIA.	L	M	S	Ø	X DIA.	BOLTS	
														SIZE	NO.
★ 2	2.50	2.50	3.39	3.50	2.61	4.75	6.25	6.25	0.75	0.62	0.44	28°	¾	⅝ x 3	2
3	3.96	2.50	4.84	4.94	4.06	6.19	7.69	7.69	0.94	0.62	0.52	28°	¾	⅝ x 3	4
4	4.80	2.50	5.92	6.02	4.90	7.50	9.12	9.12	1.00	0.75	0.65	28°	⅞	¾ x 3 ½	4
6	6.90	2.50	8.02	8.12	7.00	9.50	11.12	11.12	1.06	0.88	0.70	28°	⅞	¾ x 3 ½	6
8	9.05	2.50	10.17	10.27	9.15	11.75	13.37	13.37	1.12	1.00	0.75	28°	⅞	¾ x 3 ½	6
10	11.10	2.50	12.22	12.34	11.20	14.00	15.69	15.62	1.19	1.00	0.80	28°	⅞	¾ x 4	8
12	13.20	2.50	14.32	14.44	13.30	16.25	17.94	17.88	1.25	1.00	0.85	28°	⅞	¾ x 4	8
14	15.30	3.50	16.40	16.54	15.44	18.75	20.31	20.25	1.31	1.25	0.89	28°	⅞	¾ x 4 ½	10
16	17.40	3.50	18.50	18.64	17.54	21.00	22.56	22.50	1.38	1.31	0.97	28°	⅞	¾ x 4 ½	12
18	19.50	3.50	20.60	20.74	19.64	23.25	24.83	24.75	1.44	1.38	1.05	28°	⅞	¾ x 4 ½	12
20	21.60	3.50	22.70	22.84	21.74	25.50	27.08	27.00	1.50	1.44	1.12	28°	⅞	¾ x 4 ½	14
24	25.80	3.50	26.90	27.04	25.94	30.00	31.58	31.50	1.62	1.56	1.22	28°	⅞	¾ x 5	16
30	32.00	4.00	33.29	33.46	32.17	36.88	39.12	39.12	1.81	2.00	1.50	20°	1 ⅛	1 x 6	20
36	38.30	4.00	39.59	39.76	38.47	43.75	46.00	46.00	2.00	2.00	1.80	20°	1 ⅛	1 x 6	24
42	44.50	4.00	45.79	45.96	44.67	50.62	53.12	53.12	2.00	2.00	1.95	20°	1 ⅜	1 ¼ x 6 ½	28
48	50.80	4.00	52.09	52.26	50.97	57.50	60.00	60.00	2.00	2.00	2.20	20°	1 ⅜	1 ¼ x 6 ½	32
★ 54	{ Dimensions Available On Request }														



★Not Included in AWWA C110

REV.07
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Features and Applications:

- Sizes 3 inch through 60 inch
- Constructed of ASTM A536 Ductile Iron
- Torque Limiting Twist-Off Nuts
- MEGA-BOND®
Restraint Coating System
For more information on MEGA-BOND,
refer to www.ebaa.com
- Minimum 2 to 1 Safety Factor
- The Mechanical Joint Follower Gland is
incorporated into the restraint
- Heavy Duty thick wall design
- Support Products Available:
Split repair style available 3 inch
through 54 inch.
EBAA Series 1100SD

Solid restraint harness available for
push-on pipe bells.
EBAA Series 1700

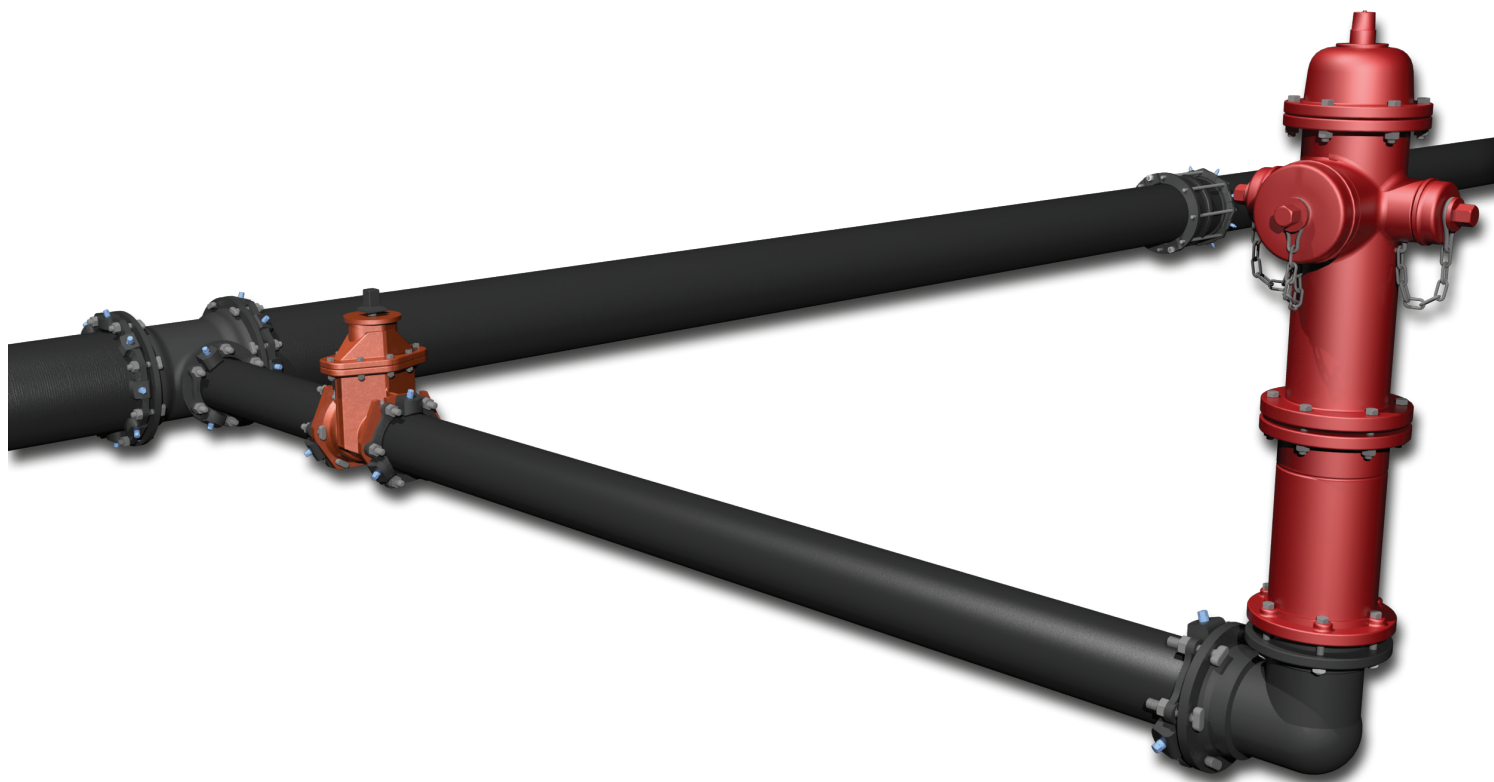
Split restraint harness available for
existing push-on bells.
EBAA Series 1100HD
- All MEGALUG and related restraint
products can be furnished as packaged
accessories complete with appropriate
restraint, gasket, lubrication, and bolt-
ing hardware
- For use on water or wastewater pipe-
lines subject to hydrostatic pressure
and tested in accordance with either
AWWA C600 or ASTM D2774

Nominal Pipe Size	Series Number	Shipping Weights	Post Assembly Deflection	Pressure Rating (PSI)
3	1103	6.1	3°	350
4	1104	7.7	3°	350
6	1106	11.9	3°	350
8	1108	14.8	3°	350
10	1110	23.9	3°	350
12	1112	31.2	3°	350
14	1114	48.5	2°	350
16	1116	56.4	2°	350
18	1118	63.1	1½°	250
20	1120	72.3	1½°	250
24	1124	133.1	1½°	250
30	1130	194.6	1°	250
36	1136	234.0	1°	250
42	1142	536.0	1°	250
48	1148	653.0	1°	250
54	1154	700.2	0.5°	250
60	1160	1,135.8	0.5°	200

**NOTE: For applications or pressures other than those shown please
contact EBAA for assistance.**



U.S. Patent Nos.
4092036, 4627774, 4779900, 4896903, 5544922



MEGALUG: THE PRODUCT OF PREFERENCE SINCE 1984

Since 1984, engineers and contractors designing and installing water and wastewater pipelines and systems have come to rely on the EBAA Series 1100 MEGALUG Mechanical Joint Restraint as the “Product of Preference” for effectively and economically restraining ductile iron pipe connections above or below ground.

MEGALUG Mechanical Joint Restraints replace external restraints such as cumbersome concrete thrust blocks and corrodible metal tie rods creating a quicker, safer and more economical installation.

Major testing laboratories agree as the 3” through 24” sizes are Underwriters Laboratories (UL) listed, and the 3” through 12” sizes are Factory Mutual (FM) approved.

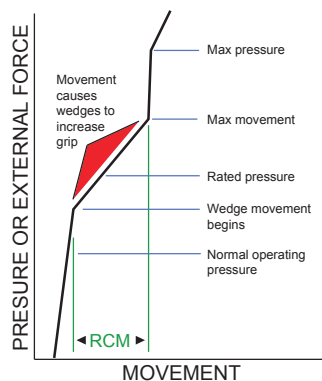
For use on all classes of ductile iron pipe (PC350 through PC150 and CL56 through CL50), for practically any application including valves, hydrants and pipe, the MEGALUG Mechanical Joint Restraint effectively and safely performs without damage to the pipe or cement linings.

THE MEGALUG GRIPPING WEDGE... PERFORMANCE PROVEN

The wedge style MEGALUG design reacts to the amount of force acting on the joint. When each wedge is set, the wedge teeth penetrate the pipe’s outer surface, and the wedge does not move on the pipe. There is very little change in this interface until the wedge movement begins inside the pocket of the main casting. Once the wedge starts moving, the formation of the buttress begins.

This “dam” of material (the wedge impression) is cold formed as the wedging action continues. If the force of pressure acting on the joint is released, the wedge moves back to near its original position. This engages the reserve-controlled movement or “RCM”. The wedge is then ready for another round.

After the wedge has moved to the back



of the pocket at the maximum pressure or load, the wedge buttress are in shear. The maximum movement is about 0.3 inch through the thirty-six inch size and 0.4 inch for forty-two and forty-eight inch.

The RCM is available even with severe

cyclic loads. This has been tested to very high-pressure differentials and the wedge impressions look the same as if a single test had been performed.

Typically, the depth of pipe wall penetration, or wedge impression at around 25,000 pounds of force per wedge (200 PSI on a six inch and 150 PSI on a twelve inch) is 0.03”. Finally, at roughly twice that force the penetration is around 0.05”. At these high pressures, there is no affect on the design thickness of ductile iron pipe made according to AWWA C150. The lack of damage to the cement lining clearly indicates that the thrust load is primarily longitudinal.

This ability to move in the pocket allows for angular flexibility as well as longitudinal flexibility.

THE ORIGINAL PATENTED GRIPPING WEDGES

Since 1964 EBAA Iron has responded aggressively to the needs of the water industry for better solutions to joint restraint problems - thus the development of the family of self actuating MEGALUG wedge action restraints.

TOOLS

MEGALUGS install using an ordinary wrench (box, ratchet, or air-driven), because the torque-limiting, twist-off nuts automatically shear during tightening when the proper torque is reached. The same 1 1/4" wrench used to tighten the T-bolts on the 4" through 24" sizes can be used to tighten and shear the twist off nuts in all sizes. If removal becomes necessary, a 5/8" hex head remains so the screws can be loosened, and retightened with a torque-indicating wrench. During removal, the wedges are held in place by retainer clips.

DEFLECTION

The MEGALUG gripping wedges provide resiliency to your pipeline design. In addition to deflecting as much as

allowed by the mechanical joint during installation, it can also deflect after assembly:

Sizes of 12" and below are capable of up to 3 degrees of deflection after installation (depending on the preset deflection.)

The 14" and 16" sizes are capable of 2 degrees deflection.

The 18" through 24" sizes are capable of 1.5 degrees deflection.

The 30" through 48" sizes are capable of 1 degree deflection.

The 54" and 60" sizes are capable of 0.5 degree deflection.

STEEL PIPE

The 1100 Series MEGALUG can be used to restrain 3" - 8" SCH 40 or 80 steel pipe when joining to mechanical appurtenances. It can also be used on steel pipe in all sizes if the pipe's outside diameter is the same as the ductile iron pipe and its thickness is equal to or greater than PC350 ductile iron pipe in sizes of 16 inch and below and PC250 ductile iron pipe 18 inches and above.

CAST IRON PIPE

Grey iron pipe diameters are often larger than ductile iron pipe diameters. The Series 1100 MEGALUG restraint may be used with grey iron pipe having standardized cast iron O.D. per AWWA C150 and C151, and with pit cast Classes "A" and "B" without modification. Use of the Series 1100 with pit cast grey iron Classes "C" and "D" will require over sizing the MEGALUG. More information on this is explained in detail in "Connections Bulletin DI-1".



MEGALUG Takes the Load

On April 11, 1997 EBAA Iron performed a remarkable force demonstration of their series 1100 MEGALUG Joint Restraint. With the use of EBAA's Series 1100 MEGALUG using a standard mechanical joint installation on 12 inch Ductile Iron Pipe, and a 80 Ton motor crane, EBAA Iron lifted a D7 Caterpillar Track Type Tractor weighing in at 50,350 lbs. Along with this, the Series 1100 MEGALUG has been tested to over 700 PSI. Concluding that EBAA's MEGALUGS can take the load.



Mechanical Joint Restraint Sample Specifications

(The text of the specifications below can be copied/pasted from www.ebaa.com/download/1100Spec.DOC)

Restraint devices for mechanical joint fittings and appurtenances conforming to either ANSI/AWWA C111/A21.11 or ANSI/AWWA C153/A21.53, shall conform to the following:

Design

Restraint devices for nominal pipe sizes 3 inch through 60 inch shall consist of multiple gripping wedges incorporated into a follower gland meeting the applicable requirements of ANSI/AWWA C110/A21.10.

The devices shall have a working pressure rating of 350 psi for 3-16 inch, 250 psi for 18-54 inch and 200 psi for the 60 inch. Ratings are for water pressure and must include a minimum safety factor of 2 to 1 in all sizes.

All EBAA products intended for installation on ductile iron pipe are designed for and limited to use on ductile iron pipes that comply with the requirements of ANSI/AWWA C151/A21.51 and have a Brinell Hardness or equivalent measurement value that does not exceed 230BHN. These requirements apply to the entire pipe wall profile at all restraining wedge engagement points and to the full penetration depth of each restraining wedge.

Material

Gland body, wedges and wedge actuating components shall be cast from grade 65-45-12 ductile iron material in accordance with ASTM A536.

For applications requiring restraint 30 inch and greater, an alternate grade of iron meeting the material requirements of ASTM A536 is acceptable, providing the device meets all end product performance requirements.

Ductile iron gripping wedges shall be heat treated within a range of 370 to 470 BHN.

Three (3) test bars shall be incrementally poured per production shift as per Underwriter's Laboratory (U.L.) Specifications

and ASTM A536. Testing for tensile, yield and elongation shall be done in accordance with ASTM E8.

Chemical and nodularity tests shall be performed as recommended by the Ductile Iron Society, on a per ladle basis.

Traceability

An identification number consisting of year, day, plant and shift (YYDDD)(plant designation)(Shift number), shall be cast into each gland body.

All physical and chemical test results shall be recorded such that they can be accessed via the identification number on the casting. These Material Traceability Records (MTR's) are to be made available, in hard copy, to the purchaser that requests such documentation and submits his gland body identification number.

Production pieces that are too small to accommodate individual numbering, such as fasteners and wedges, shall be controlled in segregate inventory until such time as all quality control tests are passed. These component parts may then be released to a general inventory for final assembly and packaging.

All components shall be manufactured and assembled in the United States. The purchaser shall, with reasonable notice, have the right to plant visitation at his/her expense.

Installation

Mechanical joint restraint shall require conventional tools and installation procedures per AWWA C600, while retaining full mechanical joint deflection during assembly

as well as allowing joint deflection after assembly.

Proper actuation of the gripping wedges shall be ensured with torque limiting twist off nuts.

Approvals

Restraint devices shall be Listed by Underwriters Laboratories (3" through 24" inch size) and Approved by Factory Mutual (3" through 12" inch size).

Mechanical joint restraint for ductile Iron pipe shall be Megalug Series 1100 produced by EBAA Iron Inc. or approved equal.

MEGA-BOND® Restraint Coating System

All wedge assemblies and related parts shall be processed through a phosphate wash, rinse and drying operation prior to coating application. The coating shall consist of a minimum of two coats of liquid thermoset epoxy coating with heat cure to follow each coat.

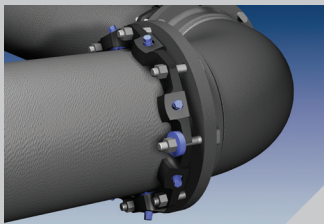
All casting bodies shall be surface pretreated with a phosphate wash, rinse and sealer before drying. The coating shall be electrostatically applied and heat cured. The coating shall be a polyester based powder to provide corrosion, impact and UV resistance.

The coating system shall be MEGA-BOND by EBAA Iron, Inc. or approved equal. Requests for approved equal must submit coating material and process details for review prior to bid.

For more information regarding MEGA-BOND, refer to the MEGA-BOND brochure or visit www.ebaa.com.

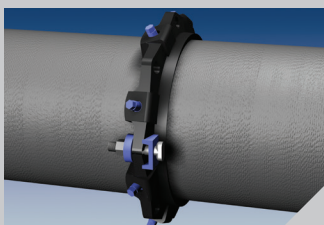
Support Products

for more information concerning these products please consult the catalog or www.ebaa.com



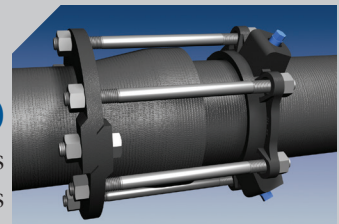
Series 1100SD

Split MEGALUG Restraint
For Existing Mechanical Joints



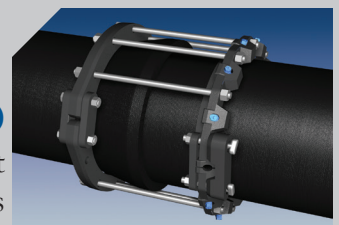
Series 1100SDB

Split MEGALUG Restraint
For Mid-Span Applications



Series 1700

MEGALUG Restraint Harness
For Push-On Bell Joints

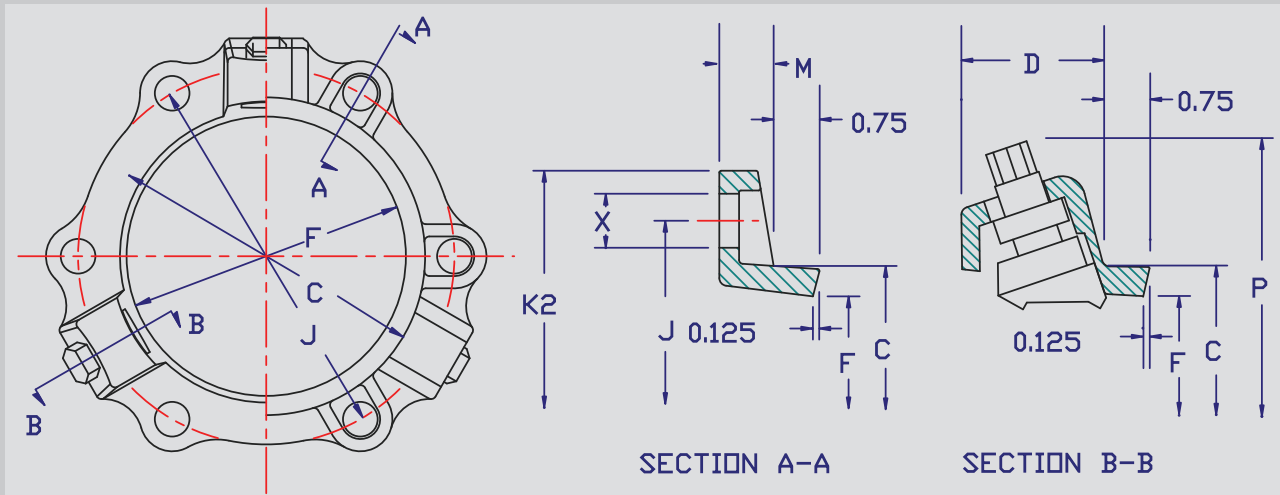


Series 1100HD

Split MEGALUG Restraint
Harness for Existing Push-On Bells

Series 1100 Submittal Reference Drawing

EBAA IRON



MADE IN USA

Nominal Pipe Size	Series Number	C	D	F	M	P*	X	J	K2	Wedge QTY.	Bolt (QTY.-Size)	Weight (LBS.)	Pressure Rating (PSI)
3	1103	4.84	2.27	4.06	0.62	9.06	0.750	6.19	7.69	2	4 - 5/8 x 3	6.1	350
4	1104	5.92	2.27	4.90	0.75	9.90	0.875	7.50	9.12	2	4 - 3/4 x 3 1/2	7.7	350
6	1106	8.02	2.27	7.00	0.88	12.00	0.875	9.50	11.12	3	6 - 3/4 x 3 1/2	11.9	350
8	1108	10.17	2.31	9.15	1.00	14.15	0.875	11.75	13.37	4	6 - 3/4 x 4	14.8	350
10	1110	12.22	2.37	11.20	1.00	16.20	0.875	14.00	15.62	8	8 - 3/4 x 4	23.9	350
12	1112	14.32	2.37	13.30	1.25	18.30	0.875	16.25	17.88	8	8 - 3/4 x 4	31.2	350
14	1114	16.40	2.69	15.44	1.50	20.94	0.875	18.75	20.25	10	10 - 3/4 x 4 1/2	48.5	350
16	1116	18.50	2.69	17.54	1.56	22.90	0.875	21.00	22.50	12	12 - 3/4 x 4 1/2	56.4	350
18	1118	20.60	2.69	19.64	1.63	25.00	0.875	23.25	24.75	12	12 - 3/4 x 4 1/2	63.1	250
20	1120	22.70	2.69	21.74	1.69	27.10	0.875	25.50	27.00	14	14 - 3/4 x 4 1/2	72.3	250
24	1124	26.90	3.20	25.94	1.81	32.64	0.875	30.00	31.50	16	16 - 3/4 x 5	133.1	250
30	1130	33.29	3.20	32.17	2.25	38.87	1.125	36.88	39.12	20	20 - 1 x 6	194.6	250
36	1136	39.59	3.20	38.47	2.25	45.17	1.125	43.75	46.00	24	24 - 1 x 6	234.0	250
42	1142	45.79	4.56	44.67	3.88	55.57	1.375	50.62	53.48	28	28 - 1 1/4 x 8 1/2	536.0	250
48	1148	52.09	4.56	50.97	3.88	61.87	1.375	57.50	60.36	32	32 - 1 1/4 x 8 1/2	653.0	250
54	1154	58.82	4.56	57.73	3.88	66.40	1.375	63.20	66.33	36	36 - 1 1/4 x 9 1/2	700.3	250
60	1160	62.90	6.40	61.78	4.23	73.00	1.375	67.72	70.70	36	36 - 1 1/4 x 11	1,135.8	200

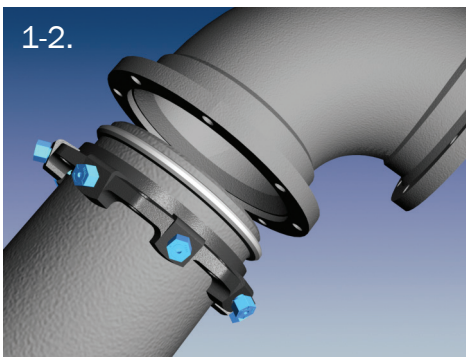
* With Twist-Off Nuts twisted off.

Important Notes

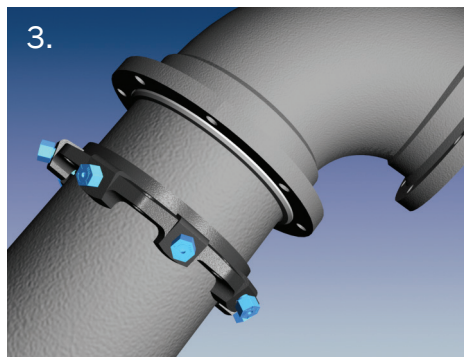
NOTE: Dimensions are in inches (±1%) and are subject to change without notice.

- The Series 1100 MEGALUG should not be used on plain end fittings.
- If encased in concrete, polyethylene wrap must be used to prevent concrete intrusion into the wedge pocket.
- For test pressures above the rated pressures shown, contact EBAA for recommendations, such as tandem restraint for high pressure applications.
- If you experience the need to install the Series 1100 MEGALUG in an unconventional manner please consult our engineering department.
- The Series 1100 MEGALUG is intended for use on ductile iron pipe. The restraint can be used on grey iron pipe if the pipe is not severely corroded and is in sound condition and has an outside diameter that can be accommodated. For more information on the use of the MEGALUG restraint on grey iron pipe ask for Connections Bulletin DI-1.
- EBAA-Seal™ Mechanical Joint Gaskets are provided with 30 inch through 60 inch MEGALUG restraints. These are required on the above referenced sizes to accommodate the pressure ratings and safety factors shown.
- Extra length T-bolts are provided with the 42 inch, 48 inch, 54 inch, and 60 inch sizes to facilitate easier assembly of the mechanical joint.
- All Series 1100 MEGALUG components are made of ductile iron conforming to ASTM A536. The wedges are heat treated to a hardness range of 370 to 470 BHN.
- LISTINGS AND APPROVALS: Sizes 3 inch through 24 inch are listed by Underwriters Laboratories, Inc. Category HJKF "Fittings, Retainer Type" with a deflection angle of 5 degrees (3 inch through 12 inch) and 2 1/2 degrees (14 inch through 24 inch). The listing file number is EX2836, Sizes 3 inch through 12 inch are Factory Mutual approved.

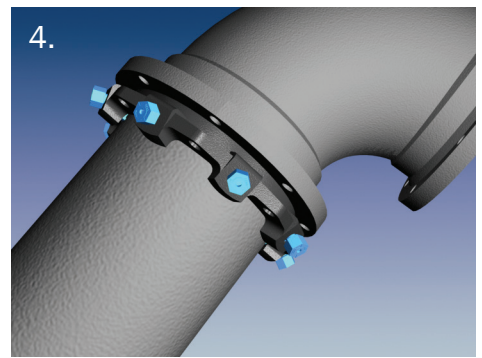
1-2.



3.



4.



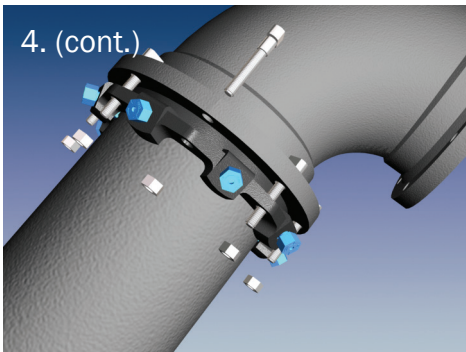
1. The Series 1100 MEGALUG® joint restraint is designed for use on ductile iron pipe conforming to ANSI/AWWA C151/A21.51 (all thickness classes) when restraining mechanical joint pipe fittings.

2. Clean the socket and the plain end. Lubrication and additional cleaning should be provided by brushing both the gasket and the plain end with soapy water or an approved pipe lubrication meeting the requirement of ANSI/AWWA C111/A21.11 just prior to slipping the gasket onto the plain end for joint assembly. Place the gland on the plain end with lip extension toward the plain end, followed by the gasket with the narrow edge of the gasket toward the plain end. [The gasket provided may be the EBAA-SEAL®

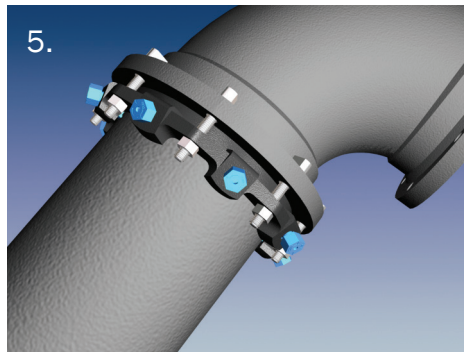
Improved Mechanical Joint Gasket, there is no narrow end as the gasket is bi-directional. In certain sizes, use of the EBAA-SEAL is required to achieve the pressure ratings of the MEGALUG.] NOTE: In cold weather it is preferable to warm the gasket to facilitate assembly of the joint.

3. Insert the pipe into the socket and press the gasket firmly and evenly into the gasket recess. Keep the joint straight during assembly.

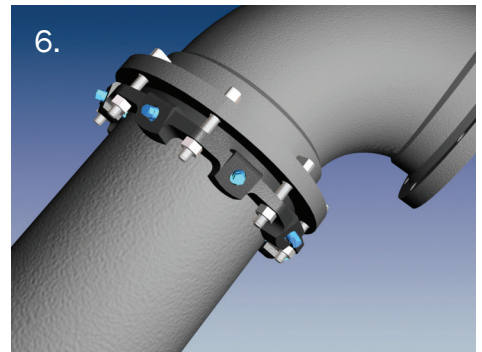
4. (cont.)



5.



6.



4. Push the gland toward the socket and center it around the pipe with the gland lip against the gasket. Insert bolts and hand tighten nuts. Make deflection after joint assembly but before tightening bolts.

5. Tighten the bolts to the normal range of torque as indicated [3-inch 45-60 ft.-lbs., 4 through 24-inch 75-90 ft.-lbs., 30 and 36-inch 100-120 ft.-lbs., and 42, 48, 54, and 60-inch 120-150 ft.-lbs.] While at all times maintaining approximately the same distance between the gland and the face of the flange at all points around the socket. This can be accomplished by partially tightening the bottom bolt first, then top bolt, next the bolts at either side, finally the remaining bolts. Repeat the process until all bolts are within the appropriate range of torque. In large

sizes (30 through 64-inch [762mm – 1,600mm]), five or more repetitions may be required. The use of a torque-indicating wrench will facilitate this procedure.

6. Tighten the torque limiting twist off nuts in a clockwise direction (direction indicated by arrow on top of nut) until all wedges are in firm contact with the pipe surface. Continue tightening in an alternating manner until all of the nuts have been twisted off.

7. If removal is necessary, utilize the $\frac{5}{8}$ " ($\frac{7}{8}$ " inch 1142 through 1154 and $1\frac{1}{8}$ " inch on the 1160) hex heads provided. If reassembly is required, assemble the joint in the same manner as above, by tightening the wedge bolts to 90 ft.-lbs (135 ft.-lbs for 1124 through 1154, and 165 ft.-lbs on the 1160). If the series 1100 restraint is removed from the pipe, be sure that all the collar bolts and wedges are in place before the restraint is reassembled.

Steps 2-5 are requirements of AWWA Standard C600-17

EBAA IRON Sales, Inc.

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contact@ebaa.com

www.ebaa.com

For More Information

For more information about MEGALUG restraints call EBAA today and request

"EBAA Connections Bulletin DI-1" concerning use of the MEGALUG restraint on grey iron pipe, or "EBAA Connections Bulletin DI-2" covering the background and operation of the MEGALUG system of restraint.

"Restraint Length Calculation" Software is available for PC/Windows applications. Support documentation about the software can be found in "EBAA Connections Bulletin PD-1 through PD-5".

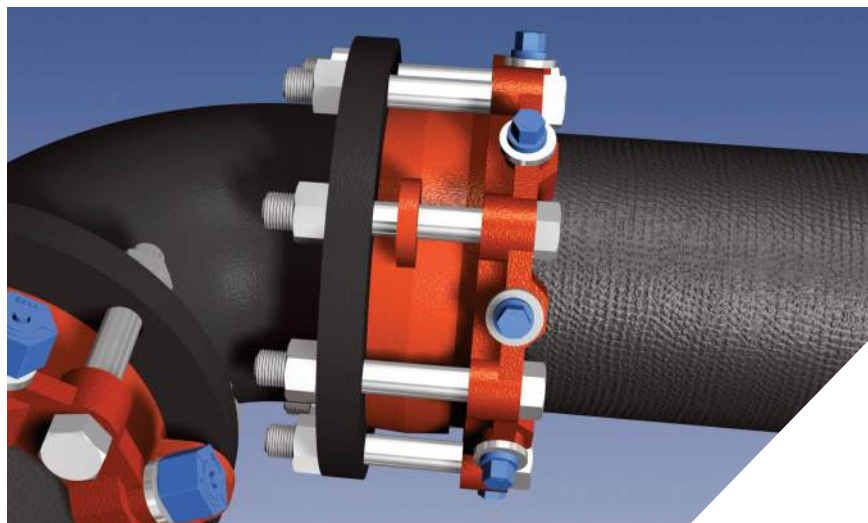


Series 2100

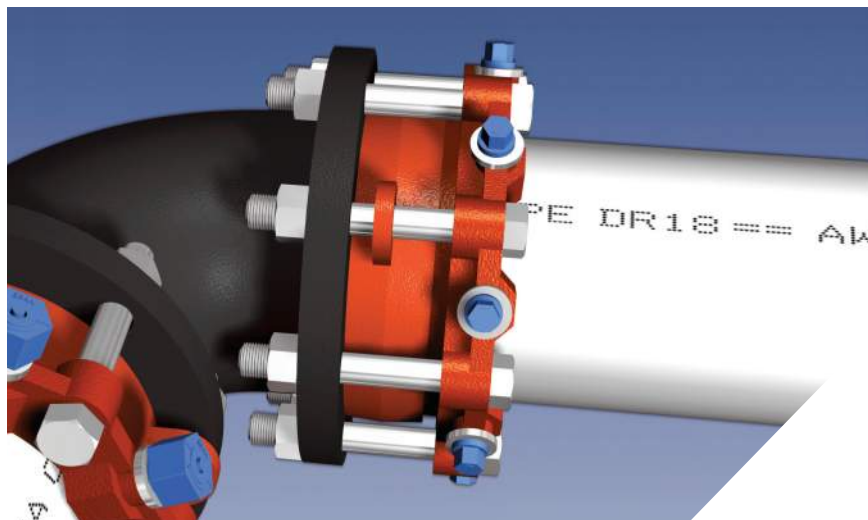
MEGAFLANGE®

Restrained Flange Adapter

U.S. Patent Nos. 4627774 and 5071175



Series 2106 on Ductile Iron Pipe



Series 2106 on C900 PVC Pipe

Features and Applications:

- MEGAFLANGE adapts and restrains plain end Ductile Iron, PVC, Carbon Steel and HDPE pipe to flanged pipe or fittings, where the flange conforms to ANSI/AWWA C111/A21.11 with flange surface facing in accordance with ANSI/AWWA C207 of the latest revision.
- Meets ANSI B16.5 Class 150/125 drilling pattern.
- Flange Bolts are zinc coated, fastener class coated bolts or stainless steel bolts are available
- Not for use on plain end fittings
- **MEGA-BOND®** Restraint Coating System
- For more information regarding MEGA-BOND, refer to our web site @ www.ebaa.com
- Minimum 2 to 1 Safety Factor
- Fully Restrained
- Constructed of ASTM A536 Ductile Iron
- **UL** listed on sizes 3 inch through 12 inch
- **FM** approved on sizes 4 inch through 12 inch on C900 Class 150 and Class 200 PVC Pipe
- Pipe can be cut to length in the field
- Joint deflection up to 5°
- Easy dismantling allows fast removal of valves, meters or fittings for replacement or repair

For use on water or wastewater pipelines subject to hydrostatic pressure and tested in accordance with either AWWA C600, C605 or ASTM D2774.

Sample Specification

Restrained flange adapters shall be used in lieu of threaded or welded flanged spool pieces. Flanged adapters shall be made of ductile iron conforming to ASTM A536 and have flange bolt circles that are compatible with ANSI/AWWA C110/A21.10 (125#/Class 150 Bolt Pattern).

Restraint for flange adapter shall consist of a plurality of individual actuated gripping wedges to maximize restraint capability. Torque limiting actuating screws shall be used to insure proper initial set of gripping wedges.

The flange adapters shall be capable of deflection during assembly or permit lengths of pipe to be field cut to allow a minimum 0.6 inch gap between the end of the pipe and the mating flange without affecting the integrity of the seal.

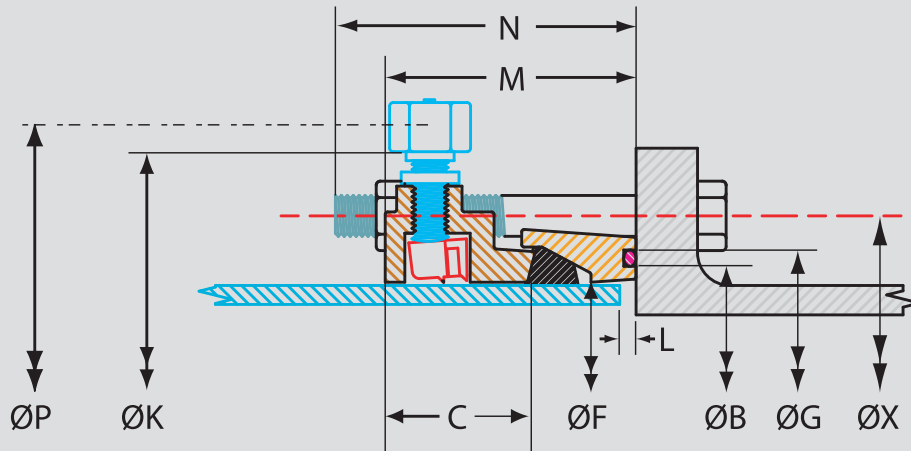
All internal surfaces of the gasket ring (wetted parts) shall be lined with a minimum of 15 mils of fusion bonded epoxy conforming to the applicable requirements of ANSI/AWWA C213. The coating shall meet ANSI/NSF-61. Exterior surfaces of the gasket ring shall be coated with a minimum of 6 mils of fusion bonded epoxy conforming to the applicable requirements of ANSI/AWWA C116/A21.16.

Restraint Ring coated with MEGA-BOND® Restraint Coating System, More information regarding MEGA-BOND can be found at www.ebaa.com.

Pressure ratings shall be a minimum of those shown in the table within current brochure.

The flange adapter shall be the Series 2100 MEGAFLANGE® Restrained Flange Adapter as produced by EBAA Iron, Inc. or approved equal.

Series 2100 Submittal Reference Drawing



Nominal Pipe Size	Series Number	Restraint Ring			Gasket Ring			Bolts				L MAX.	Assembly Deflection		P*	Ship Weight (lbs.)
		K	F	C	F	B	G	No.	Dia.	Length	X		Degrees	M		
3	2103	7.5	4.1	2.2	4.1	4.3	4.9	4	5/8	5½	6.00	0.7	5.0	4.0	9.2	14
4	2104	9.0	4.9	2.2	4.9	5.4	6.0	8	5/8	5½	7.50	0.6	5.0	4.0	10.0	20
6	2106	11.0	7.0	2.3	7.0	7.5	8.1	8	¾	6	9.50	0.8	5.0	4.3	12.1	32
8	2108	13.5	9.2	2.4	9.2	9.8	10.4	8	¾	6	11.75	0.9	5.0	4.5	14.3	38
10	2110	16.0	11.2	2.5	11.2	11.8	12.4	12	7/8	7½	14.25	1.0	3.0	4.7	16.3	65
12	2112	19.0	13.3	2.5	13.3	13.8	14.4	12	7/8	7½	17.00	1.0	3.0	4.8	18.4	73
14	2114	21.0	15.5	2.5	15.5	16.1	16.9	12	1	8	18.75	1.3	2.0	5.0	20.6	89
16	2116	23.5	17.6	2.5	17.6	18.2	19.0	16	1	8	21.25	1.3	2.0	5.0	22.6	109
18	2118	25.0	19.7	2.6	19.7	20.2	21.0	16	1½	8½	22.75	1.3	1.5	5.1	24.7	134
20	2120	27.3	21.8	2.6	21.8	22.4	23.2	20	1½	8½	25.00	1.3	1.5	5.1	26.8	157
24	2124	32.0	26.0	2.6	26.0	26.7	27.5	20	1¾	8½	29.50	1.3	1.0	5.1	31.0	192
30	2130	38.5	32.2	3.3	32.2	32.9	34.1	28	1¾	11	36.00	2.0	1.0	6.0	38.8	296
36	2136	45.5	38.5	3.3	38.5	39.2	40.4	32	1½	11**	42.75	2.0	1.0	6.0	44.6	426
42	2142	52.3	44.7	4.1	44.7	45.8	47.0	36	1½	14**	49.50	2.0	1.0	8.0	50.8	642
48	2148	58.8	51.0	4.1	51.0	52.1	53.3	44	1½	14**	56.00	2.0	1.0	8.0	57.1	797

* The "P" dimensions is measured with torque-limiting nuts twisted off.
 ** Double ended rod in lieu of bolt

Nominal Pipe Size	Minimal Distance Required To Install N
----------------------	--

3	4.75
4	4.56
6	5.00
8	4.88
10	6.31
12	6.25
14	6.62
16	6.56
18	6.94
20	6.81
24	6.62
30	8.88
36	8.63
42	11.25
48	11.38

MEGAFLANGE TESTING RESULTS

PVC TESTING

- Quick Burst Test
- DR18 tested to 755 PSI
- DR14 tested to 985 PSI
- Long Term Pressure Test
- On DR18 PVC pipe at 615 PSI for 1000 hours without failure
- Cyclic Pressure Test
- DR18 tested from 94 to 188 PSI for over 1,000,000 cycles

DUCTILE IRON AND CARBON STEEL TESTING

- Leakage Test (one minute required)
- Tested to twice rated pressure without leakage
- Hydrostatic Test (one minute required)
- 3 inch though 6 inch sizes tested to 5 times rated pressure
- 8 inch and 10 inch sizes tested to 4 times rated pressure
- 12 inch size tested to 3 times rated pressure
- Flexural Test
- Tested to withstand a bending moment based on requirements of NFPA 12-1991 "Standard for Installation of Sprinkler Systems"



APPROVED



Note: Dimensions are in inches
 and are subject to change without notice. All Dimensions are ± 1%.

Pipe Size	Ductile Iron Pipe Pressure (PSI)	Carbon Steel Pipe* Pressure (PSI)	C900 PVC Pipe				IPS PVC Pipe*		
			DR14 Pressure (PSI)	DR18 Pressure (PSI)	DR25 Pressure (PSI)	DR32.5 Pressure (PSI)	SDR17 Pressure (PSI)	SDR21 Pressure (PSI)	SDR26 Pressure (PSI)
3	350	350	-	-	-	-	250	200	160
4	350	350	305	235	165	-	250	200	160
6	350	350	305	235	165	-	250	200	160
8	350	350	305	235	165	-	250	200	160
10	350	350	305	235	165	-	250	200	160
12	350	350	305	235	165	-	250	200	160
14	350	-	-	235	165	125	-	-	-
16	350	-	-	235	165	125	-	-	-
18	300	-	-	235	165	125	-	-	-
20	250	-	-	235	165	125	-	-	-
24	200	-	-	150	165	125	-	-	-
30	150	-	-	-	-	-	-	-	-
36	150	-	-	-	-	-	-	-	-
42	150	-	-	-	-	-	-	-	-
48	150	-	-	-	-	-	-	-	-

*Transition Gasket Required

NOTE: For Application on HDPE pipe see EBAA's HDPE Restraint Catalog Sheet.

MEGAFLANGE Components

The **Series 2100 MEGAFLANGE** restrained flange adapter is comprised of two rings. The first is the restraint ring which incorporates wedges around the circumference of the ring to grip the pipe firmly and securely. The wedge style restraint offers enormous pullout strength when compared to set screw restraints. The resiliency of the wedge style restraint allows the MEGAFLANGE to withstand severe moment loads. The restraint ring and its sub-components are protected from corrosion by the MEGA-BOND® Restraint Coating System. For more information regarding MEGA-BOND see our MEGA-BOND Brochure found at www.ebaa.com.

The second ring is the gasket ring which separates the seals dedicated to each sealing surface. This ring allows pipe to be cut to lengths in the field at a tolerance of 0.6 inch or more. In addition, the gasket ring also enables the joint to deflect during assembly. The gasket ring is coated with a NSF 61 approved Fusion Bonded Epoxy (FBE) so that it may be utilized on potable drinking water systems.

DEFLECTION

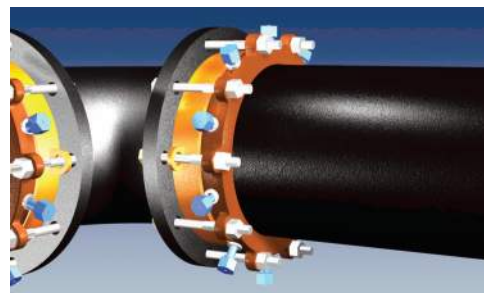
Traditional flanged joint connections require a tremendous amount of torque on the bolts to achieve a good seal. The pipe layout must be precisely planned to avoid misalignment errors due to deviations in appurtenances of pipe fabrication.

The Series 2100 MEGAFLANGE is a speedy, on-site fabrication tool which is generous in its deflection limits, from 0.5° to 5° depending on pipe size. The deflection capabilities provided by the gasket ring allow offset of almost nineteen inches of an eighteen foot length of pipe through the eight inch size.

1. Identify the pipe. The MEGAFLANGE 2100 Flange Adapter, sizes 3 inch through 12 inch, is designed for use on ductile iron pipe, PVC (C900 & IPS O.D. (ASTM D2241)) pipe, HDPE pipe, and carbon steel pipe. Check to see if the spacers under the screws are in place. If the pipe is ductile iron or C.I. O.D. PVC (C900) DO NOT REMOVE THE SPACERS. If the pipe is carbon steel or IPS O.D. PVC, REMOVE THE SPACERS (sizes 4-inch through 12-inch). The 3-inch size is designed for use on ductile iron, IPS O.D. PVC pipe. Sizes 30-inch and larger are designed for ductile iron pipe only. There are no spacers on the 3 inch and the 14 inch and larger sizes.



2. Cut the pipe to the required length. Clean the end of the pipe for a length approximately one foot using a wire brush if needed, removing all excess paint and foreign material. Also clean the opposing flange to be connected to the 2100. Place the 2100 restraint ring on the clean pipe with the lip facing the plain end.
3. Lubricate and place the EBAA-Seal™ Gasket on the clean pipe following the restraint ring. (USE A TRANSITION GASKET IN PLACE OF THE EBAA-SEAL GASKET FOR CARBON STEEL AND IPS. O.D. PVC PIPE.)
4. Place the O-ring into the groove of the 2100 Gasket Ring. (This step may have been completed at the factory, check Gasket Ring to see if O-ring is already in place.) Place the Gasket Ring on the pipe with the O-ring facing the pipe end and the gasket recess facing the EBAA-Seal (or transition) Gasket and restraint ring.



5. Bring the pipe and flanges together within the maximum assembled deflection and maximum allowable gap "L" to the flange face. Slide the gasket ring, gasket and restraint ring until contact is made with the opposing flange.
6. Insert and tighten all flange bolts. Torque all flange bolts an alternating manner to the value listed in Table 1.1. Be sure to make any necessary joint deflection before tightening the actuating screws. Joint deflection should not exceed the maximum allowable deflection. Be sure that deflection of the joint does not cause the end of the pipe to be separated from the opposing flange more than the maximum allowable gap "L".
7. Tighten the actuating screws in an alternating manner until all wedges touch the pipe. Continue tightening the nuts in an alternating pattern until all the torque-limiting nuts have been twisted off.

Table 1.1

Flange Bolt Torques

Nominal Pipe Size	Bolt Torque (ft.-lbs.)
3	45 - 60
4 - 6	75 - 90
8 - 24	90 - 110
30 - 48	110 - 130



8. If removal is necessary, utilize the 5/8 inch hex head provided. For reinstallation, repeat steps 2 through 7, torquing the actuating screws to 70 ft.-lbs. or until the hex heads bottom out on the spacers or gland.

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FOSTER ADAPTOR



BOLT-THRU MECHANICAL JOINT RESTRAINT RESTRAIN MJ VALVES AND FITTINGS TO EACH OTHER



BOLT-THROUGH
MJ RESTRAINT

Simple
Secure

PRODUCT APPLICATIONS

Valves to tees, crosses

90's, 45's other fittings



APPROXIMATE DRAW-UP LENGTHS

3 to 12-in	1-inch
14 to 24-in	2-inches
30 to 36-in	3-inches

PRODUCT BENEFITS

- 1** No pipe to saw
- 2** Space saving and light-weight
- 3** No possibility of pipe damage since no pipe connection is used
- 4** Secure bolt-through design makes for labor-saving, easy and fast installation
- 5** Made in U.S.A., ductile iron
3 to 24-inch: 350psi rating
30 to 36-inch: 250psi rating
- 6** Up to four times fewer bolts to tighten
- 7** Comes complete with all accessories
- 8** Close-coupled connections allow you to pre-assemble valves and fittings, then transport to the job site
- 9** Less chance for leakage. Two MJ's are connected using one accessory kit rather than the standard two
- 10** Available in standard asphalt or epoxy coating. Sizes 14 to 36-inch ship standard with epoxy coating

COATINGS	Asphalt or Epoxy on sizes 3 to 12-inch; 14-inch and above, epoxy is standard. Protecto 401 epoxy is available for sewer applications.
ACCESSORY PAKS	Corten included, Teflon® blue and stainless optional Special Long-bolt Accessory Paks available for C110 full-body fittings and certain butterfly valves with thicker bolt flanges.

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ENGINEERING DATA / PRODUCT CHECKLIST

FOSTER ADAPTOR



BOLT-THRU MECHANICAL JOINT RESTRAINT RESTRAIN MJ VALVES AND FITTINGS TO EACH OTHER



SIZES	3, 4, 6, 8, 10, 12, 14, 16, 18, 20, 24, 30, 36
TYPICAL USES	Restrains valves to tees and valves to crosses Restrains reducers and other fittings to tees or crosses Restrains 90's and other fittings to each other
ADVANTAGES (Product)	✓ Tough, BOLT-THROUGH Restraint ✓ Takes the place of <u>two</u> bolt-on lug type products ✓ No pipe to saw and no lug bolts to tighten ✓ Ultra-compact; saves critical space; 1 to 2-inch draw-up length ✓ Quicker to install than to saw pipe and tighten up to 4 times more bolts ✓ One accessory kit to install (included in price) ✓ Integral casting, no welds ✓ 14, 16, 18, 20, 24, 30, 36-in epoxy coated — standard
ADVANTAGES (Manufacturer) INFACIT PRODUCTS ARE AVAILABLE FROM YOUR LOCAL WATERWORKS DISTRIBUTOR	✓ Products made in U. S. A. and Infact is an American-owned company ✓ BIG inventory; Quick shipping; No extra charge to drop ship ✓ No minimum order
HYDRANT AND OTHER CONNECTIONS	1) When using 3-inch to 14-inch Fosters on both the "branch" and "run" of compact tees or crosses, the standard accessory bolts may obstruct each other. Substitution of shorter bolts should solve this. Conduct a trial connection prior to field assembly. Special short-bolt accessory paks are available. Contact Infact for more info. 2) Because of the large variety of hydrant designs, we strongly recommend that you conduct a trial connection prior to field assembly. Foster Adaptors can be assembled on many hydrant shoes that have adequate space for bolt clearance, including M&H Model 129, Mueller Centurion and Clow Medallion. Contact Infact for more info.
OPTIONS	1) NSF 61, 7-mil fusion bonded epoxy conforming to AWWA C116/A21.16-09 as well as the coating, surface preparation and application requirements of ANSI/ AWWA C550. 2) 40-mil Protecto 401 epoxy coating for Foster Adaptor sewer applications. Contact Infact for specs and availability. 3) <i>Long-bolt Paks</i> , to restrain C110 full-bodied fittings, certain valves, etc. with thicker bolt flanges.

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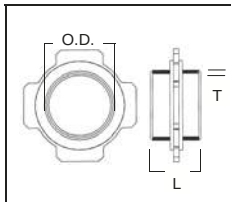
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FOSTER ADAPTOR

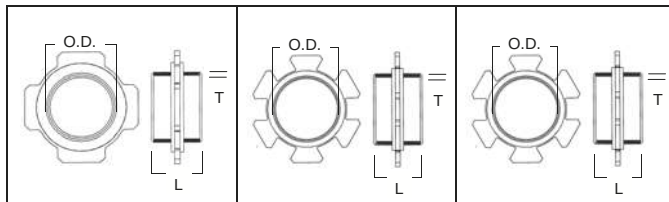


BOLT-THRU MECHANICAL JOINT RESTRAINT

RESTRAIN MJ VALVES AND FITTINGS TO EACH OTHER



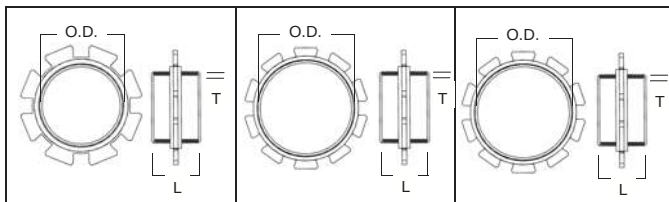
3-INCH



4-INCH

6-INCH

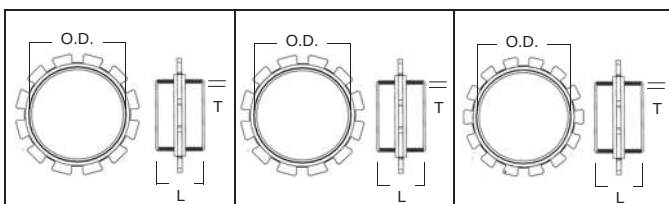
8-INCH



10-INCH

12-INCH

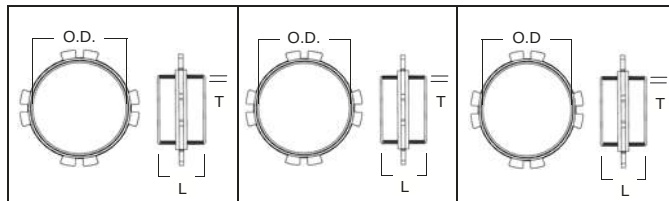
14-INCH



16-INCH

18-INCH

20-INCH



24-INCH

30-INCH

36-INCH

LATEST SIZES ADDED: 3-inch, 30-inch, and 36-inch

SIZE, inches	DIMENSIONS, inches			ADAPTOR Weight	PAK Weight	TOTAL Weight
	O. D.	T	L	lbs.	lbs.	lbs.
3	3.96	.34	4	6	5	11
4	4.80	.35	4	8	8	16
6	6.90	.37	4	16	9	25
8	9.05	.39	4	19	10	29
10	11.10	.40	4	29	13	42
12	13.20	.43	4.5	31	15	46
14	15.30	.48	6.75	55	16	71
16	17.40	.51	6.75	60	20	80
18	19.50	.55	6.75	92	16	108
20	21.60	.58	6.75	101	20	121
24	25.80	.62	6.75	117	28	135
30	32.00	.66	7.75	153	68	221
36	38.30	.75	7.75	220	86	306

Adaptor sizes, dimensions and weights are nominal



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ENGINEERING DATA / SPECIFICATIONS

FOSTER ADAPTOR



BOLT-THRU MECHANICAL JOINT RESTRAINT RESTRAIN MJ VALVES AND FITTINGS TO EACH OTHER



STANDARD ADAPTORS, STANDARD ACCESSORIES

The Infact Foster Adaptor is made in the U. S. A. It meets the ductile iron and working pressure specifications of AWWA compact fittings, ANSI/AWWA C153/A21.53 and C110/A21.10 — American National Standard for Ductile Iron Compact Fittings, 3-Inch Through 36-Inch (76mm Through 914mm) for Water Service.

SAMPLE SPECIFICATION: Mechanical joint (MJ) valves and fittings shall be connected using a bolt-through positive restraint mechanism manufactured of U. S. A. ductile iron conforming to ASTM A536, 65-45-12. The positive restraint device shall connect the valves and/or fittings at a linear distance not to exceed three (3) inches and without attachment to pipe. The device shall come complete with all accessories, including standard styrene butadiene rubber (SBR) MJ gaskets conforming to the latest revision of AWWA C111/ASTM F-477 and weathering steel (Corten) bolts conforming to AWWA C111/A21.11 and ASTM A242. Nuts for 3 through 12-inch sizes shall be SAE Grade 5 steel with black oxide coating. Nuts for 14-inch and larger adaptors shall be heavy hex Corten steel conforming to ASTM A242. Sizes 3-12-inch of the bolt-through MJ positive restraining device shall be supplied with an NSF 61 asphaltic seal coating in accordance with ANSI/AWWA C104/A21.4. Sizes 14-36-inch shall be supplied with NSF 61, 7-mil. fusion bonded epoxy conforming to AWWA C116/A21.16-09 as well as the coating, surface preparation and application requirements of ANSI/AWWA C550. For sewer installations, the device shall be supplied with 40-mil Protecto 401 epoxy. [Epoxy coating, blue Teflon® coated, and stainless steel hardware are available for all sizes.] The device shall be used with standard mechanical joint fittings (AWWA C110 or C153) and valves. The device shall be Infact Corporation FOSTER ADAPTOR or equal.

EPOXY ADAPTORS, BLUE FLUOROCARBON (TEFLON®) ACCESSORIES

The Infact Foster Adaptor is made in the U. S. A. It meets the ductile iron and working pressure specifications of AWWA compact fittings, ANSI/AWWA C153/A21.53 and C110/A21.10 — American National Standard for Ductile Iron Compact Fittings, 3-Inch Through 36-Inch (76mm Through 914mm) for Water Service.

SAMPLE SPECIFICATION: Mechanical joint (MJ) valves and fittings shall be connected using a bolt-through positive restraint mechanism manufactured of U. S. A. ductile iron conforming to ASTM A536, 65-45-12. The positive restraint device shall connect the valves and/or fittings at a linear distance not to exceed three (3) inches and without attachment to pipe. The device shall come complete with all accessories, including standard styrene butadiene rubber (SBR) MJ gaskets conforming to the latest revision of AWWA C111/ASTM F-477 and blue fluorocarbon [Teflon®] coated bolts and nuts conforming to AWWA C111/A21.11. The bolt-through MJ positive restraining device and the ductile iron spacers shall be supplied with an NSF 61, 7-mil. fusion bonded epoxy conforming to AWWA C116/A21.16-09 as well as the coating, surface preparation and application requirements of ANSI/AWWA C550. For sewer installations, the device shall be supplied with 40-mil Protecto 401 epoxy. The device shall be used with standard mechanical joint fittings (AWWA C110 or C153) and valves. The device shall be Infact Corporation FOSTER ADAPTOR or equal.

EPOXY ADAPTORS, STAINLESS STEEL ACCESSORIES

The Infact Foster Adaptor is made in the U. S. A. It meets the ductile iron and working pressure specifications of AWWA compact fittings, ANSI/AWWA C153/A21.53 and C110/A21.10 — American National Standard for Ductile Iron Compact Fittings, 3-Inch Through 36-Inch (76mm Through 914mm) for Water Service.

SAMPLE SPECIFICATION: Mechanical joint (MJ) valves and fittings shall be connected using a bolt-through positive restraint mechanism manufactured of U. S. A. ductile iron conforming to ASTM A536, 65-45-12. The positive restraint device shall connect the valves and/or fittings at a linear distance not to exceed three (3) inches and without attachment to pipe. The device shall come complete with all accessories, including standard styrene butadiene rubber (SBR) MJ gaskets conforming to the latest revision of AWWA C111/ASTM F-477. The bolt-through MJ positive restraining device and the ductile iron spacers shall be supplied with an NSF 61, 7-mil. fusion bonded epoxy conforming to AWWA C116/A21.16-09 as well as the coating, surface preparation and application requirements of ANSI/AWWA C550. For sewer installations, the device shall be supplied with 40-mil Protecto 401 epoxy. The bolts and nuts shall be ASTM A193 type 304 or 316 stainless (as requested) and the nuts shall be Teflon® coated. The device shall be used with standard mechanical joint fittings (AWWA C110 or C153) and valves. The device shall be Infact Corporation FOSTER ADAPTOR or equal.

NOTE: When using 3-inch to 14-inch Fosters on both the “branch” and “run” of compact tees or crosses, the standard accessory bolts may obstruct each other. Substitution of shorter bolts should solve this. Conduct a trial connection prior to field assembly. Special short-bolt accessory paks are available. For C110 full-body fittings and certain thick-flange valves, Infact Long-bolt Paks are available. Because of the large variety of hydrant designs, we strongly recommend that you conduct a trial connection prior to field assembly. Foster Adaptors can be assembled on many hydrant shoes that have adequate space for bolt clearance, including M&H Model 129, Mueller Centurion and Clow Medallion. Contact Infact for more info.

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IN**FACT****WORK-SAVERS
SPACE-SAVERS****Installer's Guide**

PRODUCT	DESCRIPTION	INSTALLED DRAW-UP DISTANCE	TYPE OF JOINT		RECOMMENDED APPLICATIONS		
			MJ	FLANGE	VALVES	FITTINGS	HYDRANTS
FOSTER ADAPTOR	Bolt-thru MJ Adaptor 	Sizes 3-12: 1-inch --- Sizes 14-24: 2-inches --- Sizes 30-36: 3-inches	✓		✓	✓	
FLEX T-2	Self-Restraining 2-way Tee 	0 inches --- (Direct Restraint)	✓	✓	✓	✓	
FLEX T-3	Self-Restraining 3-way Tee 	0 inches --- (Direct Restraint)	✓	✓	✓	✓	
FLEX ADAPTOR	Ultra-Compact Coupling 	8 1/2 inches	✓	✓	✓	✓	✓
FLEX REDUCER	Ultra-Compact Reducer 	8 1/2 inches	✓	✓	✓	✓	✓

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Comparison of Waterworks Products that Restrain Valves, Fittings and Hydrants

	INFAC T FOSTER ADAPTOR	LUG PRODUCT	INFAC T FLEX T-2 & T-3	ANCHOR COUPLING	INFAC T FLEX ADAPTOR	GRIP RINGS	INFAC T FLEX REDUCER
Built-in full axial, circumferential restraint. Restraining forces are spread more evenly than in products using radial bolts or lugs.	X		X	X	X	X	X
Bolt-through restraint	X						
Angular deflection for easy restraint of unaligned fittings			X		X	X	X
Full axial rotation for easy alignment of bolt holes			X	X	X	X	X
Two independent pivot points adjacent to fittings for better close quarters installation and easier connection to misaligned fittings			X		X		X
Requires ONE device to connect fittings (rather than two or more)	X		X	X	X		X
Do not have to cut and install a piece of pipe in order for device to restrain fittings	X			X	X		
Uses standard bolts and gaskets and no radial bolts on pipe that operator may forget to or improperly tighten			X	X	X	X	
Field adaptable to being connected to any length pipe		X				X	
Field adaptable to restrain either MJ or flange joints using interchangeable glands			X		X		X

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FOSTER ADAPTOR

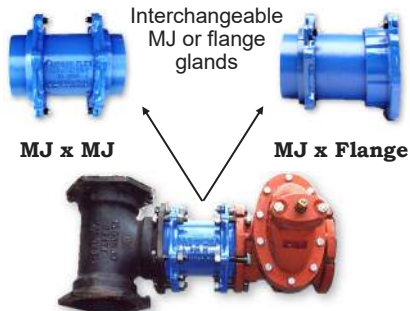
BOLT-THRU MJ RESTRAINT
FOR VALVES AND FITTINGS



- No pipe to cut
- Up to 75% fewer bolts
- Saves space
- Easier to lift and install
- Comes with all accessories

FLEX ADAPTOR

SPACE-SAVING MJ OR FLANGE RESTRAINT
FOR VALVES AND FITTINGS

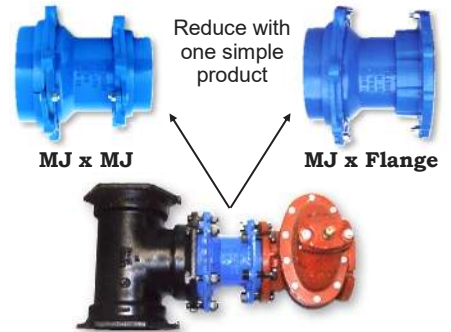


- Rotating and deflecting MJ glands make this the easiest way to align and restrain MJ or flange valves, fittings, and hydrants
- Available in 6-inch MJ x MJ, MJ x flange, flange x flange

U. S. PATENT NO. 6015170

FLEX REDUCER

SPACE-SAVING MJ OR FLANGE REDUCER
FOR VALVES AND FITTINGS



- Save up to 2-feet of space
- Faster, simpler installation
- Uses standard accessories
- EZ-change MJ or flange glands
- EZ-align MJ glands spin and deflect

U. S. PATENT NO. 6015170

FLEX T-2

DIRECT-TO-TEE RESTRAINT
FOR MJ OR FLANGE VALVES AND FITTINGS



- Restrain 2 valves or fittings to the T-2
- FAST installation — no couplings needed
- EZ-to-align MJ glands spin and deflect
- MJ x MJ, MJ x flange, or flange x flange
- Same glands fit all Flex products

U. S. PATENT NO. 6015170

FLEX T-3

DIRECT-TO-TEE RESTRAINT
FOR MJ OR FLANGE VALVES AND FITTINGS



- Restrain 3 valves or fittings to the T-3
- Choose MJ or flange glands
- Heavy-duty split glands
- Epoxy coating standard
- Direct-to-tee restraint

U. S. PATENT NO. 6015170

MUD PLUGS, INSULATION

VALVE BOX DEBRIS PLUGS
METER BOX and VAULT INSULATION



- Simple push-in, pull-out installation
- Plug compresses for tight seal
- Won't absorb water or freeze
- Flexible, non-conductive poly handle
- Custom sizes available

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Restrained Coupling

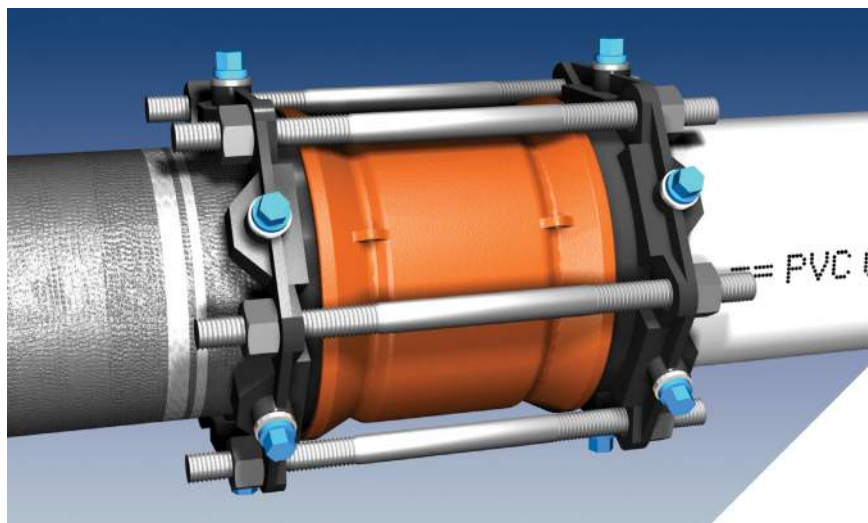


Image depicts Series 3806 on 6 inch ductile iron pipe and 6 inch PVC pipe.

The Series 3800 are designed to restrain forces based on the maximum working pressure ratings listed in the Pressure Ratings Table.

Forces experienced due to expansion/contraction of the pipe require special consideration.

For use on water or wastewater pipelines subject to hydrostatic pressure and tested in accordance with either AWWA C600 or ASTM D2774.

Features and Applications:

- For use on:
 - Ductile Iron Pipe, 4 in. - 48 in.
 - C900 or ASTM D2241 PVC Pipe, 4 in. - 12 in.
 - C905 PVC Pipe, 14 in. - 36 in.
 - Carbon Steel Pipe, 4 in. - 12 in.
 - HDPE Pipe, 4 in. - 12 in.
- Minimum 2 to 1 Safety Factor
- **MEGA-BOND®** Restraint Coating System
For more information regarding MEGA-BOND, refer to our web site at www.ebaa.com
- Constructed of ASTM A536 Ductile Iron
- Corrosion Resistant, low alloy, high strength steel bolts and nuts per ANSI/AWWA C111/A21.11
- Couplings meets or exceeds the applicable requirements of:
 - AWWA C219
 - ASTM A536
 - ANSI/AWWA C111/A21.11
 - ASTM D2000

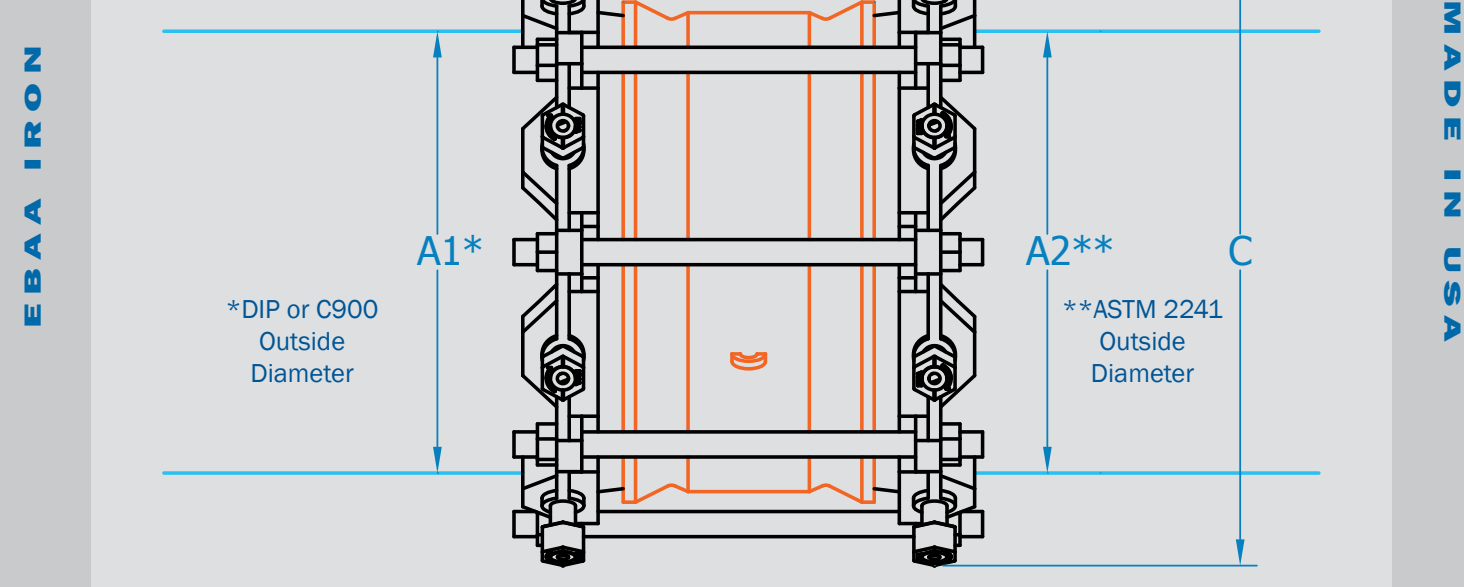
Nominal Pipe Size	Series Number	Shipping Weight	Ductile Iron Pipe	Steel Pipe	Pressure Ratings (PSI)									
					HDPE Pipe			C900 PVC Pipe			C905 PVC Pipe			
					DR11	DR13.5	DR17	DR14	DR18	DR25	DR18	DR25	DR32.5	DR41
4	3804	36.50	350	350	160	130	100	305	235	165	-	-	-	-
6	3806	49.40	350	350	160	130	100	305	235	165	-	-	-	-
8	3808	67.93	350	350	160	130	100	305	235	165	-	-	-	-
10	3810	85.46	350	350	160	130	100	305	235	165	-	-	-	-
12	3812	103.98	350	350	160	130	100	305	235	165	-	-	-	-
14	3814	157.18	350	-	-	-	-	-	-	-	235	165	125	80
16	3816	183.02	350	-	-	-	-	-	-	-	235	165	125	80
18	3818	201.64	250	-	-	-	-	-	-	-	200	165	-	-
20	3820	229.93	250	-	-	-	-	-	-	-	200	165	-	-
24	3824	364.24	250	-	-	-	-	-	-	-	165	165	125	100
30	3830	540.60	250	-	-	-	-	-	-	-	-	165	125	-
36	3836	695.90	250	-	-	-	-	-	-	-	-	125	125	-
42	3842	1,219.48	250	-	-	-	-	-	-	-	-	-	-	-
48	3848	1,449.76	250	-	-	-	-	-	-	-	-	-	-	-

Weights units are pounds and are approximate values.

Special Notes For Use On HDPE Pipe

For use on 4 inch through 12 inch HDPE only. The use of a pipe wall stiffening insert is required for use on HDPE pipe. The stiffeners must be sized to encompass the entire bearing length of the restraint devices. Pipe systems must be engineered to prevent movement causing fitting to slide or rotate.

EBAA products for HDPE are designed for underground pressurized fluid service and are pressure rated to match the pipe SDR pressure rating, de-rated as appropriate for service temperature. Maximum test pressure limited to pipe rated pressure.



Nominal Pipe Size	Series Number	A1* Pipe O.D. (Maximum)	A2** Pipe O.D. (Minimum)	B Overall Length	C† Max. Restraint O.D. (Casing Clearance)	D Thrust Bolt (Number-Size)	E Barrel Length	Maximum Deflection
4	3804	4.80	4.50	14.0	9.5	4 - ¾ x 14	7.5	4°
6	3806	6.90	6.63	14.0	12.1	6 - ¾ x 14	7.5	4°
8	3808	9.05	8.63	14.0	13.6	6 - ¾ x 14	7.5	4°
10	3810	11.10	10.75	14.0	16.0	8 - ¾ x 14	7.5	4°
12	3812	13.20	12.75	14.0	18.1	8 - ¾ x 14	7.5	4°

Sample Specification

** Requires the use of a Transition Gasket. † As installed with Twist-Off nuts off.
Note: Dimensions are in inches (± 1%) and are subject to change without notice.

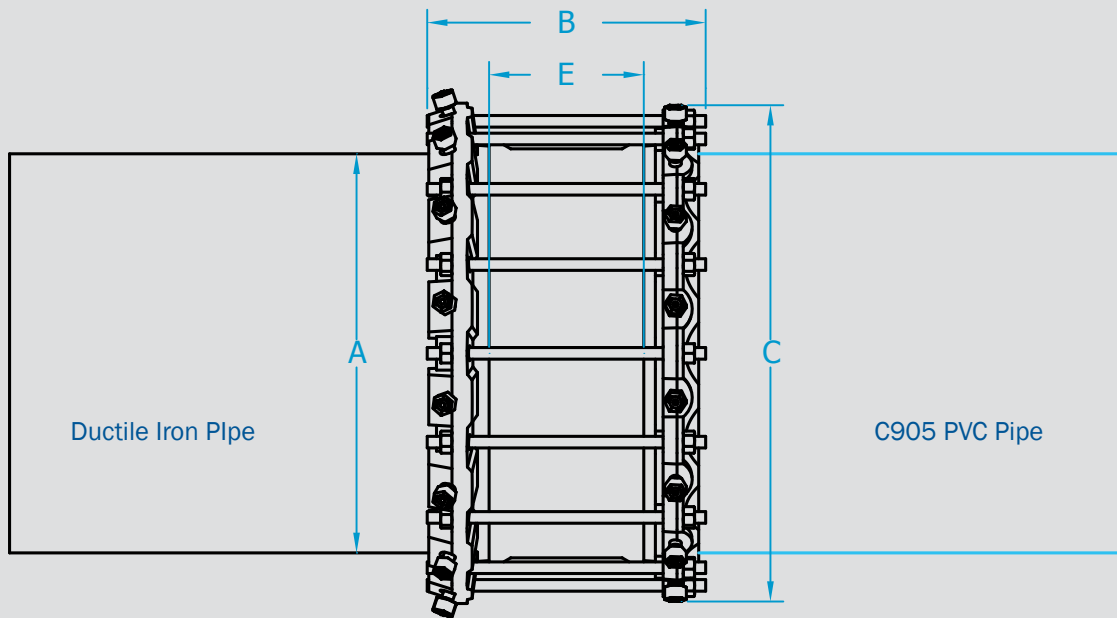
Joint restraint to prevent axial separation shall be incorporated into the design of the sleeve or coupling used to connect two plain ends of same or dissimilar materials, such as Ductile Iron pipe, steel pipe, PVC pipe (C900 or ASTM D2241) and or High Density Polyethylene (HDPE) pipe. Internal pipe wall stiffeners must be used when restraining HDPE.

The restraint mechanism shall incorporate a plurality of individually actuating gripping surfaces to maximize restraint capability, and have torque limiting twist off nuts to insure proper actuating of the restraint devices. The restraint devices shall be coated using MEGA-BOND®. For complete specifications on MEGA-BOND visit www.ebaa.com.

The coupling sleeve internal surfaces (wetted parts) shall be lined with a minimum of 15 mils of fusion bonded epoxy conforming to the applicable requirements of ANSI/AWWA C213. Sealing gaskets shall be constructed of SBR. The coating shall meet ANSI/NSF-61. Exterior surfaces shall be coated with a minimum of 6 mils of fusion bonded epoxy conforming to the applicable requirements of ANSI/AWWA C116/A21.16. Ductile Iron components shall meet or exceed the requirements of ASTM A536, and shall be tested in accordance with said standard.

The restrained joining system shall meet the applicable requirements of AWWA C219, ANSI/AWWA C111/A21.11, and ASTM D2000.

The restrained coupling system shall be Series 3800 manufactured by EBAA Iron, Inc.



Nominal Pipe Size	Series Number*	A Pipe O.D.	B Overall Length	C** Max. Restraint O.D. (Casing Clearance)	Thrust Bolt (Number-Size)	E Barrel Length	Maximum Deflection
14	3814XX	15.3	18.0	20.94	10 - ¾ x 18	10.0	4°
16	3816XX	17.4	18.0	23.00	12 - ¾ x 18	10.0	4°
18	3818XX	19.5	18.0	25.10	12 - ¾ x 18	10.0	4°
20	3820XX	21.6	18.0	27.20	14 - ¾ x 18	10.0	4°
24	3824XX	25.8	18.0	32.64	16 - ¾ x 18	10.0	4°
30	3830XX	32.0	20.0	38.87	20 - 1 x 20	10.0	3°
36	3836XX	38.3	20.0	45.17	24 - 1 x 20	10.0	3°
42	3842XX	44.5	24.0	55.57	28 - 1¼ x 24	10.0	1°
48	3848XX	50.8	24.0	61.87	32 - 1¼ x 24	10.0	1°

** As installed with Twist-Off nuts off.

*Specifying the Proper Series Number

Note: Dimensions are in inches (± 1%) and are subject to change without notice.

4 inch through 12 inch

For Nominal Pipe Sizes 4 inch through 12 inch the Series 3800 MEGA-COUPLING utilizes a common restraint ring for Ductile Iron Pipe (DIP), C900 PVC Pipe, ASTM D2241 PVC Pipe (IPS), Carbon Steel Pipe, and HDPE Pipe (with internal pipe wall stiffener for HDPE). The only item that needs to be specified during the order process for 4 inch through 12 inch is what type of gasket is required for joint assembly for the various pipe O.D.'s:

- 3800S for coupling either DIP, C900 PVC and or DIP O.D. HDPE Pipe
- 3800T for coupling either Carbon Steel and or ASTM D2241 PVC
- 3800ST for coupling either DIP, C900 PVC or DIP O.D. HDPE Pipe on one side and either Carbon Steel or ASTM D2241 PVC on the other side

14 inch through 48 inch

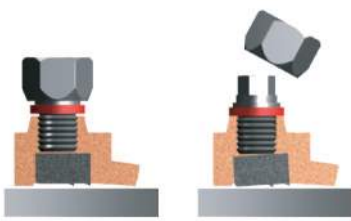
For Nominal Pipe sizes 14 inch and above the Series 3800 MEGA-COUPLING utilizes the Series 1100 MEGALUG® restraint on DIP and the Series 2000PV MEGALUG restraint on C905 PVC. Since the O.D. of both DIP and C905 PVC is the same, the EBAA-SEAL™ Improved Mechanical Joint Gasket is provided.

- 3800DI for coupling DIP
- 3800PV for coupling C905 PVC
- 3800DIPV for coupling one side DIP and other C905 PVC

Spacer Instructions (4 in. through 12 in. only)

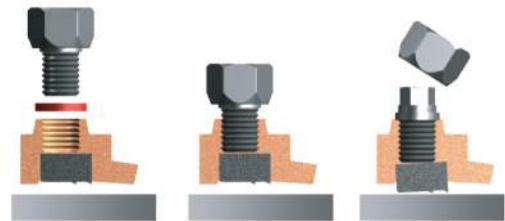
Ductile Iron or C900 PVC O.D. Sized Pipe

For installation on **Ductile Iron or C900 PVC sized pipe**, use as received and install per instructions.

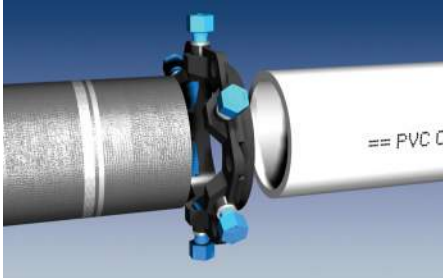


Steel or ASTM D2241 O.D. Sized Pipe (IPS O.D.)

For installation on **Steel or ASTM D2241 O.D. Sized pipe**, remove spacers and replace screws. Install per instructions.



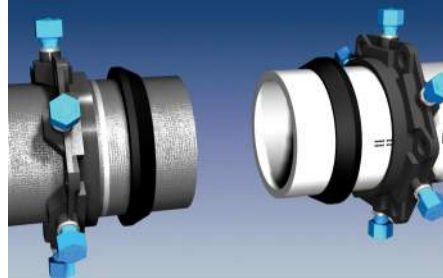
Installation Instructions



1. Identify the pipe. The spacers under the actuating screws must be removed for use on ASTM D2241 O.D. sized pipe (4 inch through 12 in. only). The spacers must remain in place for use on Ductile Iron or C900 PVC O.D. sized pipe. (See spacer instructions above)

Clean and inspect the pipe ends. Beveling of the ends is not necessary.

If used on HDPE Pipe, a pipe wall stiffener insert that encompass the entire bearing length of the restraint devices must be installed prior to step two.



2. Place the end rings on the pipes with the lip extensions toward the pipe ends. For 4 inch through 12 the restraints will be the Series 2000HPV, for 14 inch and above Ductile Iron pipe the restraint ring will be the Series 1100 MEGALUG® and for C905 PVC 14 inch and above the restraint ring will be the Series 2000PV MEGALUG. Lubricate and install the Standard Mechanical Joint Gasket.

(Standard Mechanical Joint Gaskets must be used with Ductile Iron Pipe and C900 PVC pipe 4 inch through 12 inch while the EBAA-SEAL™ Improved Mechanical Joint gasket is used on 14 inch and above. Transition gaskets must be used with Steel or ASTM D2241 pipe).



3. Center the sleeve body over the ends of the pipes while maintaining a 1/2" to 1" gap between the pipe ends. Slide the gaskets and end rings toward the sleeve body.



4. Install the threaded rods and hand tighten the nuts on each end. Gradually tighten the nuts in an alternating manner until the proper torque value has been reached, while maintaining the same distance between the rings and the ends of the body at all points around the rings. For 4 inch through 24 inch torque to 75-90 ft.-lbs., for 30 and 48 inch torque to 120-150 ft.-lbs.



5. Hand tighten the actuating screws until all wedges are touching the pipes. Continue tightening the screws in an alternating manner until the torque limiting heads twist off. The screws may bottom out during this step.



EBAA IRON Sales, Inc.

P.O. Box 857, Eastland, TX 76448

Tel: (254) 629-1731

Fax: (254) 629-8931

(800) 433-1716 within US and Canada

contact@ebaa.com

www.ebaa.com



▲ Series 1124SDB as an anchor for connecting rods.

Features and Applications:

- For use on Ductile Iron Pipe in mid-span restraint applications
- Minimum 2 to 1 Safety Factor
- Split design for ease of installation
- Constructed of ASTM A536 Ductile Iron
- For Submittal Reference Information please refer to the Series 1100 Brochure.

For use on water or wastewater pipelines subject to hydrostatic pressure and tested in accordance with either AWWA C600 or ASTM D2774.

Sample Specification

A mid span restraint anchor point on existing ductile iron pipe shall consist of the following: The restraint shall be split for ease of installation and manufactured of ductile iron conforming to ASTM A536. The restraint devices shall be coated using MEGA-BOND®. (For complete specifications on MEGA-BOND visit www.ebaa.com). The split restraint ring, incorporating a plurality of individually-actuating gripping surfaces, shall be used to grip the pipe. The device shall have a minimum working pressure rating as shown in the product brochure. The restraint shall be the Series 1100SDB, as manufactured by EBAA Iron, Inc. or approved equal.

All EBAA products intended for installation on ductile iron pipe are designed for and limited to use on ductile iron pipes that comply with the requirements of ANSI/AWWA C151/A21.51 and have a Brinell Hardness or equivalent measurement value that does not exceed 230BHN. These requirements apply to the entire pipe wall profile at all restraining wedge engagement points and to the full penetration depth of each restraining wedge.

Nominal Pipe Size	Series Number	Approximate Shipping Weight	Pressure Rating (PSI)
			Ductile Iron
3	1103SDB	9.4	350
4	1104SDB	10.7	350
6	1106SDB	15.1	350
8	1108SDB	18.8	350
10	1110SDB	38.0	300
12	1112SDB	46.7	300
14	1114SDB	65.7	300
16	1116SDB	73.6	300
18	1118SDB	80.3	200
20	1120SDB	89.5	200
24	1124SDB	151.6	200
30	1130SDB	218.6	200
36	1136SDB	258.6	200
42	1142SDB	467.2	175
48	1148SDB	554.1	175
54	1154SDB	785.0	250

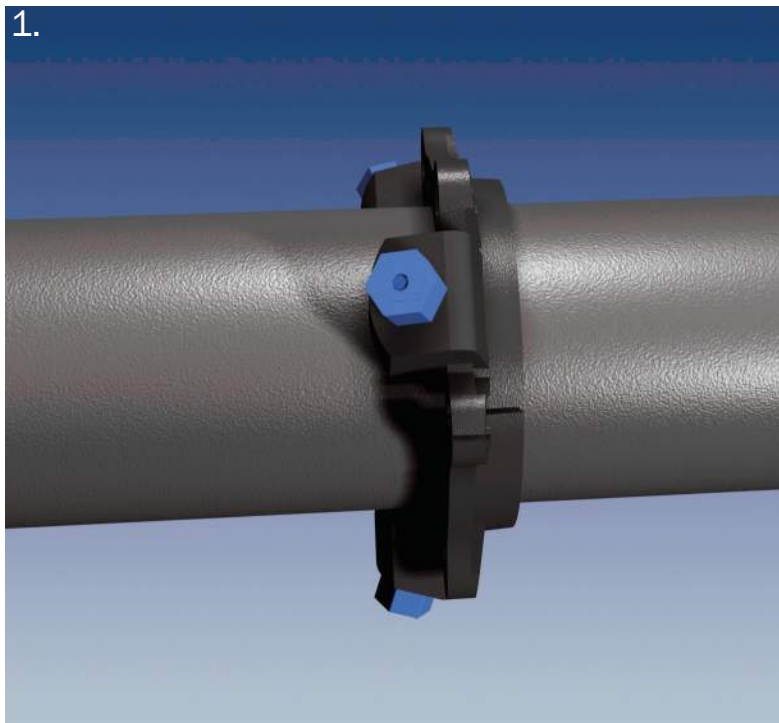
NOTE: For applications or pressures other than those shown, please contact EBAA for assistance.



▲ Two Series 1124SDB as an anchor in a concrete "Dead Man" wall (polyethylene wrap, not depicted, must be used. See reverse side.).

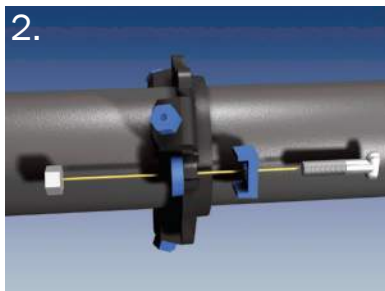
Installation Instructions for Series 1100SDB

1.

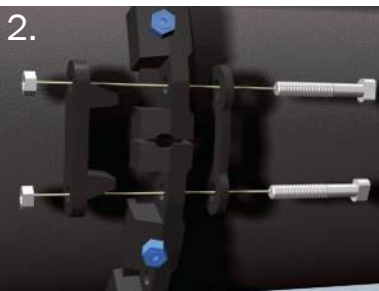


1. Remove the clamps from the split gland. Loosely assemble the halves on the pipe by assembling each clamp so that the angled surfaces of the clamp mate with the angled surfaces on each side of the split, and with the backup plate on the opposite side of the gland from the clamp. Insert the long T-bolt (provided) through the clamps and tighten hand tight.

2.



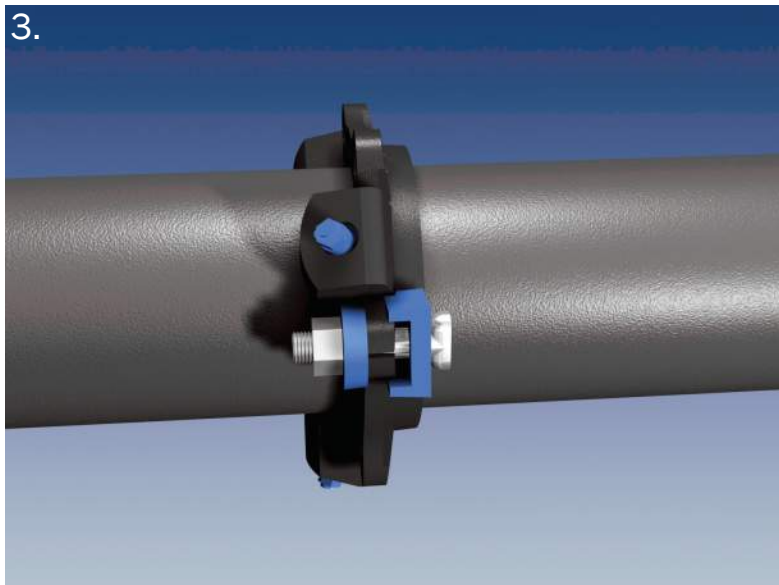
3-8 inch



10-54 inch

2. Tighten the T-bolts. Tighten the bolts to the normal range of bolt torque [45-60 ft-lbs for 3 inch, 75-90 ft-lbs for 4 inch through 24 inch, 100-120 ft-lbs 30 inch through 36 inch, and 120-150 ft-lbs for 42 inch through 54 inch]. The use of a torque indicating wrench will facilitate this procedure.
3. Tighten the torque limiting twist-off nuts in a clockwise direction (direction indicated by an arrow on top of nut) until all wedges are in firm contact with the pipe surface. Continue tightening in an alternating manner until all of the nuts have twisted off.

3.



4. If removal is necessary; use the $\frac{5}{8}$ " ($\frac{7}{8}$ " 1142SDB through 1154SDB) hex heads provided. If reassembly is required, assemble the joint in the same manner as above, tightening the wedge bolts to 90 ft-lbs (135 ft-lbs for 1124SDB through 1154SDB).



Important Note:

When used as an anchor within concrete, polyethylene wrap must be used to prevent concrete intrusion into the wedge pocket.

For Submittal Reference Information please refer to the Series 1100 Brochure.

EBAA IRON Sales, Inc.

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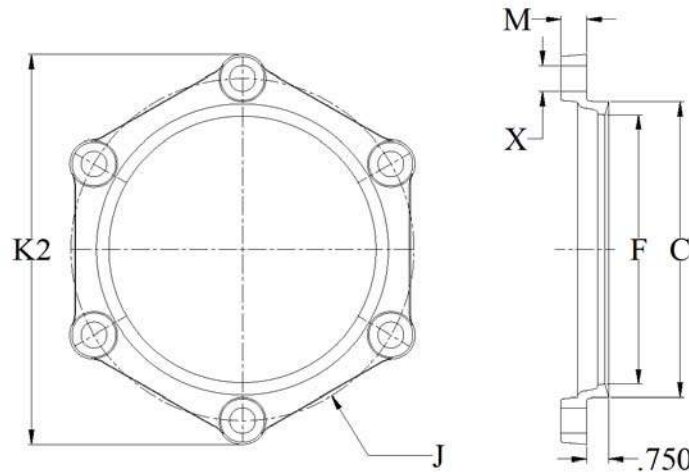
☐ DOMESTIC

☐ NON-DOMESTIC

SUBMITTAL: C153 MECHANICAL JOINT GLANDS

(Current revisions for the noted Standards apply)

SIZES:	2" - 64" (2" not included in ANSI/AWWA C153 standard)
STANDARDS:	ANSI/AWWA C153/A21.53, NFPA13/24, 3"-12" UL and 3"-12" FM listed & approved (File - Tyler Union)
MATERIAL:	Cast of ASTM A536 qualified ductile iron. Date code is cast on and required for traceability.
DEFLECTION:	Joint deflection 5° max for 2"– 12" and 3° max for 14"– 48". Reduces by 50% at nominal pipe & fitting diameters
NSF-61 & NSF372:	Meets all requirements including Annex G, Tyler Union's Underwriters Laboratory listing MH16439.
ASPHALT COATING:	Per ANSI/AWWA C104/A21.4 and ANSI/AWWA C153/A21.53.
EPOXY COATING:	Fusion bonded epoxy per ANSI/AWWA C116/A21.16. Additional coatings available upon request.
BARE FITTINGS:	Available upon request.
FASTNERS:	High strength low alloy weathering steel per ANSI/AWWA C111/A21.11 and ASTM A242
INSTALLATION:	Install per AWWA C600/C651 using pipe conforming to ANSI/AWWA C151/A21.51 or AWWA C900/905.



NOMINAL JOINT DIMENSIONS IN INCHES

Size Inches	DI Pipe Dia.	C Dia. GLAND	F Dia.	J Dia. GLAND	K ² Dia. GLAND	M GLAND	X	Size	Qty.
3	3.96	4.84	4.06	6.19	7.69	0.62	3/4	5/8x3	4
4	4.80	5.92	4.90	7.50	9.12	0.75	7/8	3/4x3-1/2	4
6	6.90	8.02	7.00	9.50	11.12	0.88	7/8	3/4x3-1/2	6
8	9.05	10.17	9.15	11.75	13.37	1.00	7/8	3/4x4	6
10	11.10	12.22	11.20	14.00	15.62	1.00	7/8	3/4x4	8
12	13.20	14.32	13.30	16.25	17.88	1.00	7/8	3/4x4	8
14	15.30	16.40	15.44	18.75	20.25	1.25	7/8	3/4x4-1/2	10
16	17.40	18.50	17.54	21.00	22.50	1.31	7/8	3/4x4-1/2	12
18	19.50	20.60	19.64	23.25	24.75	1.38	7/8	3/4x4-1/2	12
20	21.60	22.70	21.74	25.50	27.00	1.44	7/8	3/4x4-1/2	14
24	25.80	26.90	25.94	30.00	31.50	1.56	7/8	3/4x5	16
30	32.00	33.29	32.17	36.88	39.12	2.00	1-1/8	1x6	20
36	38.30	39.59	38.47	43.75	46.00	2.00	1-1/8	1x6	24
42	44.50	45.79	44.67	50.62	53.12	2.00	1-3/8	1-1/4x6-1/2	28
48	50.80	52.09	50.97	57.50	60.00	2.00	1-3/8	1-1/4x6-1/2	32
54	Available on Request								
60	Available on Request								
64	Available on Request								

Tyler Union Waterworks – Call Center Information

Anniston: (800) 226-7601

Corona: (866) 527-8471

Tyler: (800) 527-8478



SUBMITTAL

(Current revisions for the noted Standards apply)

Tyler Union Waterworks provides that our *Mechanical and Push-On joint gaskets and dimensions conform to the specifications in ANSI/AWWA C111/A21.11 (current revision). Markings include size, mold number, gasket manufacturer's mark, country where molded, and product identification letters. No markings are placed on sealing surfaces per the AWWA C111 standard.

*Note: Push-On and Mechanical Joint transition gasket design standards and markings are not addressed by ANSI/AWWA C111/A21.11 (current revision). Transition gaskets provided by Tyler Union follow the material testing standards and specifications established for ANSI/AWWA C111/A21.11 gaskets.

Gasket material is vulcanized styrene butadiene rubber (SBR). Purchaser may request special application elastomers (EPDM, Nitrile, Neoprene & FKM) which will be identified on all documentation and corresponding gaskets. Gaskets are free of foreign materials, porous areas, or other defects that make them unfit for the intended use.

Tyler Union gaskets are manufactured under quality control standards and procedures that are maintained by the gasket supplier. Appropriate documentation is maintained by the manufacturer and available for review upon request. Properties and test methods for SBR, EPDM, Nitrile, Neoprene and FKM gaskets are as provided.

Property	ASTM Test Method	Required Value
Hardness, Shore "A"	D2240-86	75 (+-5)
Minimum Tensile	D412-87	1500 psi (10MPa)
Minimum Elongation	D412-87	150 %
Minimum Aging	D572-88	60 %
Maximum Compression Set	D395-89, Method B	20 %
Resistance to surface Ozone cracking	D1149-86	No cracking

Tyler Union's approved suppliers maintain a quality assurance program that is reviewed and updated on an ongoing basis to ensure product quality. Tyler Union's gasket suppliers submit gaskets for testing and provide materials for testing to Underwriters Laboratories, Inc. Tyler Union's gasket providers are recognized under the component program (UL 194/ UL 157) of Underwriters Laboratories, Inc.. Tyler Union UL approved gaskets meet NSF-61, NSF-372 and Annex G.

Tyler Union provides that our Mechanical and Push-On joint gaskets for potable or wastewater projects will perform as designed when selected per the chart provided and installed per AWWA C600-10.

SBR (Styrene Butadiene rubber)(Buna-S) <u>Not Recommended for Hydrocarbon Service</u>	20°F to 180°F	Suitable for Water, Wastewater, most moderate chemicals, wet or dry organic acids, alcohols, ketones, and aldehydes
EPDM (Ethylene Propylene) <u>Not Recommended for Hydrocarbon Service</u>	-10°F to 250°F	Ideal for water, wastewater, ozone, & strong oxidizing chemicals May be used on steam and air within its temperature range
CR (Neoprene)	-10°F to 200°F	Recommended for moderate chemicals and acids, oil fats, greases, many solvents and air with hydrocarbons. Will not support combustion
NBR (Nitrile)(Buna-N)(Hycar)	-40°F to 250°F	Ideally suited for gasoline, petroleum products, hydrocarbons, water, mineral and vegetable oils
*FKM(Fluoroelastomer) *Check with Customer Service for availability	10°F to 425°F	Ideally suited for hydrocarbons, acids, vegetable oils & petroleum

Gasket Types Offered:	(1)Mechanical Joint std.(2) Push-On Joint std.(3)Mechanical Joint DUO (4)Mechanical & Push-on Joint Transition(5)Push-on Restraining (6)Mechanical Joint Armor Tip Conductivity(7)Compact tapping Sleeve
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Unless other wise requested by the purchaser upon order placement, all gaskets provided will be of our standard SBR material.

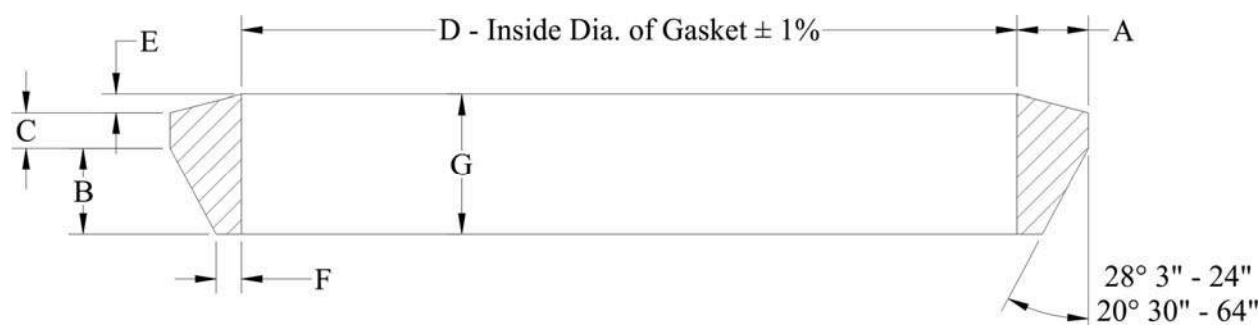
Mechanical Joint Gaskets

ANSI/AWWA C111/A21.11-12

*MJ Tru-Lock Gaskets 30-48 inch

Pipe Size	Pipe OD	A ±0.01"	B	C	D ±1%	E ±0.01%	F ±0.01"	G ±0.02"
**2	2.50	0.48	0.62	0.31	2.48	0.12	0.15	1.05
3	3.96	0.48	0.62	0.31	3.86	0.12	0.15	1.05
4	4.80	0.62	0.75	0.31	4.68	0.16	0.22	1.22
6	6.90	0.62	0.75	0.31	6.73	0.16	0.22	1.22
8	9.05	0.62	0.75	0.31	8.85	0.16	0.22	1.22
10	11.10	0.62	0.75	0.31	10.87	0.16	0.22	1.22
12	13.20	0.62	0.75	0.31	12.95	0.16	0.22	1.22
14	15.30	0.62	0.75	0.31	14.99	0.16	0.22	1.22
16	17.40	0.62	0.75	0.31	17.07	0.16	0.22	1.22
18	19.50	0.62	0.75	0.31	19.13	0.16	0.22	1.22
20	21.60	0.62	0.75	0.31	21.20	0.16	0.22	1.22
24	25.80	0.62	0.75	0.31	25.34	0.16	0.22	1.22
30	32.00	0.73	1.00/* .50	.38/* .50	31.47	0.16	.37/* .55	1.54/* 1.16
36	38.30	0.73	1.00/* .50	.38/* .50	37.67	0.16	.37/* .55	1.54/* 1.16
42	44.50	0.73	1.00/* .50	.38/* .50	43.78	0.16	.37/* .55	1.54/* 1.16
48	50.80	0.73	1.00/* .50	.38/* .50	49.98	0.16	.37/* .55	1.54/* 1.16

** Not included in AWWA C111. Manufacture's Standard does not meet AWWA C111



**** Mechanical Joint Transition Gasket Dimensions in Inches**

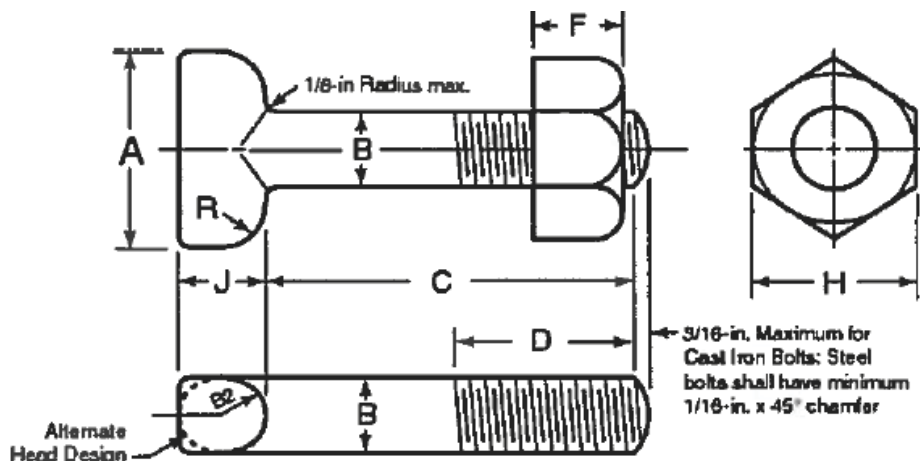
Pipe Size	A ± 0.01"	B	C	D ±1%	E	F ± 0.01"	G ± 0.02"
2	0.57	0.62	0.31	2.28	0.16	0.24	1.08
3	0.70	0.62	0.31	3.45	0.16	0.37	1.11
4	0.77	0.75	0.31	4.43	0.16	0.37	1.26
6	0.76	0.75	0.31	6.53	0.16	0.36	1.25
8	0.82	0.75	0.31	8.50	0.16	0.42	1.27
10	0.79	0.75	0.31	10.59	0.16	0.39	1.26
12	0.84	0.75	0.31	12.56	0.16	0.44	1.28

** Not included in AWWA C111. Manufacture's Standard does not meet AWWA C111

SUBMITTAL: C111 T-BOLTS AND NUTS FOR MECHANICAL JOINT

(Current revisions for all noted Standards apply)

SIZES:	5/8 X 3 through 1-1/4 X 8.5
STANDARDS:	ANSI/AWWA C111/A21.11
Material:	ASTM A242 High Strength Low Alloy Steel. Also available in ANSI 304 or 316 AISI Stainless Steel.
COATING:	Standard bolts and nuts are not coated. Stainless Steel bolt and coated Green or Red. Flouropolymer costing is available on request. Flouropolymer is Flurokote #1 (Blue for Low Alloy Steel).
FASTENERS:	High strength low alloy weathering steel per ANSI/AWWA C111/A21.11 and ASTM A242.



T-HEAD LOW ALLOY WEATHERING AND AISI STAINLESS STEEL FASTENER DIMENSION IN INCHES									
Size Inches	A ±0.05	B +0.03 / - 0.074	C +0.25 / - 0.06	F	H	J +0.15/0.03	R Mx	D*	Threads Per Inch E**
5/8x3	1.5	0.625	3.00	0.625	1.062	0.625	0.312	2.00	11
3/4x3.5	1.75	0.750	3.50	0.750	1.250	0.750	0.375	2.50	10
3/4x4	1.75	0.750	4.00	0.750	1.250	0.750	0.375	3.00	10
3/4x4.5	1.75	0.750	4.50	0.750	1.250	0.750	0.375	3.00	10
3/4x5	1.75	0.750	5.00	0.750	1.250	0.750	0.375	3.00	10
1x6	1.75	1.000	6.00	1.000	1.625	1.000	0.500	3.00	8
1x7.5	2.25	1.000	7.50	1.000	1.625	1.000	0.625	4.50	8
1-1/4x6.5	2.50	1.250	6.50	1.250	2.000	1.250	0.750	3.50	7
1-1/4x8.5	2.50	1.250	8.50	1.250	2.000	1.250	0.750	4.00	7

- NOTES:**
1. Dimension B is unthreaded shank.
 2. Draft, when required to be 6 degree maximum, may be deducted from bolt head dimensions, and radius (B/2) may be changed to suit draft.
 3. Gates, if required, may protrude a maximum of 1/8" above the top of bolt thread.
 4. Chamfer is optional if threads are rolled.
 5. If threads are rolled, the body diameter may be reduced to the approximate pitch diameter of the thread.
 6. * Tolerance +3/-0 thread.
 7. ** Number of threads per inch – course-thread series (ANSI/ASME B1.1), Class 2A, external fit UNC2A and Class 2B, UNC2B (ANSI/ASME B1.2).

ANSI/AWWA **C104/A21.4-22**
(Revision of ANSI/AWWA C104/A21.4-16)

AWWA Standard

Cement–Mortar Lining for Ductile-Iron Pipe and Fittings

Effective date: Feb. 1, 2023.

First edition approved by Board of Directors Jan. 28, 1980.

This edition approved Oct. 24, 2022.

Approved by American National Standards Institute Oct. 4, 2022.

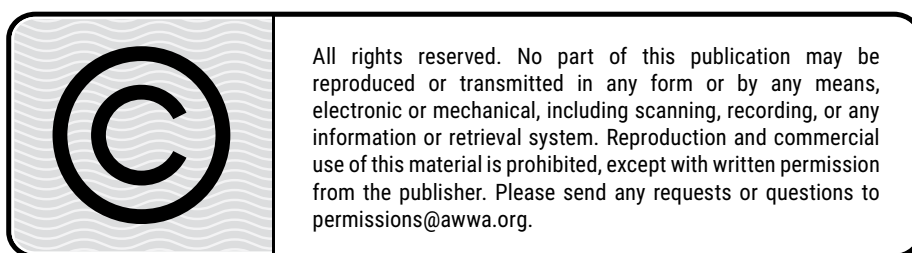
AWWA Standard

This document is an American Water Works Association (AWWA) standard. It is not a specification. AWWA standards describe minimum requirements and do not contain all of the engineering and administrative information normally contained in specifications. The AWWA standards usually contain options that must be evaluated by the user of the standard. Until each optional feature is specified by the user, the product or service is not fully defined. AWWA publication of a standard does not constitute endorsement of any product or product type, nor does AWWA test, certify, or approve any product. The use of AWWA standards is entirely voluntary. This standard does not supersede or take precedence over or displace any applicable law, regulation, or code of any governmental authority. AWWA standards are intended to represent a consensus of the water industry that the product described will provide satisfactory service. When AWWA revises or withdraws this standard, an official notice of action will be placed in the Official Notice section of *Journal AWWA*. The action becomes effective on the first day of the month following the month of *Journal AWWA* publication of the official notice.

American National Standard

An American National Standard implies a consensus of those substantially concerned with its scope and provisions. An American National Standard is intended as a guide to aid the manufacturer, the consumer, and the general public. The existence of an American National Standard does not in any respect preclude anyone, whether that person has approved the standard or not, from manufacturing, marketing, purchasing, or using products, processes, or procedures not conforming to the standard. American National Standards are subject to periodic review, and users are cautioned to obtain the latest editions. Producers of goods made in conformity with an American National Standard are encouraged to state on their own responsibility in advertising and promotional materials or on tags or labels that the goods are produced in conformity with particular American National Standards.

CAUTION NOTICE: The American National Standards Institute (ANSI) approval date on the front cover of this standard indicates completion of the ANSI approval process. This American National Standard may be revised or withdrawn at any time. ANSI procedures require that action be taken to reaffirm, revise, or withdraw this standard no later than five years from the date of ANSI approval. Purchasers of American National Standards may receive current information on all standards by calling or writing the American National Standards Institute, 25 West 43rd Street, Fourth Floor, New York, NY 10036; 212.642.4900, or emailing info@ansi.org.



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Foreword

This foreword is for information only and is not a part of ANSI/AWWA C104/A21.4.*

I. Introduction.

I.A. *Background.* The first cast-iron water mains were not lined but were installed in the same condition in which they came from the molds following cleaning. After many years, it became evident that the interior of the pipe could be affected by certain types of water. The first lining used to combat aggressive waters was a hot-dip bituminous lining that protected the pipe satisfactorily, except in areas where water penetrated the pinholes in the bituminous lining, wherein tuberculation ensued. The need for a better lining to combat tuberculation led to experiments with cement mortar as a lining material.

Cement–mortar linings prevent tuberculation by creating a high pH at the pipe wall and ultimately by providing a physical barrier to the water. Cement–mortar linings are also smooth, which results in high flow coefficients.

In 1922, the first cement–mortar-lined cast-iron pipe was installed in the water distribution system of Charleston, S.C. The pipe was lined in the field using a projectile drawn through the pipe. Since this initial installation, many improvements have been made in the production of cement–mortar-lined pipe. Cement–mortar-lined pipes are now centrifugally lined at the foundry to ensure that quality control is maintained and that a uniform thickness of mortar is distributed throughout the length of the pipe.

I.B. *History.* From 1922 to 1929, many installations of cement–mortar-lined cast-iron pipe were made under various manufacturers' specifications. In 1929, the American Standards Association (ASA) [now the American National Standards Institute (ANSI)] Committee A21 on Cast-Iron Pipe and Fittings issued a tentative standard for cement–mortar linings, which was published by AWWA as a tentative standard in 1932. After many revisions and refinements, the standard was officially adopted by ASA in 1939 under the designation A21.4, American Standard Specifications for Cement–Mortar Lining for Cast-Iron Pipe and Fittings.

From 1940 to 1952, considerable research was done to improve the quality of the cement–mortar lining. The centrifugal process of lining was improved to provide the controls and techniques necessary to ensure uniformity of lining thickness throughout the length of the pipe. Methods of curing cement mortar were investigated, resulting

* American National Standards Institute, 25 West 43rd Street, Fourth Floor, New York, NY 10036.

in the recognition that seal-coat materials could be used as a substitute for the moist-curing process. As a result of this research, a revised edition of the 1939 standard was approved and issued in 1953.

In 1958, Committee A21 was reorganized, and subcommittees were established to study each group of standards in accordance with ASA's review and revision policy. Subcommittee 4, Coatings and Linings for Cast-Iron Pipe, was organized to examine the existing ASA A21.4-53, American National Standard Specifications for Cement-Mortar Lining for Cast-Iron Pipe and Fittings. The subcommittee completed its study of A21.4-53 and submitted a proposed revision to Committee A21 in 1963. The revised third edition was approved and issued in 1964 and included a reduction in the minimum permissible thickness of lining. This reduction was based on more than 20 years of research by the Cast Iron Pipe Research Association (CIPRA) [now the Ductile Iron Pipe Research Association (DIPRA)] on experimental test lines having cement-mortar linings ranging from $\frac{1}{32}$ to $\frac{1}{4}$ in. (0.79 to 6.35 mm) in thickness and the assurance of uniformity of lining thickness using the centrifugal lining process. The 1964 revision also required the cement to meet the requirements of ASTM[†] C150, Specification for Portland Cement.

The next revision was issued in 1971. This revision incorporated a standard test for toxicity of the seal-coat material. The standard was again revised in 1974, and major changes to the section on lining quality were incorporated into that revision.

The 1980 revision to the standard included further changes to the section on lining quality, clarifying problems with some literal interpretations. Metric conversions of all dimensions and physical requirements were also included, and the projection method was added as an allowable means of lining pipe and fittings.

In the 1985 edition of the standard, all references to gray cast-iron pipe were deleted because gray cast-iron pipe was no longer being produced in the United States.

The 1990 revision of the standard included no major changes.

In 1995, the section on cement was expanded to include types of cement other than Portland, and the size range covered by the standard was expanded to include 3-in. through 64-in. (76-mm through 1,600-mm) sizes. Additionally, the section on seal coat was revised to allow the manufacturer the option of providing the cement-mortar lining with or without a seal coat unless otherwise specified; the option of using seal-coat materials other than asphaltic was deleted.

[†] ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428.

In 2003, the format was changed to the current AWWA standard style. Sec. 4.7 on the thickness of lining and Sec. 4.8 on determination of thickness were revised to delete the requirement that the thickness shall be determined while the mortar is wet.

The next edition of ANSI/AWWA C104 was approved by the AWWA Board of Directors on June 8, 2008. In 2008, wastewater and reclaimed water applications were added; water contact surfaces were allowed to be qualified for the limit of toxic substances in accordance with NSF/ANSI 61; and delivery requirements were added for Notice of Nonconformance and Affidavit of Compliance.

In 2013, raw water applications were added; testing to NSF/ANSI 61 or Food and Drug Administration (FDA) Section 175.300 of Title 21 *Code of Federal Regulations* (CFR) Resinous and Polymeric Coatings for products in contact with potable water was added; and a provision for blending other cements with ASTM C150 cement was added.

The last edition was approved on Jan. 16, 2016, and this edition was approved on Oct. 24, 2022.

I.C. *Acceptance.* In May 1985, the US Environmental Protection Agency (USEPA) entered into a cooperative agreement with a consortium led by NSF International (NSF) to develop voluntary third-party consensus standards and a certification program for direct and indirect drinking water additives. Other members of the original consortium included the Water Research Foundation (formerly AwwaRF) and the Conference of State Health and Environmental Managers (COSHEM). AWWA and the Association of State Drinking Water Administrators (ASDWA) joined later.

In the United States, authority to regulate products for use in, or in contact with, drinking water rests with individual states.[‡] Local agencies may choose to impose requirements more stringent than those required by the state. To evaluate the health effects of products and drinking water additives from such products, state and local agencies may use various references, including

1. Specific policies of the state or local agency.
2. Two standards developed under the direction of NSF[§]: NSF/ANSI/CAN[¶] 60, Drinking Water Treatment Chemicals—Health Effects, and NSF/ANSI/CAN 61, Drinking Water System Components—Health Effects.

[‡] Persons outside the United States should contact the appropriate authority having jurisdiction.

[§] NSF International, 789 North Dixboro Road, Ann Arbor, MI 48106.

[¶] Standards Council of Canada, 55 Metcalfe Street, Suite 600, Ottawa, ON K1P 6L5 Canada.

3. Other references, including AWWA standards, *Food Chemicals Codex*, *Water Chemicals Codex*,** and other standards considered appropriate by the state or local agency.

Various certification organizations may be involved in certifying products in accordance with NSF/ANSI/CAN 61. Individual states or local agencies have authority to accept or accredit certification organizations within their jurisdictions. Accreditation of certification organizations may vary from jurisdiction to jurisdiction.

Annex A, “Toxicology Review and Evaluation Procedures,” to NSF/ANSI/CAN 61 does not stipulate a maximum allowable level (MAL) of a contaminant for substances not regulated by a USEPA final Maximum Contaminant Level (MCL). The MALs of an unspecified list of “unregulated contaminants” are based on toxicity testing guidelines (noncarcinogens) and risk characterization methodology (carcinogens). Use of Annex A procedures may not always be identical, depending on the certifier.

ANSI/AWWA C104 does not address additives requirements. Thus, users of this standard should consult the appropriate state or local agency having jurisdiction in order to

1. Determine additives requirements, including applicable standards.
2. Determine the status of certifications by parties offering to certify products for contact with, or treatment of, drinking water.
3. Determine current information on product certification.

II. Special Issues.

II.A. *Use of Seal Coat.* Cement–mortar linings may be provided with or without a seal coat (Sec. 4.12.1). The purchaser of cement–mortar-lined pipe or fittings for use with water that is corrosive to cement, such as very soft water, should determine, by using the appropriate test, that a lining without a seal coat will not impart objectionable hardness or alkalinity to the water. The procedure outlined in Sec. 5.2.2.2, modified by the substitution of the water with which the pipe is to be used, is recommended. The concentration of leachates increases with the aggressiveness of the water and its residual time in the pipe and is indirectly proportional to the diameter of the pipe.

II.B. *Lining of Pipelines in Place.* This standard does not include provisions for cement–mortar lining of pipelines in place. Field lining of pipes is addressed in ANSI/AWWA C602.

** Both publications available from National Academies Press, 500 Fifth Street, NW, Washington, DC 20001.

III. Use of This Standard. It is the responsibility of the user of an AWWA standard to determine that the products described in that standard are suitable for use in the particular application being considered.

III.A. *Purchaser Options and Alternatives.* The following items should be provided by the purchaser:

1. Standard used—that is, ANSI/AWWA C104, Standard for Cement–Mortar Lining for Ductile-Iron Pipe and Fittings, of latest revision.
2. Details of federal, state, provincial, and local requirements (Sec. 4.1).
3. For applications other than potable water, whether compliance with NSF/ANSI/CAN 61, Drinking Water System Components—Health Effects, or alternatively FDA Section 175.300 of Title 21 CFR, is required (Sec. 4.1.2).
4. Type of cement (Sec. 4.2).
5. Thickness of lining (Sec. 4.8). Two thicknesses of lining are available, and purchasers who require a lining thickness twice the standard thickness have the option of so specifying.
6. Seal-coat requirements (Sec. 4.12). The purchaser has the option of specifying if the cement–mortar lining is to be provided with or without a seal coat.
7. Notice of nonconformance (Sec. 6.1).
8. Request for an affidavit of compliance (Sec. 6.2).

III.B. *Modification to Standard.* Any modifications to the provisions, definitions, or terminology in this standard must be provided by the purchaser.

IV. Major Revisions. Major revisions made to the standard in this edition include the following:

1. Clarification to acceptable water use in Sec. 4.4.
2. Clarification to test record requirements in Sec. 5.1.5.
3. Clarification to notice of nonconformance in Sec. 6.1.

V. Comments. If you have any comments or questions about this standard, please call AWWA Engineering and Technical Services at 303.794.7711; FAX at 303.795.7603; write to the department at 6666 West Quincy Avenue, Denver, CO 80235-3098; or email at standards@awwa.org.

CBGL911 GLASS LINING



REV 2022

APPLICATION

Glass lining inside of ductile iron pipe has been used successfully for over 40 years. Most commonly used for its "non stick" characteristics to convey fluids containing higher solids concentration such as sludge, scum, and grease in sewage service and wastewater treatment plant applications.

CBGL911 LINING MATERIAL

Glass lined pipe shall conform to ASTM B1000-21. The standard of quality shall be CBGL911. The glass powder shall be specifically formulated and shall be vitreous and inorganic. The lining must be proven to be successful in conveying sludge, scum, and grease in sewage service and wastewater treatment plant applications.

- The lining Applicator must have a minimum of 5 years proven experience in production of glass lined ductile iron piping conforming to ASTM B1000.
- Any request for substitution of material must be accompanied by a successful history of lining ductile iron pipe and fittings which are used in these applications.
- The lining Applicator must be an ISO 9001:2000 certified company.
- The manufacturer must be able to provide independent third party engineer sealed test reports verifying the following results.

These physical and chemical tests must have been performed on coupons from the applicators regular production lined ductile iron pipe and/or fittings.

1. ASTM D5965 test (per ASTM B1000) for density in the range of 2.5 to 3.0 g/cm³
2. Immersion testing in a solution of 8% sulfuric acid at a temperature of 148 °F. (65 °C) for a period of 10 minutes minimum. The glass shall have a negligible visible loss of surface gloss.
3. Glass shall be capable of withstanding an instantaneous thermal shock of 350 °F (177 °C) without cracking, crazing, blistering or spalling.
4. Glass lining have a minimum hardness of 5 on the MOHS scale.
5. Glass shall have maximum weight loss of 3 mg/in² when exposed to an HCl solution with pH of 3 at 125 °F for 30 minutes.
6. Glass shall have maximum weight loss of 3 mg/in² when exposed to an NaOH solution with pH of 10 at 125 °F for 30 minutes.
7. Glass shall have maximum weight loss of 3 mg/in² when exposed to a boiling citric acid solution per ASTM C283 for 10 minutes.
8. The finished lining shall be able to withstand a strain of 0.001 in/in of the base metal without visible damage to the glass



WHY USE GLASS LINING?

Glass Lined ductile iron pipe assists in keeping your systems clean and free from clogging. Glass lining also helps protect the pipe from abrasion present in grit removal and other sewer process systems.

Glass lining is a porcelain material and common sewer materials resist sticking to it like they do with epoxy lining materials.

CBGL911 glass lining is also excellent for high temperature requirements where other protective linings cannot remain in service.

Did you know that you can get process valves with CBGL911 glass lining? Many valve manufacturers currently supply CBGL911 glass lined valves



Struvite buildup in sludge piping system

CBGL911 GLASS LINING

CBGL911 LINING APPLICATION

A. Surface Preparation

The entire surface area to receive the glass compound shall be inspected for the presence of oil, grease, or any other soluble substances. Any areas found with contamination shall be solvent cleaned to completely remove these substances. After cleaning, all surface areas shall be mechanically ground so as to remove any casting imperfections leaving only a smooth surface with no raised projections. Finally, the entire surface area to be lined shall be struck with a blast media so that all remaining impurities from the cleaning operations are removed. Only tightly adhering cast-in oxides may be left on the surface.

B. Lining

After surface preparation, the interior of the pipe or fitting shall receive an application of base glass compound which has been suspended in a water carrier. After the water has evaporated, the pipe or fitting shall be oven-fired at a temperature of 1375 degrees F. (746 degrees C) in order to fuse the glass to the metal surface. The subsequent finish coats are similarly processed with a resultant lining thickness of 10 mils (.25mm) minimum.



Surface grinding prep for a ductile iron casting

INSPECTION

1. All ductile iron pipe and fittings shall be checked for thickness using a magnetic film thickness gage by the method outlined in SSPC-PA2 Film Thickness Rating.
2. The lining shall be in accordance with the Applicator's industry standard tolerance for coverage and continuity. All pipe and fittings shall be visually inspected after lining at the Applicator's facility. Each and every pipe or fitting shall be tested with a 67.5 volt Tinker Rasor electronic device with a wetted sponge to determine the presence of pinholes or voids. Excessive pinholes shall be cause for rejection at the Applicator's facility.
3. Each and every pipe or fitting shall be marked on the interior glass surface using a permanent marker; the date of application and tests, the facility location and the number of acceptable pinholes detected. Quality Assurance reports will be maintained at the Applicator's facility evidencing all work performed and data recorded.
4. The Owner/or his Representative may witness at the Applicator's facility; the glass application process, review the Quality Assurance Reports and witness the actual tests conducted upon his product, all prior to the product leaving the facility. This may be done by written Specification or by sufficient notice to the Applicator to allow coordination of the visit.

CERTIFICATION

The Applicator must be able to supply a notarized Certificate of Compliance attesting to the fact that the glass lining has met all of the requirements of this specification and that the material lined has met the applicable standards of AWWA and ANSI where noted.



CBGL911 Glass Lined Valve Components

CBGL911 GLASS LINING

HANDLING

Glass lined pipe and fittings must be handled only from the exterior surfaces; no forks, straps, hooks, etc; shall come into contact with the interior surface of glass. Pipe and fittings that have been altered after it left the Applicator's facility, such as field cuts or further fabrication, may not exhibit the same tested results as was documented at the Applicator's facility.

FIELD REPAIR

CBGL911 Glass repair Epoxy shall be used for touch-up or repair in accordance with the Manufacturer's recommendation below.

1. Remove burrs caused by field cutting of ends or handling damage and smooth out the edge of the lining if rough.
2. Remove all traces of oil, grease, asphalt, dust, dirt, etc.
3. Remove any damaged lining caused by field cutting operations' or handling and clean any exposed metal by sanding or scraping. Sandblasting or power tool cleaning roughening is also acceptable. It is recommended that any loose lining be removed by chiseling, cutting, or scraping into well-adhered lined area before patching. Be sure to overlap at least 1" of lining in the area to be repaired.
4. With the area to be sealed or repaired absolutely clean and suitably roughened, apply a coat of CBGL Glass Repair Epoxy using the following procedure:
 - a. Mixing procedure - The Repair Kit for CBGL 911 Glass Lining contains two small cans of CBGL 911 Glass Repair Epoxy. CBGL 911 Glass Repair Epoxy is a two-component epoxy and the contents of the small container shall be mixed with the contents of the large container. If less than the full contents of each can is to be mixed, the material may be mixed using the mixing ratio printed on the labels. Add Q4-5311 Part B to A4-5311 Part A. Mix thoroughly with enclosed paddle. All activated material must be used within one hour of mixing.
 - b. Application of Material- After the material has been thoroughly mixed, it can be applied to the properly prepared surface by brush. Brushing is usually best due to the fact that the areas to be repaired are usually small.
5. It is important to coat the entire freshly cut exposed metal surface of the cut pipe end. To ensure proper sealing, overlap at least one inch of the lining with this repair material.



TECHNICAL DATA CBGL 911 GLASS REPAIR EPOXY



	DESCRIPTION:	A brushable polyamidoamine epoxy designed for sealing cut ends and repairs when pipes are lined with CBGL 911 Glass Lining.
	LIMITATIONS:	CBGL911 Glass Repair Epoxy should be used over glass or bare metal for repairing cut ends of pipe or damaged areas in CBGL Glass Lining. The Glass Repair Epoxy must be used over properly prepared ductile iron surfaces or roughened glass lining.
	SURFACE PREPARATION	The surface preparation shall be equal to the specifications for the project or as outlined in the touch-up procedure. Do not apply glass repair epoxy over wet or frozen surfaces, or over glass surfaces that have not been roughened.
	DRY FILM: THICKNESS:	As outlined in specifications, usually 3.0 to 5.0 mils per coat in two coats for 6.0 to 10.0 mils dry film thickness.
	APPLICATION:	Brush
	THINNING:	Thin or cleanup with Methyl Ethyl Ketone.
	PHYSICAL DATA:	VOLATILE ORGANIC CONTENTS: 2.90 lbs. per gallon
	SAFETY DATA:	See individual product label for safety and health data information. Individual Material Safety Data Sheets are available upon request.



Allstates Coatings Co.®

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903-845-6436 • www.allstatescoatings.com

PRODUCT INFORMATION

ALT-220 Acryspphalt™

ALT-220 Acryspphalt™

USER friendly, complies with all V.O.C. emissions and E.P.A. standards.



Certified to
NSF/ANSI 61

GENERAL: A non-flammable asphalt emulsion blended with an extremely high concentration of unique polymer and plasticizers to achieve extreme adherence to the substrate, durability and flexibility. Acryspphalt™ is a black one component, sprayable or dipped coating specially designed to protect from corrosion, water and environmental elements for prime or top coat on any zinc, surface-metal, concrete, wood or brick.

SURFACE

PREPARATION: Surface must be dry, free from dirt, dust and oil. If necessary, dirt, loose pieces of coatings or chemical residue should be removed. For best results, DO NOT use when surface is wet or when temperatures are below 40°F (4.4°C). **CAUTION:** Wear adequate eye protection and breathing apparatus during cleaning and coating procedure.

APPLICATION: Mix Acryspphalt™ before using. No thinning is necessary. Apply by dipping or with adequate spray or other appropriate equipment. NSF Certified - Maximum 4 mils DFT, 96 hours cure at 77°F (25°C). For interior application plenty of ventilation and fresh air is required.

COVERAGE: For estimation purposes use 860+ sq. feet per gallon (1 mil dry film)

CAUTION: KEEP FROM FREEZING. DO NOT store in direct sunlight or where temperatures exceed 100°F (37.8°C). Container should be closed when not in use. Keep out of the reach of children. Wash tools in water. Use paint thinner if material has dried.

PACKAGING: 5 gallon pails, 55 gallon drums, 275/330 gallon tote tanks and bulk.

Material Specifications:

Requirements:

Weight per Gallon @ 77°F, ASTM D1010, (lbs)	8.30 - 8.60
Nonvolatile %, ASTM D2939	50.00 - 54.00
Viscosity @ 77°F, ASTM D562 (Kreb's Stormer Unit)	58 - 70 KU
Softening Point (0°F), ASTM D36	>250
Penetration (0.1 mm), ASTM D5	25-30
V.O.C. (lbs per gallon)	0.00-0.06
Std. Spec. for Emulsified Asphalt	
AASHTO Designation M140-70	meets spec.
ASTM Designation D977-73	meets spec.
Salt Fog, ASTM B117; 2ml/hr (FIRST appearance of rust)	198 hrs, no blistering, no loss of
.....	adhesion or re-emulsification
Resistance to water, ASTM D2939	no blistering, no loss of adhesion,
.....	no re-emulsification
Dry Time (mins) ASTM D2939 @ 5 mil wet	5-10
Shelf Life (yrs)	6 months

This information is to assist customers in determining if this product is suitable for the proposed application, and to satisfy themselves as to the suitability of the contents. Nothing herein shall constitute a warranty, express or implied, including any warranty of merchantability or fitness, nor is protection from any law or patent implied. Only products bearing the NSF Mark on the product, product packaging, and/or documentation shipped with the product are Certified. Published technical data and instructions are subject to change without notice. Contact your Allstates Coatings' representative for additional technical data and instruction.



C&B Piping Coating Submittal

Exterior Coating: Pota-Pox® Plus Series L140F-1211 Epoxy shop coating

Manufacturer: Tnemec

Surface Prep: Remove oil, grease, and other surface contaminants per NAPF 500-03-01
Abrasive blast clean per NAPF 500-03-04 for pipe, NAPF 500-03-05 for cast fittings

Thickness: 3-5 mils Nominal DFT

Notes:

- C&B Piping applies this coating per the Tnemec recommendations attached on the PDS.
- The shop coating provides some temporary protection during loading, transit, unloading, jobsite handling, and jobsite storage. After shipment from C&B, the shop coat will be subsequently affected by a combination of physical and environmental conditions outside of our control. If products are properly handled and stored on the jobsite, our shop coat may meet the standards to be used as the system primer as long as the application of subsequent field applied coatings are to be performed within 30 days of receipt of products. All subsequent field applied coatings must strictly follow the recommendation of the coatings manufacturer.
- The applicability of the shop coat to be subsequently used as the primer for a paint system, and the extent of testing and surface prep required at that time, is to be determined by the contractor or field paint subcontractor applying the paint system. Please refer to the PDS attached for maximum recoat window based on conditions.
- If the shop coating is not desired, it can be ordered as BARE, or fully removed by field coatings contractor per applicable NAPF 500-03 standards at cost by others.

Handling & Storage:

- Careful handling and effective storage are critical in limiting physical and environmental damage to the shop coating. Shop coatings exposed to weather and/or chemical exposure will limit the maximum recoat window and potentially cause adhesion issues.





POTA-POX® PLUS SERIES L140F

PRODUCT PROFILE

GENERIC DESCRIPTION	Polyamidoamine Epoxy
COMMON USAGE	Versatile epoxy coating (factory accelerated version of L140) for protection and finishing of steel. It has excellent resistance to abrasion and is suitable for immersion service. This product can be used for lining storage tanks that contain demineralized, deionized, distilled, or potable water.
COLORS	1255 Beige, 1211 Red, 158L Tank White. Note: Epoxies chalk with extended exposure to sunlight. Lack of ventilation, incomplete mixing, miscatalyzation or the use of heaters that emit carbon dioxide and carbon monoxide during application and initial stages of curing may cause yellowing to occur.
SPECIAL QUALIFICATIONS	Certified by NSF International in accordance with NSF/ANSI/CAN Std. 61 and the extraction requirements of NSF/ANSI/CAN 600. Series L140F is qualified for interior use on tanks and fittings. Reference Tnemec's certified product listing at www.nsf.org for additional details.

COATING SYSTEM

SURFACER/FILLER/PATCHER	Series 215, 217, 218
PRIMERS	Self-priming, or Series 1, 91-H ₂ O, 94-H ₂ O, N140, N140F, L140, V140, V140F, 141
TOPCOATS	Interior: Series 21, 22, FC22, L140, L140F, N140, N140F, V140, V140F, 141 Exterior: Series 22, 27, 66, L69, L69F, N69, N69F, V69, V69F, 73, 118, L140, L140F, N140, N140F, V140, V140F, 141, 156, 157, 161, 180, 181, 700, V700, 701, V701, 1026, 1028, 1029, 1078, 1078V, 1080, 1081, 1094, 1095, 1096, 1224. Note: When topcoating with Series 700, V700, 701, or V701, an intermediate coat of Series 73, 1095, or 1096 is required. Note: The following maximum recoat times apply; Series 1094, 1095, 14 days. Note: The following recoat times apply for Series L140F: Immersion Service—Surface must be scarified after 30 days. Atmospheric Service—After 30 days, scarification or an epoxy tie-coat is required. Contact your Tnemec representative for specific recommendations.

SURFACE PREPARATION

STEEL	Immersion Service: SSPC SP10/NACE 2 Near White Blast Cleaning or ISO Sa 2 1/2 Very Thorough Blast Cleaning with a minimum angular anchor profile of 1.5 mils. Non-Immersion Service: SSPC SP6/NACE 3 Commercial Blast Cleaning or ISO Sa 2 Thorough Blast Cleaning with a minimum angular anchor profile of 1.5 mils. Note: Commercial Blast Cleaning generally produces the best coating performance for this exposure. If conditions will not permit this, in moderate exposures Series L140F may be applied to SSPC SP2 or SP3 Hand or Power Tool Cleaned Surfaces (SSPC Rust Grade Condition C).
CAST/DUCTILE IRON	All external surfaces of ductile iron pipe and fittings shall be delivered to the application facility without asphalt or any other protective lining on the exterior surface. All oils, small deposits of asphalt paint, grease, and soluble deposits should be removed and uniformly abrasive blasted using angular abrasive in accordance with NAF 500-03-04: External Pipe Surface condition. When viewed without magnification, the exterior surfaces shall be free of all visible dirt, dust, loose annealing oxide, rust, mold coating and other foreign matter. Any area where rust reappears before application shall be reblasted. The surface shall contain a minimum angular anchor profile of 1.5 mils (38.1 microns) (Reference NACE RP0287 or ASTM D 4417, Method C).
CONCRETE	Allow new cast in place concrete to cure a minimum of 28 days at 75°F (24°C). Verify concrete dryness in accordance with ASTM F 1869 "Standard Test Method for Measuring Moisture Vapor Emission Rate of Concrete Subfloor Using Anhydrous Calcium Chloride" (moisture vapor transmission should not exceed three pounds per 1,000 square feet in a 24 hour period), F 2170 "Standard Test Method for Determining Relative Humidity in Concrete using in-situ Probes" (relative humidity should not exceed 80%), or D 4263 "Standard Test Method for Indicating Moisture in Concrete by the Plastic Sheet Method" (no moisture present). Prepare concrete surfaces in accordance with NACE No. 6/SSPC SP13 Joint Surface Preparation Standards and ICRI Technical Guidelines. Abrasive blast, shot blast, water jet or mechanically abrade concrete surfaces to remove laitance, curing compounds, hardeners, sealers and other contaminants and to provide an ICRI CSP 2-3 surface profile. Large cracks, voids and other surface imperfections should be filled with a recommended filler or surfacer.
PAINTED SURFACES	Non-Immersion Service: Ask your Tnemec representative for specific recommendations.
PRIMED SURFACES	Immersion Service: Scarify the Series L140F prime coat surface by abrasive-blasting with a fine abrasive before topcoating if the Series L140F prime coat has been in exterior exposure for 30 days or longer and Series 66, L69, L69F, N69, N69F, V69, N69 or 161 is the specified topcoat.
ALL SURFACES	Must be clean, dry and free of oil, grease and other contaminants.

TECHNICAL DATA

VOLUME SOLIDS	66.0 ± 2.0% (mixed) †
RECOMMENDED DFT	2.0 to 10.0 mils (50 to 255 microns) per coat. Note: Reference the NSF website at www.nsf.org for details on the maximum allowable DFT. Note: Number of coats and thickness requirements will vary with the substrate, application method, and exposure. Contact your Tnemec representative.

CURING TIME AT 5 MILS DFT

Temperature	To Handle	To Recoat	Immersion
75°F (24°C)	4 hours	5 hours	7 days
65°F (18°C)	7-8 hours	9-11 hours	8 days
55°F (13°C)	12-14 hours	16-20 hours	9-10 days
45°F (7°C)	18-22 hours	28-32 hours	12-13 days
35°F (2°C)	28-32 hours	46-50 hours	16-18 days

Curing time varies with surface temperature, air movement, humidity and film thickness. **Ventilation:** When used in enclosed areas, provide adequate ventilation during application and cure. **Note:** Refer to product listings on www.nsf.org for specific potable water return to service information.

POTA-POX® PLUS | SERIES L140F

VOLATILE ORGANIC COMPOUNDS

Unthinned: 0.75 lbs/gallon (90 grams/litre) (TBAC exempt)
Unthinned: 1.02 lbs/gallon (122 grams/litre)
Thinned 5% (No. 49 Thinner): 0.75 lbs/gallon (90 grams/litre) (TBAC exempt)
Thinned 5% (No. 49 Thinner): 1.01 lbs/gallon (122 grams/litre) †

HAPS

Unthinned: 0.1 lbs/gal solids
Thinned 5% (No. 49 Thinner): 0.1 lbs/gal solids

THEORETICAL COVERAGE

1,059 mil sq ft/gal (26.0 m²/L at 25 microns). See APPLICATION for coverage rates. †

NUMBER OF COMPONENTS

Two: Part A (amine) and Part B (epoxy) — One (Part A) to one (Part B) by volume.

PACKAGING

	Part A	Part B	Yield (mixed)
Large Kit	5 gallon pail	5 gallon pail	10 gallons (37.9 L)
Small Kit	1 gallon can	1 gallon can	2 gallons (7.6 L)

NET WEIGHT PER GALLON

14.62 ± 0.25 lbs (6.63 ± 0.11 kg) (mixed) †

STORAGE TEMPERATURE

Minimum 20°F (-7°C) Maximum 110°F (43°C)
 For optimum application properties, material temperature should be above 60°F (16°C) prior to application.

TEMPERATURE RESISTANCE

(Dry) Continuous 250°F (121°C) Intermittent 275°F (135°C)

SHELF LIFE

Part A: 24 months; Part B: 12 months at recommended storage temperature.

FLASH POINT - SETA

Part A: 98°F (37°C) Part B: 95°F (35°C)

HEALTH & SAFETY

Paint products contain chemical ingredients which are considered hazardous. Read container label warning and Material Safety Data Sheet for important health and safety information prior to the use of this product.
Keep out of the reach of children.

APPLICATION

COVERAGE RATES

	Dry Mils (Microns)	Wet Mils (Microns)	Sq Ft/Gal (m ² /Gal)
Suggested	6.0 (150)	9.0 (230)	176 (16.4)
Minimum	2.0 (50)	3.0 (75)	529 (49.2)
Maximum	10.0 (255)	15.0 (375)	106 (9.8)

Note: Roller or brush application requires two or more coats to obtain recommended film thickness. Allow for overspray and surface irregularities. Wet film thickness is rounded to the nearest 0.5 mil or 5 microns. Application of coating below minimum or above maximum recommended dry film thicknesses may adversely affect coating performance. Reference the NSF website at www.nsf.org for details on the maximum allowable DFT. †

MIXING

Start with equal amounts of Series L140F Parts A and B. Power mix contents of each container separately, making sure no pigment remains on the bottom. Pour a measured amount of Part B into a clean container large enough to hold both components. Add an equal volume of Part A to Part B while under agitation. Continue agitation until the two components are thoroughly mixed. **Note:** Both components must be above 50°F (10°C) prior to mixing. For optimum mixing and application properties, the material should be above 60°F (16°C).

Thin by volume and thoroughly mix. Failure to thoroughly mix the Part A and Part B components prior to thinning can affect product's gloss and performance. Do not use mixed material beyond pot life limits. **Note:** For application to surfaces between 35°F to 50°F (2°C to 10°C), allow mixed material to stand 30 minutes and restir before using.

THINNING

Use No. 49 Thinner. For air spray, thin up to 5% or 1/4 pint (190 mL) per gallon. No thinning necessary for airless spray. For roller or brush application, thin up to 5% or 1/4 pint (190 mL) per gallon. **Caution: Series L140F NSF certification is based on thinning with No. 49 Thinner only.** Use of any other thinner voids NSF/ANSI/CAN Std 61.

POT LIFE

2 hours at 50°F (10°C) 1 hour at 75°F (24°C) 30 minutes at 100°F (38°C)

SPRAY LIFE

30 minutes at 75°F (24°C)

Note: Spray application after listed times will adversely affect ability to achieve recommended dry film thickness.

APPLICATION EQUIPMENT

Air Spray

Gun	Fluid Tip	Air Cap	Air Hose ID	Mat'l Hose ID	Atomizing Pressure	Pot Pressure
DeVilbiss JGA	E	704	5/16" or 3/8" (7.9 or 9.5 mm)	3/8" or 1/2" (9.5 or 12.7 mm)	50-80 psi (3.4-5.5 bar)	10-20 psi (0.7-1.4 bar)

Low temperatures or longer hoses require higher pot pressure.

Airless Spray

Tip Orifice	Atomizing Pressure	Mat'l Hose ID	Manifold Filter
0.015"-0.019" (380-485 microns)	3500-5100 psi (241-351 bar)	1/4" or 3/8" (6.4 or 9.5 mm)	60 mesh (250 microns)

Use appropriate tip/atomizing pressure for equipment, applicator technique, and weather conditions.

Roller: Use a 3/8" or 1/2" (9.5 mm or 12.7 mm) synthetic woven nap roller cover. Use longer nap to obtain penetration on rough or porous surfaces.

Brush: Recommended for small areas only. Use high-quality natural or synthetic bristle brushes.

SURFACE TEMPERATURE

Minimum 35°F (2°C) Maximum 135°F (57°C)
 The surface should be dry and at least 5°F (3°C) above the dew point. Coating will not cure below minimum surface temperature.

POTA-POX[®] PLUS | SERIES L140F**CLEANUP**

Flush and clean all equipment immediately after use with the recommended thinner or MEK.

† Values may vary with color.

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